



**VIDYA VIKAS INSTITUTE OF  
ENGINEERING AND TECHNOLOGY,  
MYSORE**

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**VTU Previous Question Papers**

**BE-I & II Sem**

**(Mathematics, Physics & Chemistry Cycle)**

**Dec.2024 / Jan.2025 & 2022 Scheme**

***VVIET***

***Bannur road, Alanahalli, Mysore-570028***

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## First/Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

### Basic Electronics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*

*2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                      | M  | L  | C   |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Explain the VI characteristics of PN junction diode.                                                                                                                                                 | 06 | L2 | CO1 |
|            | b. | Explain the operation of center tapped full wave rectifier.                                                                                                                                          | 08 | L2 | CO1 |
|            | c. | Design the zener diode regulator for the following specification.<br>$I_L = 10 \text{ mA}$ , $V_0 = 5 \text{ V}$ , $I_{zmin} = 1 \text{ mA}$ , $I_{zmax} = 25 \text{ mA}$ and $V_i = 10 \text{ V}$ . | 06 | L3 | CO1 |
| OR         |    |                                                                                                                                                                                                      |    |    |     |
| Q.2        | a. | Explain the halfwave rectifier with 'C' filter.                                                                                                                                                      | 07 | L2 | CO1 |
|            | b. | Explain the operation of zener voltage regulator.                                                                                                                                                    | 07 | L2 | CO1 |
|            | c. | With relevant circuit explain the concept of Load line analysis of a diode.                                                                                                                          | 06 | L2 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                      |    |    |     |
| Q.3        | a. | Explain common emitter input and output characteristics of a BJT.                                                                                                                                    | 08 | L2 | CO2 |
|            | b. | Explain various voltages and currents of BJT.                                                                                                                                                        | 07 | L1 | CO2 |
|            | c. | With neat circuit diagram, explain DC line concept of a transistor amplifier to fix the Q point.                                                                                                     | 05 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                                      |    |    |     |
| Q.4        | a. | Explain the construction and operation of a JFET.                                                                                                                                                    | 08 | L2 | CO2 |
|            | b. | Explain the construction and operation of a enhancement MOSFET.                                                                                                                                      | 08 | L2 | CO2 |
|            | c. | Mention any four differences between BJT and JFET.                                                                                                                                                   | 04 | L1 | CO2 |
| Module – 3 |    |                                                                                                                                                                                                      |    |    |     |
| Q.5        | a. | Define (i) CMRR (ii) Slew rate (iii) Input offset voltage (iv) Input offset current.                                                                                                                 | 08 | L1 | CO2 |
|            | b. | Design a op-Amp circuit to get output voltage $V_0 = -(3V_1 + 2V_2 + 0.5V_3)$ . Assume $R_f = 10 \text{ k}\Omega$ .                                                                                  | 06 | L1 | CO2 |
|            | c. | Derive the expression for the output voltage of a non-inverting amplifier.                                                                                                                           | 06 | L2 | CO2 |
| 1 of 2     |    |                                                                                                                                                                                                      |    |    |     |

OR

|     |    |                                                                                            |    |    |     |
|-----|----|--------------------------------------------------------------------------------------------|----|----|-----|
| Q.6 | a. | Mention any six ideal characteristics of an opamp.                                         | 06 | L1 | CO2 |
|     | b. | Derive the expression for output voltage of a differentiator circuit.                      | 07 | L2 | CO2 |
|     | c. | Derive the expression for the output voltage of a three input inverting summing amplifier. | 07 | L2 | CO2 |

## Module – 4

|     |    |                                                                                                                                           |    |    |     |
|-----|----|-------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.7 | a. | Convert : (i) $(23.25)_{10} = ( )_2 = ( )_{16}$<br>(ii) $(3250)_{10} = ( )_8 = ( )_{16}$                                                  | 06 | L2 | CO3 |
|     | b. | Given the two binary numbers $X = 1010100$ and $Y = 1000011$ , perform the subtraction (i) $X - Y$ (ii) $Y - X$ , using two's complement. | 08 | L2 | CO3 |
|     | c. | Design Half adder using basic gates.                                                                                                      | 06 | L2 | CO3 |

OR

|     |    |                                                    |    |    |     |
|-----|----|----------------------------------------------------|----|----|-----|
| Q.8 | a. | Prove Demorgan's theorem for two variables.        | 06 | L2 | CO3 |
|     | b. | Design full adder using basic gates.               | 08 | L2 | CO3 |
|     | c. | Express the Boolean function $F = xy + \bar{x}z$ . | 06 | L2 | CO3 |

## Module – 5

|     |    |                                                      |    |    |     |
|-----|----|------------------------------------------------------|----|----|-----|
| Q.9 | a. | Explain the operation of LVDT.                       | 08 | L2 | CO4 |
|     | b. | Explain briefly block diagram of a communication.    | 08 | L2 | CO5 |
|     | c. | Mention the applications of optoelectric transducer. | 04 | L1 | CO2 |

OR

|      |    |                                                     |    |    |     |
|------|----|-----------------------------------------------------|----|----|-----|
| Q.10 | a. | Explain the operation of photodiode transducer.     | 06 | L2 | CO4 |
|      | b. | Explain briefly Thermistor transducers.             | 08 | L2 | CO4 |
|      | c. | Define Modulation. Explain the need for modulation. | 06 | L1 | CO5 |

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# CBCS SCHEME

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BCHEC102/202

**First/Second Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025**

## Applied Chemistry for Civil Engineering Stream

Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**

**2. M : Marks , L: Bloom's level , C: Course outcomes.**

**3. VTU formula Handbook is permuted.**

| Module – 1    |    |                                                                                                                                                                                                                                     | M | L  | C   |
|---------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 1             | a. | Describe the classification of refractory materials based on chemical composition.                                                                                                                                                  | 7 | L2 | CO1 |
|               | b. | Explain the testing of cement by EDTA method.                                                                                                                                                                                       | 7 | L3 | CO1 |
|               | c. | Explain the properties and applications of Aluminium and its alloys.                                                                                                                                                                | 6 | L2 | CO1 |
| <b>OR</b>     |    |                                                                                                                                                                                                                                     |   |    |     |
| 2             | a. | Describe the preparation of soda-line glass. Mention the properties and applications of glass.                                                                                                                                      | 7 | L2 | CO1 |
|               | b. | Describe the manufacturing of cement by wet method.                                                                                                                                                                                 | 7 | L2 | CO1 |
|               | c. | Explain the properties and applications of Iron and its alloys.                                                                                                                                                                     | 6 | L2 | CO1 |
| Module – 2    |    |                                                                                                                                                                                                                                     |   |    |     |
| 3             | a. | What are secondary batteries? Explain the construction and working of Li-ion battery.                                                                                                                                               | 7 | L2 | CO2 |
|               | b. | Explain the following types of corrosion :<br>i) Differential metal corrosion<br>ii) Caustic embrittlement - stress corrosion.                                                                                                      | 7 | L2 | CO2 |
|               | c. | Explain the construction and working of methanol – oxygen (MeOH – O <sub>2</sub> ) fuel cell.                                                                                                                                       | 6 | L2 | CO2 |
| <b>OR</b>     |    |                                                                                                                                                                                                                                     |   |    |     |
| 4             | a. | Define Photovoltaic cell. Explain the construction and applications of photovoltaic cell.                                                                                                                                           | 7 | L2 | CO2 |
|               | b. | Explain the following corrosion control methods :<br>i) Galvanization<br>ii) Sacrificial anode method.                                                                                                                              | 7 | L2 | CO2 |
|               | c. | Explain the electrochemical corrosion of steel in concrete.                                                                                                                                                                         | 6 | L2 | CO2 |
| Module – 3    |    |                                                                                                                                                                                                                                     |   |    |     |
| 5             | a. | Explain the determination of total hardness of water by EDTA method.                                                                                                                                                                | 7 | L2 | CO3 |
|               | b. | In a COD experiment, 15.4 cm <sup>3</sup> of 0.04N FAS solution is required to react with 25 cm <sup>3</sup> of water sample. The blank titre value is 28.0 cm <sup>3</sup> . Calculate chemical oxygen demand of the water sample. | 7 | L3 | CO3 |
|               | c. | Explain the synthesis of nanomaterials by sol-gel method.                                                                                                                                                                           | 6 | L2 | CO3 |
| <b>1 of 2</b> |    |                                                                                                                                                                                                                                     |   |    |     |

## OR

|   |    |                                                                                                                                                                                                                                                                                                                               |   |    |     |
|---|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 6 | a. | Explain the process of water purification by forward osmosis.                                                                                                                                                                                                                                                                 | 7 | L2 | CO3 |
|   | b. | Explain the following size dependent properties of nano materials with example :<br>i) Surface area<br>ii) Catalytic property.                                                                                                                                                                                                | 7 | L2 | CO3 |
|   | c. | 50 mL of water sample required 20 mL of 0.01 M EDTA solution using EBT indicator. In other experiment 50 mL of the water sample was boiled and precipitate was removed by filtration required 10 mL of 0.01M EDTA using EBT indicator. Calculate :<br>i) Total hardness<br>ii) Permanent hardness<br>iii) Temporary hardness. | 6 | L3 | CO3 |

## Module – 4

|   |    |                                                                                                                                                                                                       |   |    |     |
|---|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 7 | a. | Explain the synthesis, properties and applications of chloropolyvinyl chloride (CPVC).                                                                                                                | 7 | L2 | CO4 |
|   | b. | A polymer sample contains 50, 100, 200 and 300 molecules having molecular mass 1500, 2500, 3500 and 4500 respectively. calculate the number average and weight average molecular mass of the polymer. | 7 | L3 | CO4 |
|   | c. | Mention the properties and applications of Geo-polymer concrete.                                                                                                                                      | 6 | L2 | CO4 |

## OR

|   |    |                                                                                           |   |    |     |
|---|----|-------------------------------------------------------------------------------------------|---|----|-----|
| 8 | a. | Describe the synthesis, property and applications of Nylon fibers.                        | 7 | L2 | CO4 |
|   | b. | Define biodegradable polymers. Explain the synthesis and applications of polylactic acid. | 7 | L2 | CO4 |
|   | c. | What are adhesives? List the properties and applications of epoxy resin.                  | 6 | L2 | CO4 |

## Module – 5

|   |    |                                                                                                   |   |    |     |
|---|----|---------------------------------------------------------------------------------------------------|---|----|-----|
| 9 | a. | Define the following terms with example :<br>i) Phase<br>ii) Component<br>iii) Degree of freedom. | 7 | L2 | CO5 |
|   | b. | With the help of neat phase diagram, describe the lead-silver system.                             | 7 | L2 | CO5 |
|   | c. | Describe the instrumentation and applications of conductometric sensors.                          | 6 | L2 | CO5 |

## OR

|    |    |                                                                                                |   |    |     |
|----|----|------------------------------------------------------------------------------------------------|---|----|-----|
| 10 | a. | Describe the applications of conductometric sensors in estimation of acid mixtures.            | 7 | L2 | CO5 |
|    | b. | Describe the instrumentation and applications of potentiometric sensors in estimation of iron. | 7 | L2 | CO5 |
|    | c. | Explain the determination of pH of soil sample using pH sensors.                               | 6 | L2 | CO5 |

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BCHEE102/202

**First/Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025**

**Chemistry for EEE Stream**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                                                                                                                                                                                | M | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Explain the conductors, semiconductors and insulators based on band theory.                                                                                                                                                                                                                                                                                    | 7 | L2 | CO1 |
|            | b. | Describe the purification of electronic grade Silicon by Float zone method.                                                                                                                                                                                                                                                                                    | 6 | L2 | CO1 |
|            | c. | What are conducting polymers? Explain the mechanism of conduction in polyacetylene.                                                                                                                                                                                                                                                                            | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                |   |    |     |
| Q.2        | a. | What is electroless plating? Describe electroless plating of copper in the manufacture of double sided PCB.                                                                                                                                                                                                                                                    | 6 | L2 | CO1 |
|            | b. | Explain the preparation, properties and commercial applications of graphene oxide.                                                                                                                                                                                                                                                                             | 7 | L2 | CO1 |
|            | c. | Define number average and weight average molecular weights. In a sample of a polymer, 20% molecules have molecular mass 15000 g/mol, 35% molecules have molecular mass 25000 g/mol and remaining molecules have molecular mass 20000 g/mol. Calculate the number average and weight average molecular weights of the polymer. Calculate PDI and comment on it. | 7 | L3 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                                                                                                                                                                                |   |    |     |
| Q.3        | a. | What are PV cells? Explain the construction and working of a typical PV cell. Mention its advantages.                                                                                                                                                                                                                                                          | 6 | L2 | CO2 |
|            | b. | What are fuel cells? Describe the construction and working of methanol-oxygen fuel cell.                                                                                                                                                                                                                                                                       | 7 | L2 | CO2 |
|            | c. | Explain the construction and working of Lithium polymer battery. Mention its applications.                                                                                                                                                                                                                                                                     | 7 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                |   |    |     |
| Q.4        | a. | What are batteries? Describe the classification of battery with suitable examples.                                                                                                                                                                                                                                                                             | 6 | L2 | CO2 |
|            | b. | Explain the construction and working of Vanadium flow battery. Mention its applications.                                                                                                                                                                                                                                                                       | 7 | L2 | CO2 |

|                   |    |                                                                                                                                                                                                                                                                                                        |   |    |     |
|-------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|                   | c. | Explain the construction and working of sodium ion battery. Mention its applications.                                                                                                                                                                                                                  | 7 | L2 | CO2 |
| <b>Module – 3</b> |    |                                                                                                                                                                                                                                                                                                        |   |    |     |
| Q.5               | a. | Define Corrosion. Describe electrochemical theory of corrosion taking iron as an example.                                                                                                                                                                                                              | 7 | L2 | CO3 |
|                   | b. | What is anodisation? Explain anodisation of aluminium and mention its applications.                                                                                                                                                                                                                    | 7 | L2 | CO3 |
|                   | c. | Define corrosion penetration rate. A thick brass sheet of area 400 inch <sup>2</sup> is exposed to moist air. After 2 years of period, it was found to experience a weight loss of 375 g due to corrosion. If the density of brass is 8.73 gram/cm <sup>3</sup> . Calculate CPR in mpy and mmpy units. | 6 | L3 | CO3 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                                                                        |   |    |     |
| Q.6               | a. | What is differential aeration corrosion? Describe differential aeration corrosion with suitable examples.                                                                                                                                                                                              | 7 | L2 | CO3 |
|                   | b. | Describe sacrificial anodic method of corrosion control with example.                                                                                                                                                                                                                                  | 6 | L2 | CO3 |
|                   | c. | What is e-waste? Describe the ill effects of e-waste on environment and human health.                                                                                                                                                                                                                  | 7 | L2 | CO3 |
| <b>Module – 4</b> |    |                                                                                                                                                                                                                                                                                                        |   |    |     |
| Q.7               | a. | Mention the properties and application of nano sensors and nano fibers.                                                                                                                                                                                                                                | 6 | L2 | CO4 |
|                   | b. | Describe the synthesis of nanomaterial by Sol-gel method. Mention its advantages and disadvantages.                                                                                                                                                                                                    | 7 | L2 | CO4 |
|                   | c. | What are QLED's? Mention their properties and applications.                                                                                                                                                                                                                                            | 7 | L2 | CO4 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                                                                        |   |    |     |
| Q.8               | a. | What are nano materials? Explain the following size dependent properties of nano materials:<br>(i) Surface area<br>(ii) Conducting property<br>(iii) Catalytic property                                                                                                                                | 7 | L2 | CO4 |
|                   | b. | What are OLED's? Mention their properties and applications.                                                                                                                                                                                                                                            | 6 | L2 | CO4 |
|                   | c. | What are perovskites materials? Give the properties and applications of perovskites materials in optoelectronic devices.                                                                                                                                                                               | 7 | L2 | CO4 |
| <b>Module – 5</b> |    |                                                                                                                                                                                                                                                                                                        |   |    |     |
| Q.9               | a. | What are concentration cells? The emf of a cell $\text{Ag(s)}/\text{AgNO}_3(0.02\text{M})//\text{AgNO}_3(x\text{M})/\text{Ag(s)}$ found to be 0.084 V at 298 K. Write the cell reactions and calculate the value of x.                                                                                 | 6 | L3 | CO5 |
|                   | b. | Describe the principle, instrumentation and application of potentiometric sensors for the estimation of Iron.                                                                                                                                                                                          | 7 | L3 | CO5 |

|             |           |                                                                                                                    |          |           |            |
|-------------|-----------|--------------------------------------------------------------------------------------------------------------------|----------|-----------|------------|
|             | <b>c.</b> | What are reference electrodes? Explain the construction and working of Calomel electrode.                          | <b>7</b> | <b>L2</b> | <b>CO5</b> |
| <b>OR</b>   |           |                                                                                                                    |          |           |            |
| <b>Q.10</b> | <b>a.</b> | Describe the principle, instrumentation and application of conductometric sensors for the estimation of weak acid. | <b>7</b> | <b>L3</b> | <b>CO5</b> |
|             | <b>b.</b> | What are ion selective electrodes? Explain the construction and working of glass electrode.                        | <b>7</b> | <b>L2</b> | <b>CO5</b> |
|             | <b>c.</b> | Explain the principle and working of colorimetric sensors for the estimation of copper.                            | <b>6</b> | <b>L3</b> | <b>CO5</b> |

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BCHEM102/202

**First/Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025**

**Applied Chemistry for ME Stream**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*

*2. M : Marks , L: Bloom's level , C: Course outcomes.*

*3. VTU Formula Hand Book is permitted.*

| Module – 1 |    |                                                                                                                                                                                                                                                                                                                                                                                  | M | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | With a neat diagram, explain the construction working and application of photovoltaic cell.                                                                                                                                                                                                                                                                                      | 7 | L2 | CO1 |
|            | b. | Calculate the GCV and NCV of a sample of coal 0.75g of which when burnt in a bomb calorimeter raised the temperature of water from 22.5°C to 26°C. The mass of water is 1000 g and water equivalent of calorimeter is 250g. The specific heat of water is 4.187 kJ/kg/K, latent heat of steam is 2454 kJ/kg. The coal sample contains 92% carbon , 7% H <sub>2</sub> and 1% ash. | 7 | L3 | CO1 |
|            | c. | Explain the construction and working of Li-ion battery and mention its applications.                                                                                                                                                                                                                                                                                             | 6 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                  |   |    |     |
| Q.2        | a. | Explain the determination of calorific value of solid / liquid fuel using bomb calorimeter.                                                                                                                                                                                                                                                                                      | 7 | L1 | CO1 |
|            | b. | Explain the construction and working of Methanol – Oxygen fuel cell.                                                                                                                                                                                                                                                                                                             | 7 | L1 | CO1 |
|            | c. | Interpret a suitable method for the synthesis of biodiesel and mention advantages and disadvantages of biodiesel.                                                                                                                                                                                                                                                                | 6 | L1 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                                                                                                                                                                                                  |   |    |     |
| Q.3        | a. | Explain the electrochemical theory of corrosion with suitable reactions by taking iron as example.                                                                                                                                                                                                                                                                               | 7 | L2 | CO2 |
|            | b. | Calculate the CPR in both mpy and mmpy for a thick steel sheet of area 100 inch <sup>2</sup> which experiences a weight loss of 400 g after one year. (Density of steel = 7.9 g/cm <sup>3</sup> ).                                                                                                                                                                               | 7 | L3 | CO2 |
|            | c. | What is metal finishing? Mention any five technological importance of metal finishing.                                                                                                                                                                                                                                                                                           | 6 | L1 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                  |   |    |     |
| Q.4        | a. | Describe anodization and mention any two applications.                                                                                                                                                                                                                                                                                                                           | 7 | L1 | CO2 |
|            | b. | What is Electroless plating? Explain the electroless plating of Nickel.                                                                                                                                                                                                                                                                                                          | 7 | L2 | CO2 |
|            | c. | Explain the following with suitable example :<br>i) Differential metal corrosion    ii) Differential aeration corrosion.                                                                                                                                                                                                                                                         | 6 | L1 | CO2 |

| Module – 3 |    |                                                                                                                                                                                                                                                                |   |    |     |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.5        | a. | Explain the synthesis of CPVC and mention its properties and applications.                                                                                                                                                                                     | 7 | L1 | CO3 |
|            | b. | In a sample of polymer , 25% molecules have molecular mass 15000g/mol , 40% molecules have molecular mass 25000 g/mol and remaining molecule have molecular mass 20000 g/mol. Calculate the number average and weight average molecular weight of the polymer. | 7 | L3 | CO3 |
|            | c. | Explain the synthesis of Teflon and mention its applications.                                                                                                                                                                                                  | 6 | L2 | CO3 |
| OR         |    |                                                                                                                                                                                                                                                                |   |    |     |
| Q.6        | a. | Define and explain the condensation polymerization with suitable example.                                                                                                                                                                                      | 7 | L1 | CO3 |
|            | b. | Explain the synthesis of Kevlar fibers and mention the properties and applications.                                                                                                                                                                            | 7 | L2 | CO3 |
|            | c. | Write any three properties and industrial applications of :<br>i) metal matrix polymer composites ii) Lubricants.                                                                                                                                              | 6 | L1 | CO3 |
| Module – 4 |    |                                                                                                                                                                                                                                                                |   |    |     |
| Q.7        | a. | Explain the application of potentiometric sensor in the estimation of iron.                                                                                                                                                                                    | 7 | L1 | CO4 |
|            | b. | Explain the principle and working of glass electrode with neat diagram.                                                                                                                                                                                        | 7 | L2 | CO4 |
|            | c. | Define Phase , Components and Degree of freedom.                                                                                                                                                                                                               | 6 | L1 | CO4 |
| OR         |    |                                                                                                                                                                                                                                                                |   |    |     |
| Q.8        | a. | Explain the application of colorimetric sensor in the estimation of copper.                                                                                                                                                                                    | 7 | L1 | CO4 |
|            | b. | Discuss the application of phase rule to lead – silver system.                                                                                                                                                                                                 | 7 | L1 | CO4 |
|            | c. | Explain the application of glass electrode in the determination of $p^H$ of beverages.                                                                                                                                                                         | 6 | L2 | CO4 |
| Module – 5 |    |                                                                                                                                                                                                                                                                |   |    |     |
| Q.9        | a. | What are alloys? Mention the composition properties and application of stainless steel.                                                                                                                                                                        | 7 | L1 | CO5 |
|            | b. | Sol gel method is suitable for the preparation of nanomaterials. Justify with suitable example.                                                                                                                                                                | 7 | L1 | CO5 |
|            | c. | Explain any two size dependent properties of nanomaterials.                                                                                                                                                                                                    | 6 | L2 | CO5 |
| OR         |    |                                                                                                                                                                                                                                                                |   |    |     |
| Q.10       | a. | Mention the composition , properties and application of brass.                                                                                                                                                                                                 | 7 | L1 | CO5 |
|            | b. | What are Ceramics? Mention the properties and applications of Perovskites.                                                                                                                                                                                     | 7 | L1 | CO5 |
|            | c. | Write a brief note on :<br>i) CNTs ii) Graphene.                                                                                                                                                                                                               | 6 | L1 | CO5 |

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BCHES102/202

**First/Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025**

**Applied Chemistry for CSE Stream**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*

*2. VTU Formula Hand Book is permitted.*

*3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                | M | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Explain the working principle of conductometric sensors (Conductometry) and applied sensors (Colorimetry).                     | 7 | L2 | CO1 |
|            | b. | Write a note on Disposable sensors? Explain its advantages over classical sensors.                                             | 7 | L3 | CO2 |
|            | c. | Describe the construction, working and applications of Lithium-ion battery and mention its applications.                       | 6 | L4 | CO3 |
| OR         |    |                                                                                                                                |   |    |     |
| Q.2        | a. | What are Electrochemical Sensors? Explain its applications in the measurement of Dissolved Oxygen (DO).                        | 7 | L3 | CO2 |
|            | b. | What are Transducers? Explain the applications of Electrochemical gas sensors in sensing SO <sub>x</sub> and NO <sub>x</sub> . | 7 | L3 | CO5 |
|            | c. | Describe the construction, working and applications of Sodium-ion battery and mention its applications.                        | 6 | L4 | CO3 |
| Module – 2 |    |                                                                                                                                |   |    |     |
| Q.3        | a. | Explain the types of organic memory devices by taking P-type and n-type semiconductor materials.                               | 7 | L2 | CO2 |
|            | b. | What are Memory Devices? Explain the classification of electronic memory devices with examples.                                | 7 | L1 | CO2 |
|            | c. | Explain any four properties of polythiophenes (P3HT) suitable for optoelectronic devices.                                      | 6 | L2 | CO4 |
| OR         |    |                                                                                                                                |   |    |     |
| Q.4        | a. | Mention any four properties and applications of QLED.                                                                          | 7 | L2 | CO3 |
|            | b. | Mention any four properties and applications of LCD-displays.                                                                  | 7 | L2 | CO3 |
|            | c. | What are nanomaterials? Explain any four properties of poly (9-vinyl Carbazole) (PVK) suitable for optoelectronic devices.     | 6 | L2 | CO4 |
| Module – 3 |    |                                                                                                                                |   |    |     |
| Q.5        | a. | Define metallic corrosion. Describe the electrochemical theory of corrosion by taking iron as an example.                      | 7 | L3 | CO2 |

|                   |           |                                                                                                                                                                                                                                                                                                                                                            |   |    |     |
|-------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|                   | <b>b.</b> | Describe galvanizing and mention its applications.                                                                                                                                                                                                                                                                                                         | 7 | L2 | CO3 |
|                   | <b>c.</b> | Define electrolyte concentration cell. A galvanic cell is obtained by combining two copper electrodes of concentrations 0.1 and 0.5 M immersed in copper sulphate solution at 25 °C. Give the cell reaction and calculate EMF of the cell.                                                                                                                 | 6 | L3 | CO4 |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                                                                            |   |    |     |
| <b>Q.6</b>        | <b>a.</b> | Explain the construction, working and applications of Calomel Electrode.                                                                                                                                                                                                                                                                                   | 7 | L2 | CO2 |
|                   | <b>b.</b> | What is CPR? A thick sheet of area 600 cm <sup>2</sup> (93 inch <sup>2</sup> ) is exposed to air near the ocean. After a 6 months it was found to experience a weight loss of 360 g due to corrosion, if the density of the steel is 7.9 g/cm <sup>3</sup> . Calculate the corrosion penetration rate in mpy and mmpy (Given K = 534 in mpy and 87.6 mm/y) | 7 | L3 | CO4 |
|                   | <b>c.</b> | Explain : (i) Differential metal corrosion<br>(ii) Water line corrosion.                                                                                                                                                                                                                                                                                   | 6 | L2 | CO2 |
| <b>Module – 4</b> |           |                                                                                                                                                                                                                                                                                                                                                            |   |    |     |
| <b>Q.7</b>        | <b>a.</b> | In a sample of a polymer, 150 molecules have the molecular mass 100 g/mol, 200 molecules have the molecular mass 1000 g/mol, 350 molecules have the molecular mass 10,000 g/mol. Determine number average and weight average molecular mass. Find the Index of Polydispersity.                                                                             | 7 | L3 | CO4 |
|                   | <b>b.</b> | Explain the preparation, properties and commercial applications of Kevlar.                                                                                                                                                                                                                                                                                 | 7 | L2 | CO2 |
|                   | <b>c.</b> | Explain the generation of hydrogen of Alkaline Water Electrolysis.                                                                                                                                                                                                                                                                                         | 6 | L2 | CO3 |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                                                                            |   |    |     |
| <b>Q.8</b>        | <b>a.</b> | Explain the synthesis of polyacetylene and mention its applications.                                                                                                                                                                                                                                                                                       | 7 | L2 | CO2 |
|                   | <b>b.</b> | Explain the generation of hydrogen by proton exchange membrane Electrolysis of water.                                                                                                                                                                                                                                                                      | 7 | L2 | CO3 |
|                   | <b>c.</b> | Describe the construction and working of photovoltaic cells.                                                                                                                                                                                                                                                                                               | 6 | L2 | CO2 |
| <b>Module – 5</b> |           |                                                                                                                                                                                                                                                                                                                                                            |   |    |     |
| <b>Q.9</b>        | <b>a.</b> | Describe the sources and composition of e-waste materials.                                                                                                                                                                                                                                                                                                 | 7 | L2 | CO1 |
|                   | <b>b.</b> | Explain the ill effects of toxic materials used in manufacturing electrical and electronic products.                                                                                                                                                                                                                                                       | 7 | L2 | CO1 |
|                   | <b>c.</b> | Discuss the extraction of gold from e-waste.                                                                                                                                                                                                                                                                                                               | 6 | L2 | CO3 |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                                                                            |   |    |     |
| <b>Q.10</b>       | <b>a.</b> | What are e-wastes? Explain the need for e-waste management.                                                                                                                                                                                                                                                                                                | 7 | L2 | CO1 |
|                   | <b>b.</b> | Write a brief note on role of stakeholders for example : Producers, Consumers, Recyclers and Statutory bodies.                                                                                                                                                                                                                                             | 7 | L2 | CO1 |
|                   | <b>c.</b> | Explain the pyrometallurgy and direct recycling methods.                                                                                                                                                                                                                                                                                                   | 6 | L2 | CO2 |

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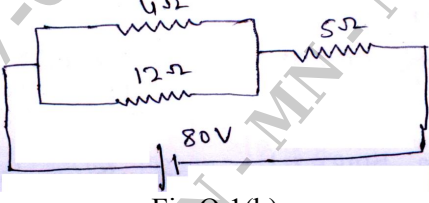
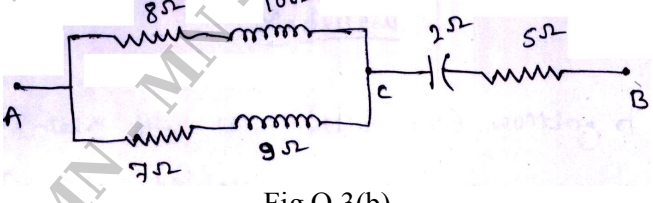
## First/Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

### Elements of Electrical Engineering

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                                             |  | M | L         | C           |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|-----------|-------------|
| Q.1        | a. | State and explain Kirchoff's law as applied to D.C. circuits.                                                                                                                                                               |  | 7 | L1,<br>L2 | CO1         |
|            | b. | For a network shown in Fig.Q.1(b), determine:<br>i) The voltage drop in each resistor<br>ii) The current in each resistor.                                                                                                  |  | 6 | L3        | CO1,<br>CO3 |
|            |    |  <p style="text-align: center;">Fig.Q.1(b)</p>                                                                                            |  |   |           |             |
|            | c. | State and explain the Ohm's law. Mention its limitations.                                                                                                                                                                   |  | 7 | L1,<br>L2 | CO1         |
| OR         |    |                                                                                                                                                                                                                             |  |   |           |             |
| Q.2        | a. | Explain statically and dynamically induced e.m.f.'s.                                                                                                                                                                        |  | 8 | L1,<br>L2 | CO1         |
|            | b. | Two coupled coils of self-inductances 0.8H and 0.2 H have a co-efficient of coupling 0.9. Find the mutual inductance and turns ratio.                                                                                       |  | 6 | L3        | CO1         |
|            | c. | Derive the energy stored in the magnetic field.                                                                                                                                                                             |  | 6 | L1,<br>L2 | CO1         |
| Module – 2 |    |                                                                                                                                                                                                                             |  |   |           |             |
| Q.3        | a. | Define form factor, peak factor obtain its value for a sinusoidal voltage.                                                                                                                                                  |  | 6 | L1        | CO2         |
|            | b. | Derive an expression for power in pure inductance circuit and draw voltage, current and power waveforms.                                                                                                                    |  | 6 | L1,<br>L2 | CO2,<br>CO3 |
|            | c. | In the Fig.Q.3(b) shown below calculate the impedances of AB and the phase angle between voltage and current. Also calculate the total power consumed if the applied voltage between A and B is $200\angle 30^\circ$ volts. |  | 8 | L3        | CO2,<br>CO3 |
|            |    |  <p style="text-align: center;">Fig.Q.3(b)</p>                                                                                          |  |   |           |             |

## OR

|            |           |                                                                                                                                                                                      |          |               |            |
|------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|------------|
| <b>Q.4</b> | <b>a.</b> | Derive the expression for R.M.S. value of current of a sinusoidally varying quantity.                                                                                                | <b>6</b> | <b>L1, L2</b> | <b>CO2</b> |
|            | <b>b.</b> | Define power factor and mention its practical importance.                                                                                                                            | <b>7</b> | <b>L1, L2</b> | <b>CO2</b> |
|            | <b>c.</b> | A series circuit with $R = 10 \Omega$ , $L = 50 \text{ mH}$ and $C = 100 \mu\text{F}$ is supplied with 200 V, 50 Hz. Find: i) The impedance ii) Current iii) Power iv) Power factor. | <b>7</b> | <b>L3</b>     | <b>CO2</b> |

## Module – 3

|            |           |                                                                                                                                                                        |          |               |                 |
|------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|-----------------|
| <b>Q.5</b> | <b>a.</b> | Mention the advantages of three-phase system over single phase system.                                                                                                 | <b>5</b> | <b>L2</b>     | <b>CO2</b>      |
|            | <b>b.</b> | Obtain the relationship between line and phase, voltage and current in a three phase balanced star connected system.                                                   | <b>7</b> | <b>L1, L2</b> | <b>CO3, CO2</b> |
|            | <b>c.</b> | A balanced 3-phase, star connected load of 150 kW takes a leading current of 100 A with line voltage of 1100 V, 50 Hz. Find the circuit constants of a load per phase. | <b>8</b> | <b>L3</b>     | <b>CO2, CO3</b> |

## OR

|            |           |                                                                                                                                                                           |          |               |                 |
|------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|-----------------|
| <b>Q.6</b> | <b>a.</b> | Show that the two Wattmeter's are sufficient to measure three phase power. Hence derive the expression for the power factor in terms of Wattmeter readings.               | <b>8</b> | <b>L1, L2</b> | <b>CO2</b>      |
|            | <b>b.</b> | Establish the relationship between the line and phase currents and voltages in a three phase balanced delta connected system.                                             | <b>6</b> | <b>L1, L2</b> | <b>CO2, CO3</b> |
|            | <b>c.</b> | A balanced delta-connected load of $(8 + j6) \Omega$ per phase is supplied from a 3-phase 440 V source. Find line current, power factor, power per phase and total power. | <b>6</b> | <b>L3</b>     | <b>CO2, CO3</b> |

## Module – 4

|            |           |                                                                 |          |               |            |
|------------|-----------|-----------------------------------------------------------------|----------|---------------|------------|
| <b>Q.7</b> | <b>a.</b> | Explain the construction and working of Kelvin's double bridge. | <b>8</b> | <b>L1, L2</b> | <b>CO4</b> |
|            | <b>b.</b> | Explain two way and three way control of lamp with truth table. | <b>6</b> | <b>L1, L2</b> | <b>CO5</b> |
|            | <b>c.</b> | Explain the construction and working of megger.                 | <b>6</b> | <b>L1, L2</b> | <b>CO5</b> |

## OR

|            |           |                                                                                                      |          |               |            |
|------------|-----------|------------------------------------------------------------------------------------------------------|----------|---------------|------------|
| <b>Q.8</b> | <b>a.</b> | Mention the difference between the current transformer and potential transformer.                    | <b>6</b> | <b>L1, L2</b> | <b>CO4</b> |
|            | <b>b.</b> | Explain the construction of Schering's bridge and derive the expression for the unknown capacitance. | <b>8</b> | <b>L1, L2</b> | <b>CO4</b> |
|            | <b>c.</b> | Write a short note on cabling and capping wiring.                                                    | <b>6</b> | <b>L1, L2</b> | <b>CO5</b> |

## Module – 5

|            |           |                                                                                 |          |               |            |
|------------|-----------|---------------------------------------------------------------------------------|----------|---------------|------------|
| <b>Q.9</b> | <b>a.</b> | What is electric shock? Give the list of preventive measures against the shock. | <b>6</b> | <b>L1, L2</b> | <b>CO5</b> |
|------------|-----------|---------------------------------------------------------------------------------|----------|---------------|------------|

| BEEE103/203 |           |                                                                                                                                                                                                                                                                                                                                                                                                       |          |                   |            |
|-------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------|------------|
|             | <b>b.</b> | Define tariff. Explain briefly the two part tariff with its advantage and disadvantage.                                                                                                                                                                                                                                                                                                               | <b>6</b> | <b>L1,<br/>L2</b> | <b>CO5</b> |
|             | <b>c.</b> | What is earthing? With neat diagram, explain pipe earthing.                                                                                                                                                                                                                                                                                                                                           | <b>8</b> | <b>L1,<br/>L2</b> | <b>CO5</b> |
| <b>OR</b>   |           |                                                                                                                                                                                                                                                                                                                                                                                                       |          |                   |            |
| <b>Q.10</b> | <b>a.</b> | With neat diagram, explain the working of RCCB and ELCB.                                                                                                                                                                                                                                                                                                                                              | <b>8</b> | <b>L1,<br/>L2</b> | <b>CO5</b> |
|             | <b>b.</b> | Write a short note on fuse.                                                                                                                                                                                                                                                                                                                                                                           | <b>5</b> | <b>L1,<br/>L2</b> | <b>CO5</b> |
|             | <b>c.</b> | In a residential house the following loads are connected :<br>i) Six lamps of 40 W each, switched on for 5 hr a day.<br>ii) Two fans of 60 W each, switched on for 12 hr a day.<br>iii) One 1000 W (1 kW) heater working for 2 hr per day.<br>iv) One refrigerator of 250 W working for 10 hr per day.<br>If each unit of energy costs Rs.1.90 what will be the total cost in the month of September. | <b>7</b> | <b>L3</b>         | <b>CO5</b> |

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# CBCS SCHEME

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BEMEM103/203

**First/Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025**

## Elements of Mechanical Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                        | M  | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Explain the role of Mechanical Engineering in industries and society.                                                                                  | 10 | L2 | CO1 |
|            | b. | Explain the formation of steam at constant pressure with T – h diagram.                                                                                | 10 | L2 | CO1 |
| OR         |    |                                                                                                                                                        |    |    |     |
| Q.2        | a. | Explain with a neat sketch, construction and working of Nuclear Power Plant.                                                                           | 10 | L2 | CO1 |
|            | b. | Explain the following : i) Latent treat of steam    ii) Dryness fraction<br>iii) Web steam    iv) Dry steam    v) Super heated steam.                  | 10 | L2 | CO1 |
| Module – 2 |    |                                                                                                                                                        |    |    |     |
| Q.3        | a. | Explain with a neat sketch, the following drilling :<br>i) Boring    ii) Reaming    iii) Tapping    iv) Counter sinking<br>v) Counter boring.          | 10 | L2 | CO2 |
|            | b. | Explain with a neat sketch, the components of CNC. List the advantages and applications of CNC.                                                        | 10 | L2 | CO2 |
| OR         |    |                                                                                                                                                        |    |    |     |
| Q.4        | a. | Explain the working and types of milling machine.                                                                                                      | 6  | L2 | CO2 |
|            | b. | Explain with a neat sketch, the following milling operations :<br>i) Plane milling    ii) End milling    iii) Slot milling.                            | 6  | L2 | CO2 |
|            | c. | Explain with a neat sketch the following lathe operations :<br>i) Turning    ii) Facing    iii) Knurling    iv) Thread cutting.                        | 8  | L2 | CO2 |
| Module – 3 |    |                                                                                                                                                        |    |    |     |
| Q.5        | a. | Explain the working of 4 – stroke Diesel engine with neat sketch.                                                                                      | 10 | L2 | CO2 |
|            | b. | Explain the desirable properties of a refrigerants.                                                                                                    | 10 | L2 | CO2 |
| OR         |    |                                                                                                                                                        |    |    |     |
| Q.6        | a. | Explain the working of VCR refrigeration system with neat figure.                                                                                      | 10 | L2 | CO2 |
|            | b. | Explain the following : i) Indicated power    ii) Brake power<br>iii) Mechanical efficiency    iv) Thermal efficiency<br>v) Specific fuel consumption. | 10 | L2 | CO2 |

| <b>Module – 4</b> |           |                                                                                                          |           |           |            |
|-------------------|-----------|----------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.7</b>        | <b>a.</b> | Briefly explain the types of gear drives with neat sketch.                                               | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|                   | <b>b.</b> | Explain with a neat sketch, gas welding process. List the advantages and disadvantages.                  | <b>10</b> | <b>L2</b> | <b>CO3</b> |
| <b>OR</b>         |           |                                                                                                          |           |           |            |
| <b>Q.8</b>        | <b>a.</b> | Define Soldering , brazing and welding. Explain the differences between soldering , brazing and welding. | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|                   | <b>b.</b> | Explain with a neat sketch, V – belt drive. List the advantages and disadvantages.                       | <b>10</b> | <b>L2</b> | <b>CO3</b> |
| <b>Module – 5</b> |           |                                                                                                          |           |           |            |
| <b>Q.9</b>        | <b>a.</b> | Briefly explain Electric and Hybrid vehicles. List the advantages and disadvantages.                     | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|                   | <b>b.</b> | Explain the applications of Robots in material handling , processing and assembly and inspection.        | <b>10</b> | <b>L2</b> | <b>CO3</b> |
| <b>OR</b>         |           |                                                                                                          |           |           |            |
| <b>Q.10</b>       | <b>a.</b> | Define Mechatronics. Briefly explain open loop and closed – loop mechatronic systems.                    | <b>10</b> | <b>L2</b> | <b>CO3</b> |
|                   | <b>b.</b> | Define a Robot. Explain Robot anatomy with a neat sketch.                                                | <b>10</b> | <b>L2</b> | <b>CO3</b> |

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**BENGK106/206/22BD17**

USN

[illegible]

**Question Paper Version : A**

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

# Communicative English

Time: 1 hr.]

[Max. Marks: 50

## INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
  2. Use only **Black ball point pen** for writing / darkening the circles.
  3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
  4. Darkening two circles for the same question makes the answer invalid.
  5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.
- 
1. When individuals have different frames of reference due to their unique experience, it can create :
    - a) Physical barriers
    - b) Cultural barriers
    - c) Semantic barriers
    - d) Perceptual barriers
  2. Which of the following is NOT a barrier to effective communication?
    - a) Cultural differences
    - b) Active listening
    - c) Language barriers
    - d) Emotional barriers
  3. In the context of organizational communication, which of the following is an example of a structural barrier?
    - a) Language barriers among employees
    - b) Differences in communication styles
    - c) Hierarchical organizational structure
    - d) Lack of feedback from supervisors
  4. Which of the following is NOT a step in the process of active listening?
    - a) Interrupting the speaker frequently
    - b) Providing feedback
    - c) Asking clarifying questions
    - d) Paraphrasing
  5. What is the significance of feedback in communication?
    - a) It ensures that the speaker talks more than the listener.
    - b) It provides an opportunity for clarification and improvement
    - c) It is unnecessary in effective communication
    - d) It should only be given by superiors to subordinates.

6. In the context of organizational communication, what does “inter” communication refer to ?
  - a) Communication between individuals within the same department
  - b) Communication between individuals from different departments or teams
  - c) Communication between employees and management
  - d) Communication between the organization and its external stakeholders.
7. Which of the following is an example of intercommunication?
  - a) A team meeting discussing project updates
  - b) A manager providing feedback to an employee
  - c) An employee preparing a report for their supervisor
  - d) An individual writing a memo to themselves for personal reminders.
8. What is the primary purpose of intra Communication?
  - a) To facilitate collaboration between departments
  - b) To disseminate information within a team or department
  - c) To negotiate with external stakeholders
  - d) To manage conflicts between individuals.
9. Which of the following is an example of a physical barrier to communication?
  - a) Language differences
  - b) Cultural differences
  - c) Noise in the environment
  - d) Lack of empathy
10. Psychological barriers to communication may include :
  - a) Language barriers
  - b) Noise pollution
  - c) Emotional bias or prejudice
  - d) Misinterpretation of messages
11. What is phonetics?
  - a) The study of speech sounds and their physiological production.
  - b) The study of written language
  - c) The study of syntax and grammar
  - d) The study of semantics
12. Which of the following is a key aspect of effective communication in a team setting?
  - a) Dominating conversations
  - b) Ignoring team members opinions
  - c) Active participation and collaboration
  - d) Avoiding conflict resolution
13. What is the International Phonetic Alphabet (IPA) used for?
  - a) Writing in different languages
  - b) Transcribing speech sounds
  - c) Encoding written texts
  - d) Creating new words
14. What is the primary difference between vowels and consonants?
  - a) Vowels are voiced, consonants are voiceless
  - b) Vowels involve airflow without significant constriction, while consonants involve constriction.
  - c) Vowels are nasal sounds, consonants are oral sounds
  - d) Vowels are always at the beginning of words, consonants are always at the end.
15. Which of the following is a consonant?
  - a) [i]
  - b) [a]
  - c) [p]
  - d) [e]

16. How many Consonants sounds are there in English?  
a) 5                                      b) 10                                      c) 24                                      d) 14
17. Which of the following is a voiced consonant?  
a) [p]                                      b) [s]                                      c) [v]                                      d) [f]
18. What is the term for the study of the physical properties of speech sounds?  
a) Phonology                              b) Phonetics                              c) Morphology                              d) Syntax
19. Which of the following is an example of a diphthong?  
a) [p]                                      b) [ae]                                      c) [ai]                                      d) [m]
20. What part of speech is used to show the relationship between a noun (or pronoun) and other words in a sentence?  
a) Conjunction                              b) Verb                                      c) Preposition                              d) Interjection
21. Which of the following sentence is correctly punctuated?  
a) I went to the store, and bought apples, bananas, and oranges  
b) I went to the store and bought apples, bananas and oranges  
c) I went to the store; and bought apples, bananas, and oranges  
d) I went to the store and bought apples, bananas, and oranges
22. Which article is used before a consonant sound?  
a) A                                      b) An                                      c) The                                      d) None of these
23. In the sentence "I need \_\_\_\_\_ pen to write", should the blank be filled with an article?  
a) A                                      b) An                                      c) The                                      d) None of these
24. Which of the following is a part of speech used to express strong emotions?  
a) Adverb                                      b) Pronoun                                      c) Interjection                                      d) Conjunction
25. "The cat is \_\_\_\_\_ the table," which preposition should fill in the blank?  
a) At                                      b) In                                      c) On                                      d) With
26. Which of the following is a correct use of a preposition?  
a) "I am in the home".                                      b) "She walked at the park".  
c) "He sat on the chair".                                      d) None of these
27. Which word contains a prefix?  
a) Happy                                      b) Cooked                                      c) Revisit                                      d) Quickly
28. Which sentence uses correct capitalization?  
a) The chief executive officer was interviewed for the magazine  
b) The Chief Executive Officer was interviewed for the magazine  
c) The chief executive officer was interviewed for the Magazine  
d) The chief Executive Officer was interviewed for the Magazine.
29. What is the term for the consonant sound produced by a complete closure in the vocal tract followed by a sudden release of air?  
a) Fricative                                      b) Approximant                                      c) Plosive                                      d) Nasal

30. Which of the following is a part of speech that describes a person , place , thing or idea?  
a) Verb                      b) Adjective                      c) Noun                      d) Conjunction
31. What part of speech is used to join words, phrases or clauses?  
a) Preposition                      b) Conjunction                      c) Adverb                      d) Pronoun
32. What is the term for the fear of heights?  
a) Acrophobia                      b) Agoraphobia                      c) Claustrophobia                      d) Hydrophobia
33. What is the term for a person who is obsessed with their own power?  
a) Egoist                      b) Narcissist                      c) Altruist                      d) Introvert
34. Which of the following words is a synonym for “ephemeral”?  
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35. Which sentence demonstrates correct usage of the word “affect”?  
a) The weather will effect the crops                      b) The weather will affect the crops  
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36. What is the antonym of the word “exquisite”?  
a) Beautiful                      b) Delicate                      c) Common                      d) Refined
37. Which word is the odd one out in the following group?  
a) Zealous                      b) Enthusiastic                      c) Apathetic                      d) Eager
38. What is the meaning of the term “vernacular”?  
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c) Foreign language                      d) Technical language
39. What is the question tag for the statement : “He didn’t eat lunch”?  
a) He didn’t eat lunch, did he?                      b) He didn’t eat lunch, does he?  
c) He didn’t eat lunch, hasn’t he?                      d) He didn’t eat lunch, isn’t he?
40. What is the question tag for the statement : “He doesn’t speak French”?  
a) He doesn’t speak French, isn’t he?                      b) He doesn’t speak French, does he?  
c) He doesn’t speak French, hasn’t he?                      d) He doesn’t speak French, can’t he?
41. Which of the following is a correct question tag for the statement : “It’s raining outside”?  
a) It’s raining outside, does it?                      b) It’s raining outside, isn’t it?  
c) It’s raining outside, can it?                      d) It’s raining outside, won’t it?
42. In the sentence “She will go to the market tomorrow,” which tense is used?  
a) Present tense                      b) Past tense  
c) Future tense                      d) Present perfect tense
43. Which of the following sentence uses the present perfect tense correctly?  
a) He is going to the gym tomorrow  
b) She has finished her homework  
c) They will visit their grandparents last weekend  
d) We were playing football yesterday

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Question Paper Version : B

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

## **Communicative English**

Time: 1 hr.]

[Max. Marks: 50

### **INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
1. Which of the following is a correct question tag for the statement : "It's raining outside"?  
a) It's raining outside, does it?                      b) It's raining outside, isn't it?  
c) It's raining outside, can it?                      d) It's raining outside, won't it?
  2. In the sentence "She will go to the market tomorrow," which tense is used?  
a) Present tense                      b) Past tense  
c) Future tense                      d) Present perfect tense
  3. Which of the following sentence uses the present perfect tense correctly?  
a) He is going to the gym tomorrow  
b) She has finished her homework  
c) They will visit their grandparents last weekend  
d) We were playing football yesterday
  4. In the sentence "They had been waiting for hours," which tense is used?  
a) Present tense                      b) Past tense  
c) Future tense                      d) Past perfect continuous tense
  5. In the sentence "She is taller than him, "What part of speech is "than"?  
a) Preposition                      b) Conjunction                      c) Adjective                      d) Pronoun
  6. What is the superlative form of the adjective "good"?  
a) Better                      b) Best                      c) Goodest                      d) Gooder



20. What is the question tag for the statement : “He doesn’t speak French”?
- a) He doesn’t speak French, isn’t he?      b) He doesn’t speak French, does he?  
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31. When individuals have different frames of reference due to their unique experience, it can create :
- a) Physical barriers      b) Cultural barriers  
c) Semantic barriers      d) Perceptual barriers
32. Which of the following is NOT a barrier to effective communication?
- a) Cultural differences      b) Active listening  
c) Language barriers      d) Emotional barriers

33. In the context of organizational communication, which of the following is an example of a structural barrier?
- a) Language barriers among employees
  - b) Differences in communication styles
  - c) Hierarchical organizational structure
  - d) Lack of feedback from supervisors
34. Which of the following is NOT a step in the process of active listening?
- a) Interrupting the speaker frequently
  - b) Providing feedback
  - c) Asking clarifying questions
  - d) Paraphrasing
35. What is the significance of feedback in communication?
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  - b) Cultural differences
  - c) Noise in the environment
  - d) Lack of empathy
40. Psychological barriers to communication may include :
- a) Language barriers
  - b) Noise pollution
  - c) Emotional bias or prejudice
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  - b) I went to the store and bought apples, bananas and oranges
  - c) I went to the store; and bought apples, bananas, and oranges
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42. Which article is used before a consonant sound?
- a) A
  - b) An
  - c) The
  - d) None of these

43. In the sentence "I need \_\_\_\_\_ pen to write", should the blank be filled with an article?  
a) A                                      b) An                                      c) The                                      d) None of these
44. Which of the following is a part of speech used to express strong emotions?  
a) Adverb                                      b) Pronoun                                      c) Interjection                                      d) Conjunction
45. "The cat is \_\_\_\_\_ the table," which preposition should fill in the blank?  
a) At                                      b) In                                      c) On                                      d) With
46. Which of the following is a correct use of a preposition?  
a) "I am in the home".                                      b) "She walked at the park".  
c) "He sat on the chair".                                      d) None of these
47. Which word contains a prefix?  
a) Happy                                      b) Cooked                                      c) Revisit                                      d) Quickly
48. Which sentence uses correct capitalization?  
a) The chief executive officer was interviewed for the magazine  
b) The Chief Executive Officer was interviewed for the magazine  
c) The chief executive officer was interviewed for the Magazine  
d) The chief Executive Officer was interviewed for the Magazine.
49. What is the term for the consonant sound produced by a complete closure in the vocal tract followed by a sudden release of air?  
a) Fricative                                      b) Approximant                                      c) Plosive                                      d) Nasal
50. Which of the following is a part of speech that describes a person, place, thing or idea?  
a) Verb                                      b) Adjective                                      c) Noun                                      d) Conjunction

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Question Paper Version : C

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

## **Communicative English**

Time: 1 hr.]

[Max. Marks: 50

### **INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
1. Which of the following sentence is correctly punctuated?  
a) I went to the store, and bought apples, bananas, and oranges  
b) I went to the store and bought apples, bananas and oranges  
c) I went to the store; and bought apples, bananas, and oranges  
d) I went to the store and bought apples, bananas, and oranges
  2. Which article is used before a consonant sound?  
a) A                                      b) An                                      c) The                                      d) None of these
  3. In the sentence "I need \_\_\_\_\_ pen to write", should the blank be filled with an article?  
a) A                                      b) An                                      c) The                                      d) None of these
  4. Which of the following is a part of speech used to express strong emotions?  
a) Adverb                                      b) Pronoun                                      c) Interjection                                      d) Conjunction
  5. "The cat is \_\_\_\_\_ the table," which preposition should fill in the blank?  
a) At                                      b) In                                      c) On                                      d) With
  6. Which of the following is a correct use of a preposition?  
a) "I am in the home".                                      b) "She walked at the park".  
c) "He sat on the chair".                                      d) None of these
  7. Which word contains a prefix?  
a) Happy                                      b) Cooked                                      c) Revisit                                      d) Quickly

8. Which sentence uses correct capitalization?
  - a) The chief executive officer was interviewed for the magazine
  - b) The Chief Executive Officer was interviewed for the magazine
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  - d) The chief Executive Officer was interviewed for the Magazine.
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10. Which of the following is a part of speech that describes a person, place, thing or idea?
  - a) Verb
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11. Which of the following is a correct question tag for the statement : “It’s raining outside”?
  - a) It’s raining outside, does it?
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  - a) Present tense
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  - c) Adjective
  - d) Pronoun
16. What is the superlative form of the adjective “good”?
  - a) Better
  - b) Best
  - c) Goodest
  - d) Gooder
17. Which of the following sentence is grammatically correct?
  - a) He don’t like coffee.
  - b) She hasn’t went to the store yet
  - c) They are swimming in the pool.
  - d) We was going to the movies.
18. Which sentence demonstrate correct subject – verb agreement?
  - a) The team are working hard to meet their deadlines
  - b) The team is working hard to meet its deadlines
  - c) The team am working hard to meet its deadlines
  - d) The team were working hard to meet their deadlines
19. Which sentence uses correct punctuation?
  - a) Please bring a pen paper and your textbook to the meeting
  - b) Please bring a pen, paper, and your textbook to the meeting.
  - c) Please bring a pen, paper and your textbook to the meeting.
  - d) Please bring a pen, paper ; and your textbook to the meeting.

20. Which of the following is an example of the correct use of the apostrophe?  
a) The cars windshield was cracked      b) The car's windshield was cracked  
c) The cars' windshield was cracked      d) The car windshield was cracked
21. What part of speech is used to join words, phrases or clauses?  
a) Preposition      b) Conjunction      c) Adverb      d) Pronoun
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Question Paper Version : D

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

## **Communicative English**

Time: 1 hr.]

[Max. Marks: 50

### **INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
1. What part of speech is used to join words, phrases or clauses?  
a) Preposition                      b) Conjunction                      c) Adverb                      d) Pronoun
  2. What is the term for the fear of heights?  
a) Acrophobia                      b) Agoraphobia                      c) Claustrophobia                      d) Hydrophobia
  3. What is the term for a person who is obsessed with their own power?  
a) Egoist                      b) Narcissist                      c) Altruist                      d) Introvert
  4. Which of the following words is a synonym for "ephemeral"?  
a) Eternal                      b) Temporary                      c) Continuous                      d) Enduring
  5. Which sentence demonstrates correct usage of the word "affect"?  
a) The weather will effect the crops                      b) The weather will affect the crops  
c) The weather will effect the crops                      d) The weather will affect on the crops
  6. What is the antonym of the word "exquisite"?  
a) Beautiful                      b) Delicate                      c) Common                      d) Refined
  7. Which word is the odd one out in the following group?  
a) Zealous                      b) Enthusiastic                      c) Apathetic                      d) Eager
  8. What is the meaning of the term "vernacular"?  
a) Formal language                      b) Slang or informal language  
c) Foreign language                      d) Technical language

9. What is the question tag for the statement : “He didn’t eat lunch”?
- a) He didn’t eat lunch, did he?
  - b) He didn’t eat lunch, does he?
  - c) He didn’t eat lunch, hasn’t he?
  - d) He didn’t eat lunch, isn’t he?
10. What is the question tag for the statement : “He doesn’t speak French”?
- a) He doesn’t speak French, isn’t he?
  - b) He doesn’t speak French, does he?
  - c) He doesn’t speak French, hasn’t he?
  - d) He doesn’t speak French, can’t he?
11. When individuals have different frames of reference due to their unique experience, it can create :
- a) Physical barriers
  - b) Cultural barriers
  - c) Semantic barriers
  - d) Perceptual barriers
12. Which of the following is NOT a barrier to effective communication?
- a) Cultural differences
  - b) Active listening
  - c) Language barriers
  - d) Emotional barriers
13. In the context of organizational communication, which of the following is an example of a structural barrier?
- a) Language barriers among employees
  - b) Differences in communication styles
  - c) Hierarchical organizational structure
  - d) Lack of feedback from supervisors
14. Which of the following is NOT a step in the process of active listening?
- a) Interrupting the speaker frequently
  - b) Providing feedback
  - c) Asking clarifying questions
  - d) Paraphrasing
15. What is the significance of feedback in communication?
- a) It ensures that the speaker talks more than the listener.
  - b) It provides an opportunity for clarification and improvement
  - c) It is unnecessary in effective communication
  - d) It should only be given by superiors to subordinates.
16. In the context of organizational communication, what does “inter” communication refer to ?
- a) Communication between individuals within the same department
  - b) Communication between individuals from different departments or teams
  - c) Communication between employees and management
  - d) Communication between the organization and its external stakeholders.
17. Which of the following is an example of intercommunication?
- a) A team meeting discussing project updates
  - b) A manager providing feedback to an employee
  - c) An employee preparing a report for their supervisor
  - d) An individual writing a memo to themselves for personal reminders.
18. What is the primary purpose of intra Communication?
- a) To facilitate collaboration between departments
  - b) To disseminate information within a team or department
  - c) To negotiate with external stakeholders
  - d) To manage conflicts between individuals.

19. Which of the following is an example of a physical barrier to communication?  
a) Language differences                      b) Cultural differences  
c) Noise in the environment                d) Lack of empathy
20. Psychological barriers to communication may include :  
a) Language barriers                          b) Noise pollution  
c) Emotional bias or prejudice              d) Misinterpretation of messages
21. Which of the following is a correct question tag for the statement : “It’s raining outside”?  
a) It’s raining outside, does it?            b) It’s raining outside, isn’t it?  
c) It’s raining outside, can it?              d) It’s raining outside, won’t it?
22. In the sentence “She will go to the market tomorrow,” which tense is used?  
a) Present tense                                b) Past tense  
c) Future tense                                 d) Present perfect tense
23. Which of the following sentence uses the present perfect tense correctly?  
a) He is going to the gym tomorrow  
b) She has finished her homework  
c) They will visit their grandparents last weekend  
d) We were playing football yesterday
24. In the sentence “They had been waiting for hours,” which tense is used?  
a) Present tense                                b) Past tense  
c) Future tense                                 d) Past perfect continuous tense
25. In the sentence “She is taller than him, “What part of speech is “than”?  
a) Preposition                                b) Conjunction                                c) Adjective                                d) Pronoun
26. What is the superlative form of the adjective “good”?  
a) Better                                        b) Best                                        c) Goodest                                d) Gooder
27. Which of the following sentence is grammatically correct?  
a) He don’t like coffee.                      b) She hasn’t went to the store yet  
c) They are swimming in the pool.        d) We was going to the movies.
28. Which sentence demonstrate correct subject – verb agreement?  
a) The team are working hard to meet their deadlines  
b) The team is working hard to meet its deadlines  
c) The team am working hard to meet its deadlines  
d) The team were working hard to meet their deadlines
29. Which sentence uses correct punctuation?  
a) Please bring a pen paper and your textbook to the meeting  
b) Please bring a pen, paper, and your textbook to the meeting.  
c) Please bring a pen, paper and your textbook to the meeting.  
d) Please bring a pen, paper ; and your textbook to the meeting.
30. Which of the following is an example of the correct use of the apostrophe?  
a) The cars windshield was cracked        b) The car’s windshield was cracked  
c) The cars’ windshield was cracked        d) The car windshield was cracked

31. Which of the following sentence is correctly punctuated?  
 a) I went to the store, and bought apples, bananas, and oranges  
 b) I went to the store and bought apples, bananas and oranges  
 c) I went to the store; and bought apples, bananas, and oranges  
 d) I went to the store and bought apples, bananas, and oranges
32. Which article is used before a consonant sound?  
 a) A                                      b) An                                      c) The                                      d) None of these
33. In the sentence "I need \_\_\_\_\_ pen to write", should the blank be filled with an article?  
 a) A                                      b) An                                      c) The                                      d) None of these
34. Which of the following is a part of speech used to express strong emotions?  
 a) Adverb                                      b) Pronoun                                      c) Interjection                                      d) Conjunction
35. "The cat is \_\_\_\_\_ the table," which preposition should fill in the blank?  
 a) At                                      b) In                                      c) On                                      d) With
36. Which of the following is a correct use of a preposition?  
 a) "I am in the home".                                      b) "She walked at the park".  
 c) "He sat on the chair".                                      d) None of these
37. Which work contains a prefix?  
 a) Happy                                      b) Cooked                                      c) Revisit                                      d) Quickly
38. Which sentence uses correct capitalization?  
 a) The chief executive officer was interviewed for the magazine  
 b) The Chief Executive Officer was interviewed for the magazine  
 c) The chief executive officer was interviewed for the Magazine  
 d) The chief Executive Officer was interviewed for the Magazine.
39. What is the term for the consonant sound produced by a complete closure in the vocal tract followed by a sudden release of air?  
 a) Fricative                                      b) Approximant                                      c) Plosive                                      d) Nasal
40. Which of the following is a part of speech that describes a person , place , thing or idea?  
 a) Verb                                      b) Adjective                                      c) Noun                                      d) Conjunction
41. What is phonetics?  
 a) The study of speech sounds and their physiological production.  
 b) The study of written language  
 c) The study of syntax and grammar  
 d) The study of semantics
42. Which of the following is a key aspect of effective communication in a team setting?  
 a) Dominating conversations                                      b) Ignoring team members opinions  
 c) Active participation and collaboration                                      d) Avoiding conflict resolution
43. What is the International Phonetic Alphabet (IPA) used for?  
 a) Writing in different languages                                      b) Transcribing speech sounds  
 c) Encoding written texts                                      d) Creating new words

44. What is the primary difference between vowels and consonants?  
a) Vowels are voiced, consonants are voiceless  
b) Vowels involve airflow without significant constriction, while consonants involve constriction.  
c) Vowels are nasal sounds, consonants are oral sounds  
d) Vowels are always at the beginning of words, consonants are always at the end.
45. Which of the following is a consonant?  
a) [i]                      b) [a]                      c) [p]                      d) [e]
46. How many Consonants sounds are there in English?  
a) 5                      b) 10                      c) 24                      d) 14
47. Which of the following is a voiced consonant?  
a) [p]                      b) [s]                      c) [v]                      d) [f]
48. What is the term for the study of the physical properties of speech sounds?  
a) Phonology              b) Phonetics              c) Morphology              d) Syntax
49. Which of the following is an example of a diphthong?  
a) [p]                      b) [ae]                      c) [ai]                      d) [m]
50. What part of speech is used to show the relationship between a noun (or pronoun) and other words in a sentence?  
a) Conjunction              b) Verb                      c) Preposition              d) Interjection

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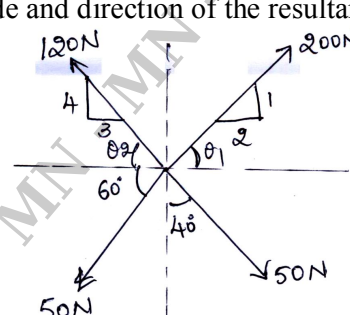
BESCK104A/BESCKA104

## First Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Introduction to Civil Engineering

Time: 3 hrs.

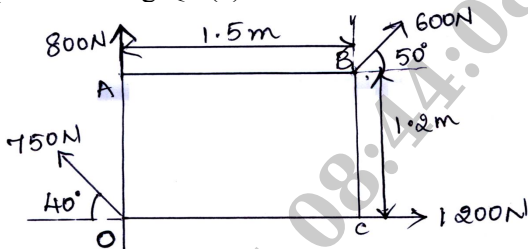
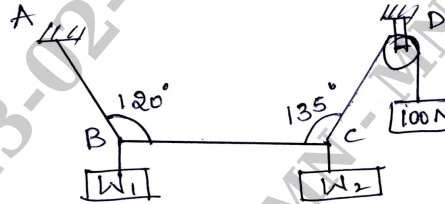
Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.*

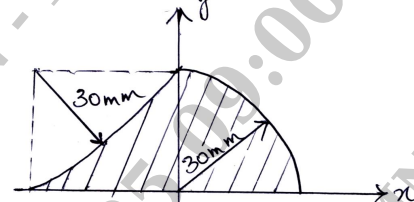
| Module – 1 |    |                                                                                                                                                                                | M  | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | List out different branches of civil engineering and explain any 4 branches briefly.                                                                                           | 10 | L1 | CO1 |
|            | b. | What are cement and Mortor? What are their uses?                                                                                                                               | 10 | L1 | CO1 |
| OR         |    |                                                                                                                                                                                |    |    |     |
| Q.2        | a. | Explain briefly the structural elements of a building.                                                                                                                         | 10 | L1 | CO1 |
|            | b. | What is RCC? Explain its advantages and disadvantages.                                                                                                                         | 10 | L1 | CO1 |
| Module – 2 |    |                                                                                                                                                                                |    |    |     |
| Q.3        | a. | Write a short note on:<br>i) Smart city concept<br>ii) Clean city concept.                                                                                                     | 10 | L1 | CO2 |
|            | b. | Why landfills are important? Explain the advantages and disadvantages of land fills.                                                                                           | 10 | L1 | CO2 |
| OR         |    |                                                                                                                                                                                |    |    |     |
| Q.4        | a. | Explain the importance of water supply and sanitary system.                                                                                                                    | 10 | L1 | CO2 |
|            | b. | What is solid waste management? Explain the sources and origin of solid wastes.                                                                                                | 10 | L1 | CO2 |
| Module – 3 |    |                                                                                                                                                                                |    |    |     |
| Q.5        | a. | Explain the principles of superposition and transmissibility of force system.                                                                                                  | 5  | L2 | CO3 |
|            | b. | State and prove Varignon’s theorem of moments.                                                                                                                                 | 5  | L2 | CO3 |
|            | c. | Compute the magnitude and direction of the resultant force in Fig.Q.5(c).<br><br>Fig.Q.5(c) | 10 | L3 | CO3 |

1 of 3

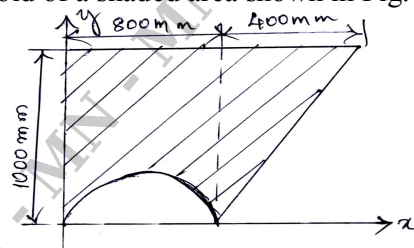
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| Q.6 | a. Find the magnitude, direction and point of application of the resultant force with respect to point O in Fig.Q.6(a).                                                                                                                                                                                          | 12 | L3 | CO3 |
|     |  <p>Fig.Q.6(a)</p>                                                                                                                                                                                                             |    |    |     |
|     | <p>b. In the Fig.Q.6(b) the portion BC of the string is horizontal and pulley is frictionless. Determine the tension in different parts of the string. Also find <math>w_1</math> and <math>w_2</math>.</p>  <p>Fig.Q.6(b)</p> | 8  | L3 | CO3 |

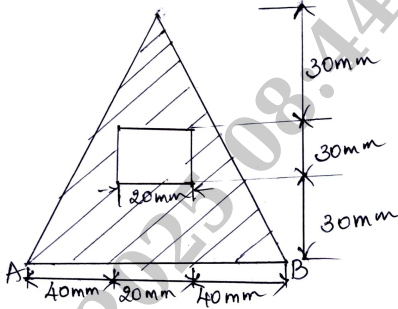
Module – 4

|     |                                                                                                                                                                           |    |    |     |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.7 | a. Define centroid and centre of gravity.                                                                                                                                 | 4  | L1 | CO4 |
|     | b. Find the centroid of a semicircle using first principle.                                                                                                               | 6  | L3 | CO4 |
|     | <p>c. Locate the centroid of a shaded area shown in Fig.Q.7(c).</p>  <p>Fig.Q.7(c)</p> | 10 | L3 | CO4 |

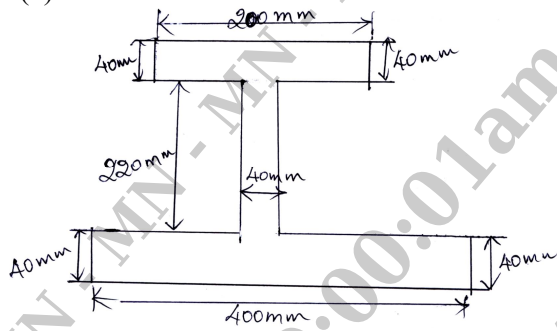
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|     |                                                                                                                                                                              |    |    |     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.8 | a. Find the centroid of a triangle from first principle.                                                                                                                     | 8  | L3 | CO4 |
|     | <p>b. Determine the centroid of a shaded area shown in Fig.Q.8(b).</p>  <p>Fig.Q.8(b)</p> | 12 | L3 | CO4 |

Module – 5

|     |    |                                                                                                                                                                                                       |    |    |     |
|-----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.9 | a. | State and prove parallel and perpendicular axis theorem.                                                                                                                                              | 10 | L2 | CO5 |
|     | b. | Determine the moment of inertia and radius of gyration of the shaded area shown in Fig.Q.9(b) about the base AB.<br> | 10 | L3 | CO5 |

OR

|      |    |                                                                                                                                                                                         |    |    |     |
|------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.10 | a. | Derive an expression for moment of inertia of a circle along its centroidal axis ( $I_{XX}$ and $I_{YY}$ ).                                                                             | 10 | L2 | CO5 |
|      | b. | Determine the polar radius of gyration about the horizontal centroidal axis for the Fig.Q.10(b).<br> | 10 | L3 | CO5 |

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# CBCS SCHEME

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BESCK104B

## First Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Introduction to Electrical Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1                                                                                                                          |    |                                                                                                                                                                                                                    | M | L  | C   |
|-------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1                                                                                                                                 | a. | With neat single line diagram, explain the various steps of electrical power transmission and distribution system.                                                                                                 | 8 | L2 | CO1 |
|                                                                                                                                     | b. | State and explain Kirchoff's current and voltage law.                                                                                                                                                              | 6 | L2 | CO1 |
|                                                                                                                                     | c. | A resistance R is connected in series with a parallel circuit comprising of two resistance $12\Omega$ and $8\Omega$ . The total power in the circuit is 70W, when the applied voltage is 20V. Calculate R.         | 6 | L3 | CO1 |
| <b>OR</b>                                                                                                                           |    |                                                                                                                                                                                                                    |   |    |     |
| Q.2                                                                                                                                 | a. | With block diagram, explain Hydel Power generation.                                                                                                                                                                | 6 | L2 | CO1 |
|                                                                                                                                     | b. | State and explain Ohm's law with its limitation.                                                                                                                                                                   | 6 | L3 | CO1 |
|                                                                                                                                     | c. | For the circuit shown in Fig. Q2(c), find the current in $2\Omega$ resistor.                                                                                                                                       | 8 | L3 | CO1 |
| <div style="display: flex; align-items: center; justify-content: center;"> <div style="margin-right: 20px;">Fig. Q2(c)</div> </div> |    |                                                                                                                                                                                                                    |   |    |     |
| Module – 2                                                                                                                          |    |                                                                                                                                                                                                                    |   |    |     |
| Q.3                                                                                                                                 | a. | Define the following terms applied to alternating current wave :<br>i) RMS value      ii) Average value      iii) Form factor<br>iv) Peak factor      v) Phase      vi) Phase difference.                          | 6 | L3 | CO2 |
|                                                                                                                                     | b. | Show that the current through purely capacitive circuit leads the applied voltage by $90^\circ$ and average power consumed is zero. Draw the wave shapes of current voltage and power.                             | 8 | L3 | CO2 |
|                                                                                                                                     | c. | An Inductive coil takes a current of 10A from a supply of 100V, 50Hz and lags the voltage of $30^\circ$ . Calculate i) Parameters of the circuit<br>ii) Power factor      iii) Active reactive and apparent power. | 6 | L2 | CO2 |
| <b>OR</b>                                                                                                                           |    |                                                                                                                                                                                                                    |   |    |     |
| Q.4                                                                                                                                 | a. | Explain the generation of three phase A.C. and list the advantage.                                                                                                                                                 | 6 | L2 | CO2 |

|                   |           |                                                                                                                                                                                                                                                                                                                                              |          |           |            |
|-------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|------------|
|                   | <b>b.</b> | A circuit consists of a resistance of $20\Omega$ and inductance of $0.05H$ , connected in series. A single phase supply of $230V$ , $50Hz$ is applied across the circuit. Find i) Impedance ii) Current iii) Power factor iv) Power consumed by the circuit v) Voltage drop across R & L.                                                    | <b>8</b> | <b>L3</b> | <b>CO2</b> |
|                   | <b>c.</b> | Three coils having resistance of $10\Omega$ and inductance of $0.02H$ are connected in star across $440V$ , $50Hz$ , $3 - \phi$ supply. Calculate the line current, p.f and total power consumed.                                                                                                                                            | <b>6</b> | <b>L3</b> | <b>CO2</b> |
| <b>Module – 3</b> |           |                                                                                                                                                                                                                                                                                                                                              |          |           |            |
| <b>Q.5</b>        | <b>a.</b> | Derive E.M.F equation of the DC generator.                                                                                                                                                                                                                                                                                                   | <b>6</b> | <b>L3</b> | <b>CO3</b> |
|                   | <b>b.</b> | Explain the characteristics of a D.C. shunt motor.                                                                                                                                                                                                                                                                                           | <b>7</b> | <b>L2</b> | <b>CO3</b> |
|                   | <b>c.</b> | A 4 pole, $1500\text{ rpm}$ , D.C generator has a lap wound armature having 32 slots and 8 conductors per slot. If the flux per pole is $0.04\text{ wb}$ . Calculate the emf induced in the armature. What would be the emf induced if the winding is wave connected.                                                                        | <b>7</b> | <b>L3</b> | <b>CO3</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                                                              |          |           |            |
| <b>Q.6</b>        | <b>a.</b> | Derive Torque equation of the D.C motor.                                                                                                                                                                                                                                                                                                     | <b>6</b> | <b>L3</b> | <b>CO3</b> |
|                   | <b>b.</b> | Explain the various methods used to control the speed of DC series motor.                                                                                                                                                                                                                                                                    | <b>8</b> | <b>L2</b> | <b>CO3</b> |
|                   | <b>c.</b> | A 4 – pole D.C shunt motor takes $25A$ from a $250V$ supply. The armature and field resistances are $0.5\Omega$ and $12\Omega$ respectively. The wave wound armature has 30 slots and each slot containing 10 conductors, if the flux per pole is $0.02\text{ wb}$ . Calculate i) speed ii) torque developed iii) power developed.           | <b>6</b> | <b>L3</b> | <b>CO3</b> |
| <b>Module – 4</b> |           |                                                                                                                                                                                                                                                                                                                                              |          |           |            |
| <b>Q.7</b>        | <b>a.</b> | Derive the emf equation of a transformer and hence obtain the voltage and current transformation ratios.                                                                                                                                                                                                                                     | <b>8</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b.</b> | With neat diagram, explain the types of $3 - \phi$ induction motor.                                                                                                                                                                                                                                                                          | <b>7</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>c.</b> | A transformer is rated at $100KVA$ . At full load its copper loss is $1200W$ and its iron loss is $960W$ . Calculate the following :<br>i) the efficiency at full load, VPF ii) the efficiency at half load, $0.8\text{ pf}$ .<br>iii) the load KVA at which maximum efficiency will occur.<br>iv) maximum efficiency at $0.85\text{ p.f}$ . | <b>5</b> | <b>L3</b> | <b>CO4</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                                                                                                                              |          |           |            |
| <b>Q.8</b>        | <b>a.</b> | Explain the various losses in transformer, how to minimize them?                                                                                                                                                                                                                                                                             | <b>8</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b.</b> | With diagrams, explain the concept of rotating magnetic field.                                                                                                                                                                                                                                                                               | <b>8</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>c.</b> | A three phase induction motor with 4 pole is supplied from the alternator having 6 poles running at $1000\text{ rpm}$ . Calculate Synchronous speed, rotor speed of the induction motor when slip is $0.04$ and frequency of the rotor emf when the speed is $600\text{ rpm}$ .                                                              | <b>4</b> | <b>L3</b> | <b>CO4</b> |

| Module – 5 |    |                                                                                                                                            |   |    |     |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.9        | a. | With neat circuit diagram and switching table explain two way and three way control of load.                                               | 8 | L2 | CO5 |
|            | b. | Explain working principle of fuse and miniature circuit breaker.                                                                           | 7 | L2 | CO5 |
|            | c. | What is earthing? With neat diagram, explain any one type of earthing.                                                                     | 5 | L2 | CO5 |
| OR         |    |                                                                                                                                            |   |    |     |
| Q.10       | a. | What is Electric shock? Give the list of preventive measures against shock.                                                                | 6 | L2 | CO5 |
|            | b. | What are the desirable characteristics of a tariff and explain two part tariff.                                                            | 6 | L2 | CO5 |
|            | c. | List out the power rating of home hold appliances including air conditioners , PCs, laptops , printers etc. Find the total power consumed. | 8 | L3 | CO5 |

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BESCK104E

**First Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025**

## Introduction C Programming

Time: 3 hrs.

Max. Marks:100

**Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.

| Module – 1        |    |                                                                                                                      | M | L  | C   |
|-------------------|----|----------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 1                 | a. | Classify the different types of input devices and briefly explain the wireless keyboard.                             | 7 | L2 | CO1 |
|                   | b. | Provide a brief explanation of the process involved in compiling and executing a C program.                          | 7 | L2 | CO1 |
|                   | c. | Explain the structure of a C program, emphasizing key components through an example.                                 | 6 | L2 | CO1 |
| <b>OR</b>         |    |                                                                                                                      |   |    |     |
| 2                 | a. | Describe the working of a printer and an LCD monitor with detailed sketches.                                         | 8 | L2 | CO1 |
|                   | b. | What are variables? Provide an example of how to declare and initialize a variable of different datatypes.           | 6 | L2 | CO1 |
|                   | c. | Write a C program that demonstrates the use of standard libraries for performing input and output operations.        | 6 | L3 | CO1 |
| <b>Module – 2</b> |    |                                                                                                                      |   |    |     |
| 3                 | a. | Classify the different types of binary operators supported in C language. Explain the bitwise logical operator.      | 7 | L2 | CO2 |
|                   | b. | Use a simple if statement to determine if a person's age qualifies them to vote.                                     | 6 | L3 | CO2 |
|                   | c. | Write a C program to calculate the sum of numbers from m to n.                                                       | 7 | L3 | CO2 |
| <b>OR</b>         |    |                                                                                                                      |   |    |     |
| 4                 | a. | Write a program to convert integer number into the corresponding floating-point number.                              | 6 | L3 | CO2 |
|                   | b. | With a flow chart, explain the concept of switch statement.                                                          | 7 | L2 | CO2 |
|                   | c. | Write a C program to generate the pattern<br>*<br>* *<br>* * *<br>* * * *                                            | 7 | L3 | CO2 |
| <b>Module – 3</b> |    |                                                                                                                      |   |    |     |
| 5                 | a. | What is the difference between a function definition and a function prototype? Illustrate with an example code.      | 7 | L2 | CO3 |
|                   | b. | Name the different storage classes and explain the purpose of each.                                                  | 6 | L2 | CO3 |
|                   | c. | Write a program to read n numbers into an array from the user and arrange them in ascending order using bubble sort. | 7 | L3 | CO3 |

OR

|   |    |                                                                                                                                                                                                                             |   |    |     |
|---|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 6 | a. | What is the difference between pass-by-value and pass-by-reference in function arguments? Provide an example for each.                                                                                                      | 7 | L3 | CO2 |
|   | b. | Write a program to call the function addNumbers(int a, int b) with values a = 5 and b = 10. Print the result inside the main function.                                                                                      | 6 | L3 | CO3 |
|   | c. | Write a C program that prompts the user to enter a key element and searches for it with in an array. The program should return the index of the element if found, or indicate that the element is not present in the array. | 7 | L3 | CO3 |

Module – 4

|   |    |                                                                                                                                                |   |    |     |
|---|----|------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 7 | a. | Write a program to transpose the elements of a $3 \times 3$ matrix.                                                                            | 7 | L4 | CO3 |
|   | b. | Write a C program to input two m X n matrices and then calculate the sum of their corresponding elements and store it in a third m X n matrix. | 8 | L4 | CO3 |
|   | c. | State the importance of using scanf to read a string. Explain.                                                                                 | 5 | L4 | CO2 |

OR

|   |    |                                                                                                              |    |    |     |
|---|----|--------------------------------------------------------------------------------------------------------------|----|----|-----|
| 8 | a. | Write a program that multiplies two 2D matrices.                                                             | 10 | L3 | CO4 |
|   | b. | Discuss the different functions available for reading and writing strings along with syntax or example code. | 10 | L2 | CO4 |

Module – 5

|   |    |                                                               |   |    |     |
|---|----|---------------------------------------------------------------|---|----|-----|
| 9 | a. | Briefly explain string taxonomy.                              | 6 | L2 | CO5 |
|   | b. | Write a C program to calculate area of circle using pointers. | 7 | L3 | CO5 |
|   | c. | Describe a structure to store customer information.           | 7 | L2 | CO5 |

OR

|    |    |                                                                                                                   |   |    |     |
|----|----|-------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 10 | a. | Write a program to find the length of a string without using the library function.                                | 7 | L3 | CO5 |
|    | b. | Define a pointer. With an example show the declaration of pointers and their usage to assign value to a variable. | 7 | L2 | CO5 |
|    | c. | Write the syntax and example to declare a structure.                                                              | 6 | L2 | CO5 |

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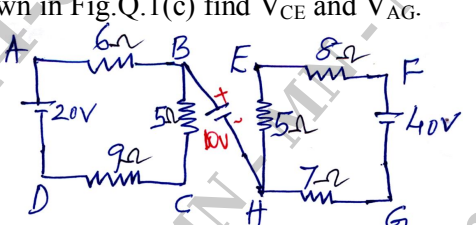
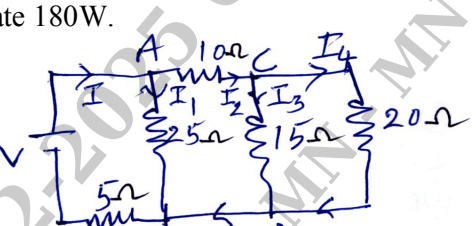
BESCK204B

## Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Introduction to Electrical Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                                 | M | L  | C   |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Explain a typical electric power system with the help of a labeled single line diagram.                                                                                                                         | 8 | L2 | CO1 |
|            | b. | Draw and explain the working of solar photovoltaic power generation system.                                                                                                                                     | 6 | L2 | CO1 |
|            | c. | For the circuit shown in Fig.Q.1(c) find $V_{CE}$ and $V_{AG}$ .<br><br>Fig.Q.1(c)                                           | 6 | L3 | CO1 |
| OR         |    |                                                                                                                                                                                                                 |   |    |     |
| Q.2        | a. | List the advantages and disadvantages of conventional and non conventional energy sources.                                                                                                                      | 7 | L2 | CO1 |
|            | b. | State and explain Kirchoff's laws as applicable to DC circuits.                                                                                                                                                 | 5 | L2 | CO1 |
|            | c. | In the circuit of Fig.Q.2(c). Find the value of supply voltage V so that 20Ω resistor can dissipate 180W.<br><br>Fig.Q.2(c) | 8 | L3 | CO2 |
| Module – 2 |    |                                                                                                                                                                                                                 |   |    |     |
| Q.3        | a. | Define the following terms applied to quantity AC:<br>i) Average value<br>ii) RMS value<br>iii) Form factor<br>iv) Frequency                                                                                    | 6 | L1 | CO2 |

1 of 3

| BESCK204B         |           |                                                                                                                                                                                                                                           |          |               |
|-------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|
|                   | <b>b.</b> | Establish the relation between voltage and current in ac circuit containing RL in series. Draw the vector diagram.                                                                                                                        | <b>8</b> | <b>L3 CO2</b> |
|                   | <b>c.</b> | List the advantages and limitations of 3 $\phi$ AC over 1 $\phi$ AC.                                                                                                                                                                      | <b>6</b> | <b>L2 CO2</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                           |          |               |
| <b>Q.4</b>        | <b>a.</b> | Draw the circuit diagram for three phase star and delta connection. Write the relation between line and phase voltage and currents.                                                                                                       | <b>6</b> | <b>L2 CO2</b> |
|                   | <b>b.</b> | Define the terms active power, reactive power and apparent power.                                                                                                                                                                         | <b>6</b> | <b>L1 CO2</b> |
|                   | <b>c.</b> | A series circuit with $R = 10\Omega$ , $L = 50 \text{ mH}$ and $C = 100 \mu\text{F}$ is provided with 200 V, 50 Hz supply. Find: i) Impedance ii) Current iii) Power iv) Power factor.                                                    | <b>8</b> | <b>L3 CO2</b> |
| <b>Module – 3</b> |           |                                                                                                                                                                                                                                           |          |               |
| <b>Q.5</b>        | <b>a.</b> | With a neat sketch, explain the principle and operation of a DC generator.                                                                                                                                                                | <b>8</b> | <b>L2 CO3</b> |
|                   | <b>b.</b> | Derive the EMF equation of DC generator.                                                                                                                                                                                                  | <b>6</b> | <b>L3 CO3</b> |
|                   | <b>c.</b> | Calculate the emf generated by a 6 pole DC generator having 480 conductors and driven at a speed of 1200 rpm. The flux/pole is 0.012 wb. Assume the generator to be i) Lap wound ii) Wave wound.                                          | <b>6</b> | <b>L3 CO3</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                           |          |               |
| <b>Q.6</b>        | <b>a.</b> | Define back emf and explain its significance.                                                                                                                                                                                             | <b>6</b> | <b>L2 CO3</b> |
|                   | <b>b.</b> | Derive expression for torque developed in a DC motor.                                                                                                                                                                                     | <b>6</b> | <b>L2 CO3</b> |
|                   | <b>c.</b> | Explain speed control of DC motor by<br>i) Armature voltage control method<br>ii) Flux control method.                                                                                                                                    | <b>8</b> | <b>L2 CO3</b> |
| <b>Module – 4</b> |           |                                                                                                                                                                                                                                           |          |               |
| <b>Q.7</b>        | <b>a.</b> | Derive EMF equation of 1 $\phi$ transformer.                                                                                                                                                                                              | <b>6</b> | <b>L3 CO4</b> |
|                   | <b>b.</b> | List the various losses occurring in a transformer, explain how copper loss varies with load.                                                                                                                                             | <b>8</b> | <b>L2 CO4</b> |
|                   | <b>c.</b> | 500 KVA, 1 $\phi$ transformer has $R_1 = 0.4 \Omega$ , $R_2 = 0.001 \Omega$ , $V_1 = 6600 \text{ V}$ , $V_2 = 400 \text{ V}$ , Iron loss = 3 kW, full load copper loss = 3.858 kW. Determine: i) Efficiency at full load, 0.8 pf lagging. | <b>6</b> | <b>L3 CO4</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                           |          |               |
| <b>Q.8</b>        | <b>a.</b> | With a neat sketch, explain how rotating magnetic field is produced in a 3 $\phi$ induction motor.                                                                                                                                        | <b>8</b> | <b>L2 CO4</b> |
|                   | <b>b.</b> | With a neat sketch, explain squirrel cage and wound rotor of a 3 $\phi$ induction motor.                                                                                                                                                  | <b>6</b> | <b>L2 CO4</b> |
| 2 of 3            |           |                                                                                                                                                                                                                                           |          |               |

| BESCK204B         |    |                                                                                                                                                             |   |    |     |
|-------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|                   | c. | Define slip explain its significance.                                                                                                                       | 6 | L2 | CO4 |
| <b>Module – 5</b> |    |                                                                                                                                                             |   |    |     |
| <b>Q.9</b>        | a. | With neat circuit and switching table explain the 2 way and 3 way control of lamp.                                                                          | 8 | L2 | CO5 |
|                   | b. | With a neat diagram, explain the working of fuse.                                                                                                           | 6 | L2 | CO5 |
|                   | c. | What is earthing? With a neat diagram explain the pipe earthing.                                                                                            | 6 | L2 | CO5 |
| <b>OR</b>         |    |                                                                                                                                                             |   |    |     |
| <b>Q.10</b>       | a. | Define tariff, explain 2 part tariff with its advantages and disadvantages.                                                                                 | 8 | L2 | CO5 |
|                   | b. | What is electric shock? Give the preventive measures against the shock.                                                                                     | 6 | L2 | CO5 |
|                   | c. | List the power rating of house hold appliances including air conditioners PC's, Laptops, printers, LED bulbs etc. Find the total load installed in a house. | 6 | L2 | CO5 |

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# CBCS SCHEME

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BESCK204D

## Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Introduction to Mechanical Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                        | M  | L  | C   |
|------------|----|------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Explain the role of Mechanical engineering in Industries and society.                                                  | 8  | L2 | CO1 |
|            | b. | With a neat sketch, explain the working of hydel power plant.                                                          | 8  | L2 | CO1 |
|            | c. | Write a note on Biofuels.                                                                                              | 4  | L1 | CO1 |
| OR         |    |                                                                                                                        |    |    |     |
| Q.2        | a. | With a neat sketch, explain the working of wind mill or wind power plant along with its advantages and disadvantages.  | 8  | L2 | CO1 |
|            | b. | Explain the working of solar flat plate collector with neat sketch.                                                    | 8  | L2 | CO1 |
|            | c. | Write short note on Global warming and Ozone depletion.                                                                | 4  | L1 | CO1 |
| Module – 2 |    |                                                                                                                        |    |    |     |
| Q.3        | a. | Explain the following machine tool operations with sketch :<br>i) Turning      ii) Boring      iii) Slot milling.      | 12 | L2 | CO2 |
|            | b. | Explain 3D printing process with its advantages and disadvantages.                                                     | 8  | L2 | CO2 |
| OR         |    |                                                                                                                        |    |    |     |
| Q.4        | a. | Explain the components of CNC with block diagram along with its application and advantages.                            | 12 | L2 | CO2 |
|            | b. | Explain the working principles of lathe machine and milling machine.                                                   | 8  | L2 | CO2 |
| Module – 3 |    |                                                                                                                        |    |    |     |
| Q.5        | a. | Explain the working of 4 – stroke petrol engine with P – V diagram.                                                    | 10 | L2 | CO3 |
|            | b. | Describe the components of I.C. Engine with a neat sketch.                                                             | 10 | L2 | CO3 |
| OR         |    |                                                                                                                        |    |    |     |
| Q.6        | a. | Explain the components and working principle of following vehicles :<br>i) Electric vehicles      ii) Hybrid vehicles. | 12 | L2 | CO3 |
|            | b. | What are the advantages and disadvantages of Electric vehicles and Hybrid vehicles?                                    | 8  | L1 | CO3 |

| Module – 4 |    |                                                                                      |    |    |     |
|------------|----|--------------------------------------------------------------------------------------|----|----|-----|
| Q.7        | a. | Explain three types of ferrous and three types of non-ferrous metals.                | 12 | L2 | CO4 |
|            | b. | Write a note on the following materials :<br>i) Ceramics                  ii) Glass. | 8  | L1 | CO4 |
| OR         |    |                                                                                      |    |    |     |
| Q.8        | a. | Explain the working principle of Arc welding with a neat sketch.                     | 8  | L2 | CO4 |
|            | b. | Explain soldering and brazing process.                                               | 6  | L2 | CO4 |
|            | c. | Explain the three types of flames used in gas welding.                               | 6  | L2 | CO4 |
| Module – 5 |    |                                                                                      |    |    |     |
| Q.9        | a. | Explain open loop and closed loop mechatronic system.                                | 8  | L2 | CO5 |
|            | b. | Explain the four types of robotic configurations with 2D sketch.                     | 12 | L2 | CO5 |
| OR         |    |                                                                                      |    |    |     |
| Q.10       | a. | Describe the three types of Automation system.                                       | 6  | L2 | CO5 |
|            | b. | What are the basic elements in an Automated system? Show that in a block diagram.    | 4  | L1 | CO5 |
|            | c. | Explain the characteristics and physical design of IoT.                              | 10 | L2 | CO5 |

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BESCK204E

## Second Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Introduction to C Programming

Time: 3 hrs.

Max. Marks:100

**Note:** 1. Answer any *FIVE* full questions, choosing *ONE* full question from each module.  
2. *M* : Marks , *L*: Bloom's level, *C*: Course outcomes.

| Module – 1 |    |                                                                                              | M | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------|---|----|-----|
| 1          | a. | With a neat diagram, explain the basic organization of a computer.                           | 6 | L2 | CO1 |
|            | b. | Explain the different characteristic features of stored program concept.                     | 6 | L2 | CO1 |
|            | c. | Explain the basic structure of a C program.                                                  | 8 | L2 | CO1 |
| <b>OR</b>  |    |                                                                                              |   |    |     |
| 2          | a. | Explain the classification of computers.                                                     | 8 | L2 | CO1 |
|            | b. | What is an identifier? What are the rules to be followed to form an identifier?              | 6 | L2 | CO2 |
|            | c. | Draw the flow chart to calculate the sum of first ten natural numbers.                       | 6 | L3 | CO2 |
| Module – 2 |    |                                                                                              |   |    |     |
| 3          | a. | Explain the different bitwise operators in C with an example for each.                       | 8 | L2 | CO2 |
|            | b. | Write a program to find whether the given number is odd or even.                             | 5 | L3 | CO2 |
|            | c. | Explain the switch statement with syntax and example.                                        | 7 | L2 | CO2 |
| <b>OR</b>  |    |                                                                                              |   |    |     |
| 4          | a. | Differentiate between while and do-while loops.                                              | 6 | L2 | CO2 |
|            | b. | Write program to check whether the given number is palindrome or not.                        | 8 | L3 | CO2 |
|            | c. | Write a program to generate and print the first 'n' Fibonacci numbers.                       | 6 | L3 | CO2 |
| Module – 3 |    |                                                                                              |   |    |     |
| 5          | a. | What is a function? Why are functions needed?                                                | 6 | L2 | CO5 |
|            | b. | Explain the different methods of passing parameters to functions giving an example for each. | 8 | L3 | CO5 |
|            | c. | Write a program to find the factorial of a number using recursion.                           | 6 | L3 | CO5 |
| 1 of 2     |    |                                                                                              |   |    |     |

OR

|   |    |                                                                                      |    |    |     |
|---|----|--------------------------------------------------------------------------------------|----|----|-----|
| 6 | a. | Discuss the different operations that can be performed on arrays.                    | 10 | L2 | CO3 |
|   | b. | Write a program to sort the given 'n' elements in ascending order using bubble sort. | 10 | L3 | CO3 |

## Module – 4

|   |    |                                                                                               |    |    |     |
|---|----|-----------------------------------------------------------------------------------------------|----|----|-----|
| 7 | a. | Explain the declaration and initialization of two-dimensional array with an example for each. | 8  | L2 | CO3 |
|   | b. | Write a program to multiply two matrices.                                                     | 12 | L3 | CO3 |

OR

|   |    |                                                                                          |    |    |     |
|---|----|------------------------------------------------------------------------------------------|----|----|-----|
| 8 | a. | Explain the different functions used to read and write strings with an example for each. | 12 | L3 | CO3 |
|   | b. | Write a program to find the length of a string without using library functions.          | 8  | L3 | CO3 |

## Module – 5

|   |    |                                                                                                                                        |    |    |     |
|---|----|----------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 9 | a. | Explain the following string manipulation functions with an example :<br>i) strcat()<br>ii) strcmp()<br>iii) strstr()<br>iv) strcpy(). | 10 | L3 | CO3 |
|   | b. | Define pointer. Explain with example the pointer declaration and initialization.                                                       | 5  | L2 | CO4 |
|   | c. | Write a program to swap two numbers using pointer.                                                                                     | 5  | L3 | CO4 |

OR

|    |    |                                                                                                                                       |    |    |     |
|----|----|---------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 10 | a. | Write a program to read and display the details of 'n' students using structure. The details include roll no, name, branch and marks. | 10 | L3 | CO4 |
|    | b. | Write a program using pointers to compute the sum mean and standard deviation of all elements stored in an array of 'n' real numbers. | 10 | L3 | CO4 |

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BETCK105B

**First Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025**

## Green Buildings

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*

*2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1        |    |                                                                                                              | M  | L  | C   |
|-------------------|----|--------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1               | a. | Define Green Building. Explain the necessity of alternate materials for conventional buildings materials.    | 10 | L2 | CO1 |
|                   | b. | Briefly explain the applications of FRP composites.                                                          | 10 | L2 | CO1 |
| <b>OR</b>         |    |                                                                                                              |    |    |     |
| Q.2               | a. | Briefly explain stabilized mud blocks and laterite Block as alternate materials.                             | 10 | L2 | CO1 |
|                   | b. | Explain the major environmental concerns due to quarrying of building materials.                             | 10 | L2 | CO1 |
| <b>Module – 2</b> |    |                                                                                                              |    |    |     |
| Q.3               | a. | With neat sketches, explain the Rat Trap Bond wall construction.                                             | 10 | L2 | CO2 |
|                   | b. | Write short note on :<br>i) Precast and ready to use building elements<br>ii) Composite beam and roof panels | 10 | L2 | CO2 |
| <b>OR</b>         |    |                                                                                                              |    |    |     |
| Q.4               | a. | Explain the role of Nirmiti Kendra Towards Low Cost housing.                                                 | 10 | L2 | CO2 |
|                   | b. | Explain the ferrocement construction process and list the various ferrocement products.                      | 10 | L2 | CO2 |
| <b>Module – 3</b> |    |                                                                                                              |    |    |     |
| Q.5               | a. | Define Global Warming. Explain the effects of Global warming.                                                | 10 | L2 | CO3 |
|                   | b. | Define Carbon footprint. Explain the Global efforts to reduce the carbon footprint emissions.                | 10 | L2 | CO3 |
| <b>OR</b>         |    |                                                                                                              |    |    |     |
| Q.6               | a. | Differentiate between the initial cost involved in Green Building and Conventional Building.                 | 10 | L2 | CO3 |
|                   | b. | List and explain various benefits of Green Building in detail.                                               | 10 | L2 | CO3 |
| <b>Module – 4</b> |    |                                                                                                              |    |    |     |
| Q.7               | a. | Explain the necessity of green building ratings.                                                             | 10 | L2 | CO4 |
|                   | b. | Explain the key highlights of GRIHA.                                                                         | 10 | L2 | CO4 |

| OR         |    |                                                                               |    |    |     |
|------------|----|-------------------------------------------------------------------------------|----|----|-----|
| Q.8        | a. | What is LEED? What are the key areas for certification of buildings by LEED?  | 10 | L2 | CO4 |
|            | b. | Write a note on different characteristics of sustainable buildings.           | 10 | L2 | CO4 |
| Module – 5 |    |                                                                               |    |    |     |
| Q.9        | a. | Explain the concept of solar passive cooling and heating design of buildings. | 10 | L2 | CO5 |
|            | b. | Explain the effective methods for reducing water usage in buildings.          | 10 | L2 | CO5 |
| OR         |    |                                                                               |    |    |     |
| Q.10       | a. | Explain solid waste management disposal techniques.                           | 10 | L2 | CO5 |
|            | b. | Explain the merits and demerits of solar powered building concepts.           | 10 | L2 | CO5 |

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BETCK205B

## Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

### Green Buildings

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                | M  | L  | C   |
|------------|----|------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Explain stabilized mud block and list stabilizers used in mud construction.                    | 10 | L2 | CO1 |
|            | b. | Explain any two types of cost effective construction material with their advantages.           | 10 | L2 | CO1 |
| OR         |    |                                                                                                |    |    |     |
| Q.2        | a. | Explain the need for reuse and recycle of building materials along with its benefits.          | 10 | L2 | CO1 |
|            | b. | Explain the various environmental issues due to quarrying of building materials.               | 10 | L2 | CO1 |
| Module – 2 |    |                                                                                                |    |    |     |
| Q.3        | a. | Explain features of cavity wall.                                                               | 10 | L2 | CO2 |
|            | b. | Explain Rat-trap bond with its advantages and disadvantages.                                   | 10 | L2 | CO2 |
| OR         |    |                                                                                                |    |    |     |
| Q.4        | a. | Explain the ferrocement construction process and list the different ferrocement products.      | 10 | L2 | CO2 |
|            | b. | Briefly explain the contribution of Nirmitti Kendra for cost effective technology development. | 10 | L2 | CO2 |
| Module – 3 |    |                                                                                                |    |    |     |
| Q.5        | a. | Write a note on causes and effects of global warming.                                          | 10 | L2 | CO3 |
|            | b. | Explain the environmental benefits of green buildings.                                         | 10 | L2 | CO3 |
| OR         |    |                                                                                                |    |    |     |
| Q.6        | a. | Explain the various means of reducing carbon emissions.                                        | 10 | L2 | CO3 |
|            | b. | Differentiate between conventional building and green building.                                | 10 | L2 | CO3 |
| Module – 4 |    |                                                                                                |    |    |     |
| Q.7        | a. | Explain the features and benefits of GRIHA.                                                    | 10 | L2 | CO4 |
|            | b. | Briefly explain the BREEAM assessment category.                                                | 10 | L2 | CO4 |
| OR         |    |                                                                                                |    |    |     |
| Q.8        | a. | Write a note on different characteristics of sustainable buildings.                            | 10 | L2 | CO4 |
|            | b. | Write a note on objectives of green building design.                                           | 10 | L2 | CO4 |
| Module – 5 |    |                                                                                                |    |    |     |
| Q.9        | a. | Explain the concept of solar passive heating in buildings.                                     | 10 | L2 | CO5 |
|            | b. | Explain the advantages and disadvantages of solar powered building concepts.                   | 10 | L2 | CO5 |
| OR         |    |                                                                                                |    |    |     |
| Q.10       | a. | Describe the process of management of solid waste.                                             | 10 | L2 | CO5 |
|            | b. | Explain water utilization in green buildings.                                                  | 10 | L2 | CO5 |

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Question Paper Version : A

**First / Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025**

**Indian Constitution**

Time: 1 hr.]

[Max. Marks: 50

**INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
1. The idea of the Constitution of India was flashed for the first time by :  
a) M.N. Roy                      b) Motilal Nehru              c) B. R. Ambedkar              d) M.K. Gandhi
  2. The first session of the Constitution Assembly was held on :  
a) 25 December 1949              b) 26 January 1949  
c) 09 December 1946              d) 11 November 1946
  3. Which one of the following acted as the Provisional President of Constituent Assembly?  
a) Dr. Rajendra Prasad              b) Jawaharlal Nehru  
c) K.M. Munshi                      d) Dr. Sachidananda Sinha
  4. The maximum influence on the Constitution of India was exercised by :  
a) The Constitution of USA              b) The Constitution of UK  
c) The Government of India Act 1935              d) The Constitution of Ireland
  5. \_\_\_\_\_ is the key to open the minds of the makers of the Constitution.  
a) Fundamental Rights              b) Directive principles of State Policy  
c) Preamble                              d) Fundamental Duties
  6. This is not the right of the arrested person  
a) To produce before the magistrate immediately  
b) To inform him the grounds of his arrest  
c) To consult his lawyer  
d) To product before the magistrate within 24 hours of his arrest.

7. Under which Amendment, a new Article 21 – A was inserted and it provides for “Right to Education” was made a Fundamental Rights.  
a) 76<sup>th</sup> Amendment    b) 86<sup>th</sup> Amendment    c) 91<sup>st</sup> Amendment    d) 42<sup>nd</sup> Amendment
8. The concept of secular state implies  
a) No religion    b) Dictatorship  
c) Neutrality of religion    d) Adoption of a single religion
9. The federal features of the Indian Constitution provides for :  
a) Distribution of legislative powers between the Union Government and the State Government.  
b) Division of powers between Executive and Judiciary  
c) Distribution of powers between Lok Sabha and Rajya Sabha  
d) Distribution of powers between Prime Minister and Cabinet.
10. How many times the Preamble of the Indian Constitution has been amended?  
a) Once    b) Twice    c) Trice    d) Not amended
11. The Original constitution classified the Fundamental Rights, into seven categories but not there are only  
a) Three categories    b) Four categories    c) Five categories    d) Six categories
12. India has which system of Government?  
a) Presidential    b) Monarchical    c) Parliamentary    d) Autocratic
13. The Government of India Act introduced a system of diarchy in the provinces ; Dyouchy means a system of  
a) Responsible Government    b) Dictatorship  
c) Double Government    d) Aristocratic Government
14. Reservation in promotion to SC’s and ST’s permitted in  
a) Article 14    b) Article 15    c) Article 16    d) Article 18
15. Which of the following is not covered under Article 20, protection in respect of conviction of offence?  
a) Right to livelihood    b) No ex – post – facto  
c) No double jeopardy    d) No self - incrimination
16. Article 196(1) (a) guarantees freedom of speech and expression to  
a) All citizen of India    b) All Indian and foreigners  
c) Only person below 21 years of age    d) Only person attained 60 years
17. Cultural and Educational Rights are dealt in  
a) Article 30 and 31    b) Article 27 and 28    c) Article 29 and 30    d) Article 14 and 15
18. A new chapter IVA on Fundamental Duties was inserted in the Indian Constitution in  
a) 1972    b) 1976    c) 1980    d) 1984
19. Which one of the following Fundamental Right was described by Dr. A. R. Ambedkar as the heart and soul of Constitution?  
a) Right to Equality    b) Right to Religion  
c) Right to Constitutional remedies    d) All of these

20. Uniform civil code for all citizen is  
 a) Directive principles  
 b) Fundamental Duties  
 c) Fundamental Rights  
 d) None of these
21. The Directive principles of State Policy  
 a) Cannot be enforced in any count  
 b) Can be enforced in High Courts only  
 c) Can be enforced in Supreme Court only  
 d) Can be enforced in both Supreme Court and High Court.
22. The Chief justice of Supreme Court of India is appointed by  
 a) Prime Minister  
 b) President  
 c) Council of Ministers  
 d) Parliament
23. To whom the Indian Constitution has given the power to pardon the sentence of death?  
 a) Chief Justice of Supreme Court  
 b) Governor of State Government  
 c) President of Union Government  
 d) Both (b) and (c)
24. Fundamental duties are applicable to all \_\_\_\_\_  
 a) States  
 b) Foreigners  
 c) Citizens  
 d) All of these
25. President can resign to his office by giving his resignation to  
 a) Chief Justice of India  
 b) Prime Minister  
 c) Speaker of Lok Sabha  
 d) Vice President
26. The Vice President has  
 a) Executive powers  
 b) Legislative powers  
 c) Right to preside over Rajya Sabha  
 d) Right to grant pardon
27. The members of Lok Sabha are elected for a term of  
 a) 4 years  
 b) 6 years  
 c) 12 years  
 d) 5 years
28. Who appoints the Chairman of the Union Public Service Commission?  
 a) President  
 b) Prime Minister  
 c) Parliament  
 d) Chief Justice of India
29. The Supreme Court has original jurisdiction to decide  
 a) Dispute between two or more states  
 b) Dispute between India and Pakistan  
 c) Dispute arises at different levels of self government  
 d) Criminal cases filed directly to Supreme Court by any citizen.
30. President of India is  
 a) Real executive  
 b) Head of cabinet  
 c) Head of the Government  
 d) Nominal Executive
31. The Chief Justice and other Judges of High Court continue in office until they attain the age of  
 a) 60 Years  
 b) 62 Years  
 c) 58 Years  
 d) 65 Years

32. What is the system of legislature in the State of Karnataka?  
a) Bicameral                      b) Unicameral                      c) Tricameral                      d) None of these
33. In our Country, the amendment of the Constitution can be initiated by the  
a) Parliament                      b) People                      c) President                      d) Supreme Court
34. To declare national emergency, a decision must be taken by  
a) Chief Justice                      b) Lok Sabha                      c) Rajya Sabha                      d) Union Cabinet
35. The speaker of Lok Sabha shall be elected by the  
a) Member of Rajya Sabha                      b) President  
c) Members of Lok Sabha                      d) Vice - President
36. The size of the ministry in a State is decided by the  
a) Governor                      b) Chief Minister  
c) Chief Justice of High Court                      d) Members of Legislative Assembly
37. Who presides over the joint session of Parliament?  
a) Speaker of Lok Sabha                      b) President  
c) Vice – President                      d) Prime Minister
38. The maximum number of Minister, including the Chief Minister in a State shall not be more than of Vidhan Sabha's strength.  
a) 10 %                      b) 12 %                      c) 15 %                      d) 20 %
39. Which amendment of the Indian Constitution lowered the voting age from 21 to 18?  
a) 42<sup>nd</sup> Amendment                      b) 44<sup>th</sup> Amendment                      c) 73<sup>rd</sup> Amendment                      d) 61<sup>st</sup> Amendment
40. Which one of the following Amendments to the Indian Constitution is called a Mini Constitution?  
a) 24<sup>th</sup> Amendment                      b) 38<sup>th</sup> Amendment                      c) 39<sup>th</sup> Amendment                      d) 42<sup>nd</sup> Amendment
41. Which Article of Indian Constitution lays down the method of Amendment?  
a) Article 360                      b) Article 368                      c) Article 370                      d) Article 376
42. National Emergency proclamation under Article 352 is issued in case of  
a) War                      b) External Aggression  
c) Armed rebellion                      d) All of these
43. Governor addresses his resignation to  
a) The Prime Minister                      b) The President  
c) Vice President                      d) Chief Minister
44. Election Commission of India conducts election to  
a) Parliament  
b) State Legislatures  
c) Office of the President and Vice President  
d) All of these

\* \* \* \* \*

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Question Paper Version : B

**First / Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025  
Indian Constitution**

Time: 1 hr.]

[Max. Marks: 50

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- 
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  3. Governor addresses his resignation to  
a) The Prime Minister                      b) The President  
c) Vice President                      d) Chief Minister
  4. Election Commission of India conducts election to  
a) Parliament  
b) State Legislatures  
c) Office of the President and Vice President  
d) All of these
  5. The President is empowered to dissolve :  
a) Only Lok Sabha                      b) Only Rajya Sabha  
c) Both Lok Sabha and Rajya Sabha                      d) None of these
  6. On what grounds the Judges of Supreme Court are removed?  
a) Unconstitutional                      b) Judicial Review  
c) Proved Misbehaviour , incapacity                      d) None of these
  7. How many times the President of India can seek re – election to his post?  
a) Once                                      b) Three times  
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8. Collectively the cabinet is responsible to the  
a) Prime Minister      b) Lok Sabha      c) Parliament      d) President
9. This is not a Fundamental duty  
a) Respect to National Flag and National Anthem  
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c) Respect to elders and teachers  
d) To preserve culture and heritage of India.
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a) First session      b) Second session      c) Third session      d) None of these
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35. \_\_\_\_\_ is the key to open the minds of the makers of the Constitution.

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  - b) President
  - c) Council of Ministers
  - d) Parliament
43. To whom the Indian Constitution has given the power to pardon the sentence of death?

  - a) Chief Justice of Supreme Court
  - b) Governor of State Government
  - c) President of Union Government
  - d) Both (b) and (c)
44. Fundamental duties are applicable to all \_\_\_\_\_

  - a) States
  - b) Foreigners
  - c) Citizens
  - d) All of these
45. President can resign to his office by giving his resignation to

  - a) Chief Justice of India
  - b) Prime Minister
  - c) Speaker of Lok Sabha
  - d) Vice President

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Question Paper Version : C

**First / Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025**

**Indian Constitution**

Time: 1 hr.]

[Max. Marks: 50

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    - d) Can be enforced in both Supreme Court and High Court.
  2. The Chief justice of Supreme Court of India is appointed by
    - a) Prime Minister
    - b) President
    - c) Council of Ministers
    - d) Parliament
  3. To whom the Indian Constitution has given the power to pardon the sentence of death?
    - a) Chief Justice of Supreme Court
    - b) Governor of State Government
    - c) President of Union Government
    - d) Both (b) and (c)
  4. Fundamental duties are applicable to all \_\_\_\_\_
    - a) States
    - b) Foreigners
    - c) Citizens
    - d) All of these
  5. President can resign to his office by giving his resignation to
    - a) Chief Justice of India
    - b) Prime Minister
    - c) Speaker of Lok Sabha
    - d) Vice President
  6. The Vice President has
    - a) Executive powers
    - b) Legislative powers
    - c) Right to preside over Rajya Sabha
    - d) Right to grant pardon
  7. The members of Lok Sabha are elected for a term of
    - a) 4 years
    - b) 6 years
    - c) 12 years
    - d) 5 years

8. Who appoints the Chairman of the Union Public Service Commission?
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  - b) Only Rajya Sabha
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  - a) Unconstitutional
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17. How many times the President of India can seek re – election to his post?
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  - c) Parliament
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  - d) To preserve culture and heritage of India.

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30. Which one of the following Amendments to the Indian Constitution is called a Mini Constitution?  
a) 24<sup>th</sup> Amendment      b) 38<sup>th</sup> Amendment      c) 39<sup>th</sup> Amendment      d) 42<sup>nd</sup> Amendment
31. The Original constitution classified the Fundamental Rights, into seven categories but not there are only  
a) Three categories      b) Four categories      c) Five categories      d) Six categories
32. India has which system of Government?  
a) Presidential      b) Monarchical      c) Parliamentary      d) Autocratic
33. The Government of India Act introduced a system of diarchy in the provinces ; Dyouchy means a system of  
a) Responsible Government      b) Dictatorship  
c) Double Government      d) Aristocratic Government

34. Reservation in promotion to SC's and ST's permitted in  
a) Article 14                      b) Article 15                      c) Article 16                      d) Article 18
35. Which of the following is not covered under Article 20, protection in respect of conviction of offence?  
a) Right to livelihood                      b) No ex – post – facto  
c) No double jeopardy                      d) No self - incrimination
36. Article 196(1) (a) guarantees freedom of speech and expression to  
a) All citizen of India                      b) All Indian and foreigners  
c) Only person below 21 years of age                      d) Only person attained 60 years
37. Cultural and Educational Rights are dealt in  
a) Article 30 and 31    b) Article 27 and 28    c) Article 29 and 30    d) Article 14 and 15
38. A new chapter IVA on Fundamental Duties was inserted in the Indian Constitution in  
a) 1972                      b) 1976                      c) 1980                      d) 1984
39. Which one of the following Fundamental Right was described by Dr. A. R. Ambedkar as the heart and soul of Constitution?  
a) Right to Equality                      b) Right to Religion  
c) Right to Constitutional remedies                      d) All of these
40. Uniform civil code for all citizen is  
a) Directive principles                      b) Fundamental Duties  
c) Fundamental Rights                      d) None of these
41. The idea of the Constitution of India was flashed for the first time by :  
a) M.N. Roy                      b) Motilal Nehru                      c) B. R. Ambedkar                      d) M.K. Gandhi
42. The first session of the Constitution Assembly was held on :  
a) 25 December 1949                      b) 26 January 1949  
c) 09 December 1946                      d) 11 November 1946
43. Which one of the following acted as the Provisional President of Constituent Assembly?  
a) Dr. Rajendra Prasad                      b) Jawaharlal Nehru  
c) K.M. Munshi                      d) Dr. Sachidananda Sinha
44. The maximum influence on the Constitution of India was exercised by :  
a) The Constitution of USA                      b) The Constitution of UK  
c) The Government of India Act 1935                      d) The Constitution of Ireland
45. \_\_\_\_\_ is the key to open the minds of the makers of the Constitution.  
a) Fundamental Rights                      b) Directive principles of State Policy  
c) Preamble                      d) Fundamental Duties
46. This is not the right of the arrested person  
a) To produce before the magistrate immediately  
b) To inform him the grounds of his arrest  
c) To consult his lawyer  
d) To product before the magistrate within 24 hours of his arrest.

47. Under which Amendment, a new Article 21 – A was inserted and it provides for “Right to Education” was made a Fundamental Rights.  
a) 76<sup>th</sup> Amendment    b) 86<sup>th</sup> Amendment    c) 91<sup>st</sup> Amendment    d) 42<sup>nd</sup> Amendment
48. The concept of secular state implies  
a) No religion    b) Dictatorship  
c) Neutrality of religion    d) Adoption of a single religion
49. The federal features of the Indian Constitution provides for :  
a) Distribution of legislative powers between the Union Government and the State Government.  
b) Division of powers between Executive and Judiciary  
c) Distribution of powers between Lok Sabha and Rajya Sabha  
d) Distribution of powers between Prime Minister and Cabinet.
50. How many times the Preamble of the Indian Constitution has been amended?  
a) Once    b) Twice    c) Trice    d) Not amended

\* \* \* \* \*

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Question Paper Version : D

**First / Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025  
Indian Constitution**

Time: 1 hr.]

[Max. Marks: 50

**INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
1. The Chief Justice and other Judges of High Court continue in office until they attain the age of  
a) 60 Years                      b) 62 Years                      c) 58 Years                      d) 65 Years
  2. What is the system of legislature in the State of Karnataka?  
a) Bicameral                      b) Unicameral                      c) Tricameral                      d) None of these
  3. In our Country, the amendment of the Constitution can be initiated by the  
a) Parliament                      b) People                      c) President                      d) Supreme Court
  4. To declare national emergency, a decision must be taken by  
a) Chief Justice                      b) Lok Sabha                      c) Rajya Sabha                      d) Union Cabinet
  5. The speaker of Lok Sabha shall be elected by the  
a) Member of Rajya Sabha                      b) President  
c) Members of Lok Sabha                      d) Vice - President
  6. The size of the ministry in a State is decided by the  
a) Governor                      b) Chief Minister  
c) Chief Justice of High Court                      d) Members of Legislative Assembly
  7. Who presides over the joint session of Parliament?  
a) Speaker of Lok Sabha                      b) President  
c) Vice – President                      d) Prime Minister
  8. The maximum number of Minister, including the Chief Minister in a State shall not be more than of Vidhan Sabha's strength.  
a) 10 %                      b) 12 %                      c) 15 %                      d) 20 %

9. Which amendment of the Indian Constitution lowered the voting age from 21 to 18?  
a) 42<sup>nd</sup> Amendment   b) 44<sup>th</sup> Amendment   c) 73<sup>rd</sup> Amendment   d) 61<sup>st</sup> Amendment
10. Which one of the following Amendments to the Indian Constitution is called a Mini Constitution?  
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d) Distribution of powers between Prime Minister and Cabinet.
20. How many times the Preamble of the Indian Constitution has been amended?  
a) Once   b) Twice   c) Trice   d) Not amended



33. To whom the Indian Constitution has given the power to pardon the sentence of death?  
 a) Chief Justice of Supreme Court                      b) Governor of State Government  
 c) President of Union Government                      d) Both (b) and (c)
34. Fundamental duties are applicable to all \_\_\_\_  
 a) States                      b) Foreigners                      c) Citizens                      d) All of these
35. President can resign to his office by giving his resignation to  
 a) Chief Justice of India                      b) Prime Minister  
 c) Speaker of Lok Sabha                      d) Vice President
36. The Vice President has  
 a) Executive powers                      b) Legislative powers  
 c) Right to preside over Rajya Sabha                      d) Right to grant pardon
37. The members of Lok Sabha are elected for a term of  
 a) 4 years                      b) 6 years                      c) 12 years                      d) 5 years
38. Who appoints the Chairman of the Union Public Service Commission?  
 a) President                      b) Prime Minister  
 c) Parliament                      d) Chief Justice of India
39. The Supreme Court has original jurisdiction to decide  
 a) Dispute between two or more states  
 b) Dispute between India and Pakistan  
 c) Dispute arises at different levels of self government  
 d) Criminal cases filed directly to Supreme Court by any citizen.
40. President of India is  
 a) Real executive                      b) Head of cabinet  
 c) Head of the Government                      d) Nominal Executive
41. The Original constitution classified the Fundamental Rights, into seven categories but not there are only  
 a) Three categories                      b) Four categories                      c) Five categories                      d) Six categories
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 c) Double Government                      d) Aristocratic Government
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 a) Article 14                      b) Article 15                      c) Article 16                      d) Article 18
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 c) No double jeopardy                      d) No self - incrimination

46. Article 19(1) (a) guarantees freedom of speech and expression to
  - a) All citizen of India
  - b) All Indian and foreigners
  - c) Only person below 21 years of age
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  - c) Right to Constitutional remedies
  - d) All of these
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  - b) Fundamental Duties
  - c) Fundamental Rights
  - d) None of these

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Question Paper Version : A

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

## **Innovation and Design Thinking**

Time: 1 hr.]

[Max. Marks: 50

### **INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
1. Design thinking is a process of \_\_\_\_  
a) Thinking about design  
b) Designing ways in which people think  
c) Asking users to solve problems  
d) Defining framing and solving problems from user's prospective.
  2. Design thinking typically helps in \_\_\_\_  
a) Innovation  
b) Data analytics  
c) Financial planning  
d) Operation efficiency
  3. Design thinking principle DO NOT include  
a) Feasibility  
b) Viability  
c) Desirability  
d) Credibility
  4. Design thinker in an organization are  
a) People  
b) Employees  
c) Managers  
d) All of these
  5. What are the steps of Design thinking process?  
a) Understand > Draw > Ideate > Create > Test  
b) Empathize > Define > Ideate > Prototype > Test  
c) Empathize > Design > Implement > Produce > Test  
d) Understand > Define > Ideate > Produce > Try

6. Design thinking is a linear process  
a) True                      b) False                      c) Cannot say                      d) None of these
7. Which of these is a reason that companies might Implement design thinking?  
a) It relies on risk - taking                      b) It creates more problem  
c) It accelerates effectiveness                      d) It eliminates mistakes
8. Tools of Design thinking are \_\_\_\_  
a) Visualization                      b) Experience mapping  
c) Rapid concept development                      d) All of these
9. Collaboration in design thinking for strategic innovation includes.  
a) Collaboration with all the members of the organization for shared solution for complex problem.  
b) Collaboration with design thinking team  
c) Collaboration with design thinking consultant  
d) Collaboration with operation team.
10. What does MVP stand for \_\_\_\_  
a) Most viable product                      b) Maximum viable product  
c) Minimum viable product                      d) None of these
11. Journey Mapping is a \_\_\_\_  
a) Geographic research                      b) Ethnographic research  
c) Both (a) & (b)                      d) None of these
12. Which of the following are NOT tools of Design thinking?  
a) Co - creation                      b) Prototyping  
c) Online Marketing                      d) Mind Mapping
13. \_\_\_\_ is a way of collaboration where in participants , regardless of their location , work together to reach a certain goal.  
a) Distributed work                      b) Distributed collaboration  
c) Multiple perspective                      d) All of these
14. A Rapid concept Development can be used before the progress of \_\_\_\_  
a) 25%                      b) 50%                      c) 30%                      d) 35%
15. Mind Maps are used to \_\_\_\_ ideas.  
a) Generate                      b) Visualize                      c) Organize                      d) All of these
16. Brain storming session is \_\_\_\_  
a) There is no rule, it is UP to the brainstorming facilitator to decide  
b) A mix of individual and collective activities  
c) A collective activity only  
d) An individual activity only
17. Preparing and representing project information in the form of stories , maps and images is \_\_\_\_  
a) Journey mapping                      b) Visualization  
c) Value chain analysis                      d) Prototyping

18. What is way to narrow down the thoughts to reach at the final solution?  
a) Convergent thinking                      b) Divergent thinking  
c) Linear thinking                            d) All of these
19. To Empathize in design thinking means  
a) To understand the user                      b) To have sympathy  
c) To be socially responsible                      d) To understand the Technology
20. \_\_\_\_\_ is used with the objective of identifying the needs of the customers which the customers themselves are unable to articulate.  
a) Mind mapping                            b) Experience mapping  
c) Story telling                                d) Rapid concept development
21. The three “I”s of design thinking DO NOT includes.  
a) Interest                      b) Implementation                      c) Inspiration                      d) Ideation
22. What is the characteristic for the location of a virtual team?  
a) In the same building                      b) Remotely  
c) In the same industry                      d) In the same country
23. Design thinking follows  
a) Waterfall Model                            b) Agile methodology  
c) Both of these                                d) None of these
24. Collaborative team work is essential in design thinking for  
a) Equal importance to all members                      b) Solving multifaceted problems  
c) Unbiased selection of ideas                      d) Better failure management
25. BPM stands for  
a) Building Product Management                      b) Business Product Management  
c) Business Process Management                      d) Basic Product Management
26. \_\_\_\_\_ is an iterative and incremental method of managing development and design.  
a) Agile Methodology                            b) Waterfall Model  
c) Cyclic Methodology                            d) All of these
27. Which of the following sequences is correct for Waterfall Methodology?  
a) Define – Design – Develop – Test – Deploy  
b) Define – Develop – Design – Test – Deploy  
c) Define – Design – Develop – Deploy – Test  
d) Design – Define – Develop – Test - Deploy
28. Representation of prototypes by  
a) Story board                      b) Scenarios                      c) Screen shots                      d) All of these
29. Scope of Strategic Innovation includes  
a) Managed Innovation                            b) Strategic Alignment  
c) Industry Foresight                                d) All of these

30. What is the term for the phenomenon that a new IT system is implemented across the whole organization simultaneously?
- a) Immersed approach
  - b) Parallel approach
  - c) Plunge approach
  - d) Instant approach
31. “Products are shaped by stories that people tell about them”. The phrase indicates the importance of \_\_\_\_\_
- a) Importance of product
  - b) User experience
  - c) Importance of stories
  - d) Importance of Narration
32. The reconfiguration of existing product technologies is known as
- a) Modular Innovation
  - b) Radical Innovation
  - c) Architectural Innovation
  - d) Incremental Innovation
33. What is the goal of value chain analysis?
- a) To create a better value for customer in the value chain
  - b) To visualize ideas and organize them
  - c) To develop new business opportunities
  - d) To test prototypes of proposed solutions
34. A model of a proposed solution looks as close as possible to the final design is \_\_\_\_\_
- a) High fidelity
  - b) Low fidelity
  - c) No fidelity
  - d) MVP
35. Which tool is used to represent how ideas are linked to main idea?
- a) Visualization
  - b) Journey mapping
  - c) Mind mapping
  - d) Rapid concept development
36. What is the core belief behind co-reaction?
- a) The presence of users is essential to the creative process
  - b) Rapid development of new business ideas
  - c) Testing assumptions about a new business idea
  - d) Visualizing ideas and organizing them
37. What is the purpose of learning launches?
- a) To test prototype of proposed solutions
  - b) To explore the fundamental assumptions of market potential for new growth
  - c) To analyze the value chain of an organization
  - d) To create a visual representation of a business idea
38. What is Business Process Modelling?
- a) A way of visualization business strategies
  - b) A method for quality management
  - c) A method for time and motion study
  - d) A way of dividing business processes into basic components and functions.
39. Frank Robinson defined and coined the term
- a) Design thinking
  - b) Mind mapping
  - c) MVP
  - d) Hypothesis

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Question Paper Version : B

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

## **Innovation and Design Thinking**

Time: 1 hr.]

[Max. Marks: 50

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- 
1. Outsourcing of innovation globally is more likely where :  
a) Innovations are autonomous      b) Innovations are systematic  
c) Innovations are systematic or autonomous  
d) Innovations are made by service sector firms
  2. Which of the below firm is associated the most with design thinking?  
a) Ikea      b) Ideo      c) Idea      d) Ikei
  3. Design thickners create low fidelity prototypes to  
a) Validate concepts for the market      b) Build production ready products  
c) Estimate the price of production  
d) Test concepts quickly and cheaply with potential users
  4. Extension of a successful brand or iteration a current period is  
a) Revolutionary Innovation      b) Dynamic Innovation  
c) Incremental Innovation      d) Evolutionary Innovation
  5. Benefits of Agile in virtual collaboration are  
a) Allow openers      b) Rhythm building  
c) Visualize the whole thing      d) All of these

6. Which famous design school become most vital in framing design thinking as a core subject
  - a) Trinity College
  - b) Business School, Ahmadabad
  - c) Massachusetts University
  - d) Stanford d. school
7. The word design thinking is defined by
  - a) Tim Brown
  - b) Rober Mckin
  - c) Peter Rowes
  - d) Balter Moore
8. David is testing his prototype, what should his next move be?
  - a) Ideate and come up with more idea
  - b) Research the people he is designing for
  - c) Collect feedback from the testers to evaluate his idea
  - d) Change his problem statement
9. What element of user experience design would be the design strategy full under?
  - a) Interaction design
  - b) Experience strategy
  - c) User Research
  - d) Information architecture
10. What purpose does point of view (POV) serve in design thinking?
  - a) It serves as the guide for developing the prototype
  - b) It illustrates what your competitors are doing better
  - c) It is used in the testing phase for receiving feedback
  - d) It provides focus for the design team is brain storming.
11. “Products are shaped by stories that people tell about them”. The phrase indicates the importance of \_\_\_\_
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19. Frank Robinson defined and coined the term  
a) Design thinking      b) Mind mapping      c) MVP      d) Hypothesis
20. Experience economy has resulted in  
a) Active participation      b) Passive consumption  
c) Minimum consumption      d) None of these
21. Journey Mapping is a \_\_\_\_\_  
a) Geographic research      b) Ethnographic research  
c) Both (a) & (b)      d) None of these
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  - d) Operation efficiency
33. Design thinking principle DO NOT include
  - a) Feasibility
  - b) Viability
  - c) Desirability
  - d) Credibility
34. Design thinker in an organization are
  - a) People
  - b) Employees
  - c) Managers
  - d) All of these
35. What are the steps of Design thinking process?
  - a) Understand > Draw > Ideate > Create > Test
  - b) Empathize > Define > Ideate > Prototype > Test
  - c) Empathize > Design > Implement > Produce > Test
  - d) Understand > Define > Ideate > Produce > Try
36. Design thinking is a linear process
  - a) True
  - b) False
  - c) Cannot say
  - d) None of these
37. Which of these is a reason that companies might Implement design thinking?
  - a) It relies on risk - taking
  - b) It creates more problem
  - c) It accelerates effectiveness
  - d) It eliminates mistakes
38. Tools of Design thinking are \_\_\_\_\_
  - a) Visualization
  - b) Experience mapping
  - c) Rapid concept development
  - d) All of these
39. Collaboration in design thinking for strategic innovation includes.
  - a) Collaboration with all the members of the organization for shared solution for complex problem.
  - b) Collaboration with design thinking team
  - c) Collaboration with design thinking consultant
  - d) Collaboration with operation team.
40. What does MVP stand for \_\_\_\_\_
  - a) Most viable product
  - b) Maximum viable product
  - c) Minimum viable product
  - d) None of these

41. The three “I”s of design thinking DO NOT includes.  
a) Interest                      b) Implementation      c) Inspiration                      d) Ideation
42. What is the characteristic for the location of a virtual team?  
a) In the same building                      b) Remotely  
c) In the same industry                      d) In the same country
43. Design thinking follows  
a) Waterfall Model                      b) Agile methodology  
c) Both of these                      d) None of these
44. Collaborative team work is essential in design thinking for  
a) Equal importance to all members                      b) Solving multifaceted problems  
c) Unbiased selection of ideas                      d) Better failure management
45. BPM stands for  
a) Building Product Management                      b) Business Product Management  
c) Business Process Management                      d) Basic Product Management
46. \_\_\_\_\_ is an iterative and incremental method of managing development and design.  
a) Agile Methodology                      b) Waterfall Model  
c) Cyclic Methodology                      d) All of these
47. Which of the following sequences is correct for Waterfall Methodology?  
a) Define – Design – Develop – Test – Deploy  
b) Define – Develop – Design – Test – Deploy  
c) Define – Design – Develop – Deploy – Test  
d) Design – Define – Develop – Test - Deploy
48. Representation of prototypes by  
a) Story board      b) Scenarios      c) Screen shots                      d) All of these
49. Scope of Strategic Innovation includes  
a) Managed Innovation                      b) Strategic Alignment  
c) Industry Foresight                      d) All of these
50. What is the term for the phenomenon that a new IT system is implemented across the whole organization simultaneously?  
a) Immersed approach                      b) Parallel approach  
c) Plunge approach                      d) Instant approach

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# CBGS SCHEME

BIDTK158/258/22BD16

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Question Paper Version : C

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

## **Innovation and Design Thinking**

Time: 1 hr.]

[Max. Marks: 50

### **INSTRUCTIONS TO THE CANDIDATES**

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
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5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- 
1. "Products are shaped by stories that people tell about them". The phrase indicates the importance of \_\_\_\_  
a) Importance of product                      b) User experience  
c) Importance of stories                      d) Importance of Narration
  2. The reconfiguration of existing product technologies is known as  
a) Modular Innovation                      b) Radical Innovation  
c) Architectural Innovation                      d) Incremental Innovation
  3. What is the goal of value chain analysis?  
a) To create a better value for customer in the value chain  
b) To visualize ideas and organize them  
c) To develop new business opportunities  
d) To test prototypes of proposed solutions
  4. A model of a proposed solution looks as close as possible to the final design is \_\_\_\_  
prototype  
a) High fidelity                      b) Low fidelity                      c) No fidelity                      d) MVP
  5. Which tool is used to represent how ideas are linked to main idea?  
a) Visualization                      b) Journey mapping  
c) Mind mapping                      d) Rapid concept development

6. What is the core belief behind co-reaction?
  - a) The presence of users is essential to the creative process
  - b) Rapid development of new business ideas
  - c) Testing assumptions about a new business idea
  - d) Visualizing ideas and organizing them
7. What is the purpose of learning launches?
  - a) To test prototype of proposed solutions
  - b) To explore the fundamental assumptions of market potential for new growth
  - c) To analyze the value chain of an organization
  - d) To create a visual representation of a business idea
8. What is Business Process Modelling?
  - a) A way of visualization business strategies
  - b) A method for quality management
  - c) A method for time and motion study
  - d) A way of dividing business processes into basic components and functions.
9. Frank Robinson defined and coined the term
  - a) Design thinking
  - b) Mind mapping
  - c) MVP
  - d) Hypothesis
10. Experience economy has resulted in
  - a) Active participation
  - b) Passive consumption
  - c) Minimum consumption
  - d) None of these
11. Design thinking is a process of \_\_\_\_
  - a) Thinking about design
  - b) Designing ways in which people think
  - c) Asking users to solve problems
  - d) Defining framing and solving problems from user's prospective.
12. Design thinking typically helps in \_\_\_\_
  - a) Innovation
  - b) Data analytics
  - c) Financial planning
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  - a) Feasibility
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20. What does MVP stand for \_\_\_\_\_  
a) Most viable product                              b) Maximum viable product  
c) Minimum viable product                      d) None of these
21. Outsourcing of innovation globally is more likely where :  
a) Innovations are autonomous                  b) Innovations are systematic  
c) Innovations are systematic or autonomous  
d) Innovations are made by service sector firms
22. Which of the below firm is associated the most with design thinking?  
a) Ikea                                                  b) Ideo                                                  c) Idea                                                  d) Ikei
23. Design thickners create low fidelity prototypes to  
a) Validate concepts for the market                  b) Build production ready products  
c) Estimate the price of production  
d) Test concepts quickly and cheaply with potential users
24. Extension of a successful brand or iteration a current period is  
a) Revolutionary Innovation                          b) Dynamic Innovation  
c) Incremental Innovation                          d) Evolutionary Innovation
25. Benefits of Agile in virtual collaboration are  
a) Allow openers                                      b) Rhythm building  
c) Visualize the whole thing                      d) All of these
26. Which famous design school become most vital in framing design thinking as a core subject  
a) Trinity College                                      b) Business School, Ahmadabad  
c) Massachusetts University                      d) Standford d. school
27. The word design thinking is defined by  
a) Tim Brown                                          b) Rober Mckin                                          c) Peter Rowes                                          d) Balter Moore

28. David is testing his prototype, what should his next move be?
  - a) Ideate and come up with more idea
  - b) Research the people he is designing for
  - c) Collect feedback from the testers to evaluate his idea
  - d) Change his problem statement
29. What element of user experience design would be the design strategy full under?
  - a) Interaction design
  - b) Experience strategy
  - c) User Research
  - d) Information architecture
30. What purpose does point of view (POV) serve in design thinking?
  - a) It serves as the guide for developing the prototype
  - b) It illustrates what your competitors are doing better
  - c) It is used in the testing phase for receiving feedback
  - d) It provides focus for the design team is brain storming.
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  - a) Story board
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# CBGS SCHEME

BIDTK158/258/22BD16

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Question Paper Version : D

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,  
Dec.2024/Jan.2025**

## **Innovation and Design Thinking**

Time: 1 hr.]

[Max. Marks: 50

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a) Agile Methodology                                  b) Waterfall Model  
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a) Design thinking      b) Mind mapping      c) MVP      d) Hypothesis
30. Experience economy has resulted in  
a) Active participation      b) Passive consumption  
c) Minimum consumption      d) None of these
31. Journey Mapping is a \_\_\_\_\_  
a) Geographic research      b) Ethnographic research  
c) Both (a) & (b)      d) None of these
32. Which of the following are NOT tools of Design thinking?  
a) Co - creation      b) Prototyping  
c) Online Marketing      d) Mind Mapping
33. \_\_\_\_\_ is a way of collaboration where in participants , regardless of their location , work together to reach a certain goal.  
a) Distributed work      b) Distributed collaboration  
c) Multiple perspective      d) All of these
34. A Rapid concept Development can be used before the progress of \_\_\_\_\_  
a) 25%      b) 50%      c) 30%      d) 35%
35. Mind Maps are used to \_\_\_\_\_ ideas.  
a) Generate      b) Visualize      c) Organize      d) All of these
36. Brain storming session is \_\_\_\_\_  
a) There is no rule, it is UP to the brainstorming facilitator to decide  
b) A mix of individual and collective activities  
c) A collective activity only  
d) An individual activity only
37. Preparing and representing project information in the form of stories , maps and images is \_\_\_\_\_  
a) Journey mapping      b) Visualization  
c) Value chain analysis      d) Prototyping
38. What is way to narrow down the thoughts to reach at the final solution?  
a) Convergent thinking      b) Divergent thinking  
c) Linear thinking      d) All of these
39. To Empathize in design thinking means  
a) To understand the user      b) To have sympathy  
c) To be socially responsible      d) To understand the Technology
40. \_\_\_\_\_ is used with the objective of identifying the needs of the customers which the customers themselves are unable to articulate.  
a) Mind mapping      b) Experience mapping  
c) Story telling      d) Rapid concept development

41. Design thinking is a process of \_\_\_\_  
 a) Thinking about design  
 b) Designing ways in which people think  
 c) Asking users to solve problems  
 d) Defining framing and solving problems from user's prospective.
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Question Paper Version : A

**First/Second Semester B.E./B.Tech Degree Examination,  
Dec.2024/Jan.2025  
Balake Kannada**

Time: 1 hr.]

[Max. Marks: 50

## INSTRUCTIONS TO THE CANDIDATES

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### Translate the following Kannada words into English :

1. Maga  
a) Father                      b) Sister                      c) Daughter                      d) Son
2. Mane  
a) Temple                      b) House                      c) Church                      d) Hospital
3. Kempu BaNNa  
a) Black colour                      b) Red colour                      c) White colour                      d) Green colour

### Translate the following English words into Kannada :

4. We  
a) Naavu                      b) Namma                      c) Namage                      d) Nimage
5. This  
a) Adu                      b) Alli                      c) Idu                      d) Both (a) and (b)
6. He  
a) Avanu                      b) Avalu                      c) Naanu                      d) Neenu

Match the following using Table given below :

|    |          |      |             |
|----|----------|------|-------------|
| a) | White    | i)   | Hudugi      |
| b) | Girl     | ii)  | Sihi        |
| c) | Daughter | iii) | Granthalaya |
| d) | Sweet    | iv)  | Bili        |
| e) | Library  | v)   | Magalu      |

7. White  
a) = i                      b) = ii                      c) = iii                      d) = iv
8. Girl  
a) = i                      b) = ii                      c) = iii                      d) = v
9. Daughter  
a) = i                      b) = ii                      c) = iv                      d) = v
10. Sweet  
a) = i                      b) = ii                      c) = iii                      d) = iv
11. Library  
a) = i                      b) = ii                      c) = iii                      d) = iv

Write appropriate words for the following :

12. Mother  
a) Huduga                      b) Hudugi                      c) Tayi                      d) Tande
13. Festival  
a) Maduve                      b) Habba                      c) Mane                      d) Raste
14. Bitter  
a) Sihi                      b) Khara                      c) Uppu                      d) Kahi
15. Holiday  
a) Mara                      b) Raje                      c) Nagu                      d) Ooru
16. Town  
a) Halli                      b) Angadi                      c) Nagara                      d) Rajya

Translate the following Kannada words as per the given model [Q.No. 17 and 18]  
Example : Kacheri – Kacheriyalli.

17. Mane  
a) Manadalli                      b) Maneyinda                      c) Maneyalli                      d) Mane
18. Angadi  
a) Angadiyalli                      b) Angadihattira                      c) Angadiyage                      d) Angadiyolage

19. How would you write "This is my Book" in Kannada?  
 a) Adu Nanna pustaka b) idu nanna pustaka  
 c) Idu Avaru pustaka d) idu nannadu
20. How would you write "what is this" in Kannada?  
 a) Idu elli b) Idu yarige c) Idu yavudu d) Idu Avaradu
21. How would you write Fruit in Kannada  
 a) Hoovu b) Ele c) HaNNu d) BaNNa
22. How would you write "why sir" in Kannada?  
 a) Elli sir b) Yaake Saar c) Hogi Saar d) Banni Saar
23. What is the meaning of "Student" in Kannada?  
 a) Huduga b) Vidyarthi c) Jana d) Shale
24. How would you write "don't" in Kannada?  
 a) Beku b) Beke c) Beda d) Yaake
25. How would you write "Marriage" in Kannada?  
 a) Habba b) Maduve c) Sabe d) Prarthane

**Write the English word for the following :**

26. Chikka  
 a) Big b) Huge c) Small d) House
27. PaaTa  
 a) Story b) Lesson c) Book d) Drink
28. aNNa  
 a) Sister b) Mother c) Father d) Brother
29. Ketta  
 a) Good b) Bad c) New d) Big
30. Huduga  
 a) Son b) Boy c) Daughter d) Sister

**Match the following using the Table given below :**

|    |         |      |        |
|----|---------|------|--------|
| a) | Fruit   | i)   | Vaidya |
| b) | Tree    | ii)  | Desha  |
| c) | Flower  | iii) | Mara   |
| d) | Doctor  | iv)  | Hannu  |
| e) | Country | v)   | Hoovu  |

31. Fruit  
 a) = i b) = ii c) = iii d) = iv

32. Tree  
a) = i b) = ii c) = iii d) = iv
33. Flower  
a) = i b) = iii c) = iv d) = v
34. Doctor  
a) = i b) = ii c) = iii d) = v
35. Country  
a) = i b) = ii c) = iv d) = v

**Translate the following English sentence into Kannada sentence :**

36. Who are they?  
a) Neenu yaaru b) Avaru yaaru c) Neenu yelli d) Avaru Enu
37. What is her Name?  
a) Avalu yaru b) Avana hesarenu  
c) Avala hesarenu d) Ninna hesarenu
38. This is my Nation  
a) Idu Avara desha b) Idu Nanna desha  
c) Adu Ninna desha d) Idu Ninna desha
39. What is your name?  
a) Avala Hesaru Enu? b) Avana Hesaru Enu?  
c) Ninna Hesaru Enu? d) Nanna Hesaru Enu?
40. This is my pen.  
a) Idu nanna pustaka b) Idu Avara vastu  
c) Idu Avara Lekhani d) Idu Nanna Lekhani

**Fill in the blank for the given English words into Kannada meaning :**

41. Small  
a) Udda b) agala c) Dappa d) Chikka
42. Salty  
a) huLi b) Uppu c) Khara d) Sihi
43. Simple  
a) Khatina b) Tappu c) Sarala d) Khasagi
44. Daily  
a) Nale b) Nenne c) Nadiddu d) Dina Nitya

45. Dignity  
a) Samasye b) Ghanathe c) Sari d) Olleya

**Write the Kannada Vocabulary for the following English words :**

46. Child \_\_\_\_\_  
a) navu b) Avaru c) Magu d) vidyarthi
47. Plant \_\_\_\_\_  
a) Mara b) Gida c) Hoova d) Rembe
48. Story \_\_\_\_\_  
a) Kate b) Angadi c) Mane d) Kavana
49. Where \_\_\_\_\_  
a) Enu b) Eke c) yaaru d) Elli
50. Bad \_\_\_\_\_  
a) hosa b) Naataka c) Ketta d) Olleya

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Question Paper Version : B

**First/Second Semester B.E./B.Tech Degree Examination,  
Dec.2024/Jan.2025  
Balake Kannada**

Time: 1 hr.]

[Max. Marks: 50

## INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners on the OMR sheets are strictly prohibited.**

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**Translate the following English sentence into Kannada sentence :**

1. Who are they?  
a) Neenu yaaru      b) Avaru yaaru      c) Neenu yelli      d) Avaru Enu
2. What is her Name?  
a) Avalu yaru      b) Avana hesarenu  
c) Avala hesarenu      d) Ninna hesarenu
3. This is my Nation  
a) Idu Avara desha      b) Idu Nanna desha  
c) Adu Ninna desha      d) Idu Ninna desha
4. What is your name?  
a) Avala Hesaru Enu?      b) Avana Hesaru Enu?  
c) Ninna Hesaru Enu?      d) Nanna Hesaru Enu?
5. This is my pen.  
a) Idu nanna pustaka      b) Idu Avara vastu  
c) Idu Avara Lekhani      d) Idu Nanna Lekhani

**Fill in the blank for the given English words into Kannada meaning :**

6. Small  
a) Udda      b) agala      c) Dappa      d) Chikka

7. Salty  
a) huLi b) Uppu c) Khara d) Sihi
8. Simple  
a) Khatina b) Tappu c) Sarala d) Khasagi
9. Daily  
a) Nale b) Nenne c) Nadiddu d) Dina Nitya
10. Dignity  
a) Samasye b) Ghanathe c) Sari d) Olleya

**Write the Kannada Vocabulary for the following English words :**

11. Child \_\_\_\_\_  
a) navu b) Avaru c) Magu d) vidyarthi
12. Plant \_\_\_\_\_  
a) Mara b) Gida c) Hoova d) Rembe
13. Story \_\_\_\_\_  
a) Kate b) Angadi c) Mane d) Kavana
14. Where \_\_\_\_\_  
a) Enu b) Eke c) yaaru d) Elli
15. Bad \_\_\_\_\_  
a) hosa b) Naataka c) Ketta d) Olleya

**Translate the following Kannada words into English :**

16. Maga  
a) Father b) Sister c) Daughter d) Son
17. Mane  
a) Temple b) House c) Church d) Hospital
18. Kempu BaNNa  
a) Black colour b) Red colour c) White colour d) Green colour

**Translate the following English words into Kannada :**

19. We  
a) Naavu b) Namma c) Namage d) Nimage
20. This  
a) Adu b) Alli c) Idu d) Both (a) and (b)
21. He  
a) Avanu b) Avalu c) Naanu d) Neenu

Match the following using Table given below :

|    |          |      |             |
|----|----------|------|-------------|
| a) | White    | i)   | Hudugi      |
| b) | Girl     | ii)  | Sihi        |
| c) | Daughter | iii) | Granthalaya |
| d) | Sweet    | iv)  | Bili        |
| e) | Library  | v)   | Magalu      |

22. White  
a) = i                      b) = ii                      c) = iii                      d) = iv
23. Girl  
a) = i                      b) = ii                      c) = iii                      d) = v
24. Daughter  
a) = i                      b) = ii                      c) = iv                      d) = v
25. Sweet  
a) = i                      b) = ii                      c) = iii                      d) = iv
26. Library  
a) = i                      b) = ii                      c) = iii                      d) = iv

Write appropriate words for the following :

27. Mother  
a) Huduga                      b) Hudugi                      c) Tayi                      d) Tande
28. Festival  
a) Maduve                      b) Habba                      c) Mane                      d) Raste
29. Bitter  
a) Sihi                      b) Khara                      c) Uppu                      d) Kahi
30. Holiday  
a) Mara                      b) Raje                      c) Nagu                      d) Ooru
31. Town  
a) Halli                      b) Angadi                      c) Nagara                      d) Rajya

Translate the following Kannada words as per the given model [Q.No. 32 and 33]  
Example : Kacheri – Kacheriyalli.

32. Mane  
a) Manadalli                      b) Maneyinda                      c) Maneyalli                      d) Mane
33. Angadi  
a) Angadiyalli                      b) Angadihattira                      c) Angadiyage                      d) Angadiyolage
34. How would you write “This is my Book” in Kannada?  
a) Adu Nanna pustaka                      b) idu nanna pustaka  
c) Idu Avaru pustaka                      d) idu nannadu

35. How would you write “what is this” in Kannada?  
a) Idu elli                      b) Idu yarige                      c) Idu yavudu                      d) Idu Avaradu
36. How would you write Fruit in Kannada  
a) Hoovu                      b) Ele                      c) HaNNu                      d) BaNNa
37. How would you write “why sir” in Kannada?  
a) Elli sir                      b) Yaake Saar                      c) Hogi Saar                      d) Banni Saar
38. What is the meaning of “Student” in Kannada?  
a) Huduga                      b) Vidyarthi                      c) Jana                      d) Shale
39. How would you write “don’t” in Kannada?  
a) Beku                      b) Beke                      c) Beda                      d) Yaake
40. How would you write “Marriage” in Kannada?  
a) Habba                      b) Maduve                      c) Sabe                      d) Prarthane

**Write the English word for the following :**

41. Chikka  
a) Big                      b) Huge                      c) Small                      d) House
42. PaaTa  
a) Story                      b) Lesson                      c) Book                      d) Drink
43. aNNa  
a) Sister                      b) Mother                      c) Father                      d) Brother
44. Ketta  
a) Good                      b) Bad                      c) New                      d) Big
45. Huduga  
a) Son                      b) Boy                      c) Daughter                      d) Sister

**Match the following using the Table given below :**

|    |         |      |        |
|----|---------|------|--------|
| a) | Fruit   | i)   | Vaidya |
| b) | Tree    | ii)  | Desha  |
| c) | Flower  | iii) | Mara   |
| d) | Doctor  | iv)  | Hannu  |
| e) | Country | v)   | Hoovu  |

46. Fruit  
a) = i                      b) = ii                      c) = iii                      d) = iv
47. Tree  
a) = i                      b) = ii                      c) = iii                      d) = iv

48. Flower  
a) = i                      b) = iii                      c) = iv                      d) = v
49. Doctor  
a) = i                      b) = ii                      c) = iii                      d) = v
50. Country  
a) = i                      b) = ii                      c) = iv                      d) = v

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Question Paper Version : C

**First/Second Semester B.E./B.Tech Degree Examination,  
Dec.2024/Jan.2025  
Balake Kannada**

Time: 1 hr.]

[Max. Marks: 50

## INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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**Write the English word for the following :**

1. Chikka  
a) Big                      b) Huge                      c) Small                      d) House
2. PaaTa  
a) Story                      b) Lesson                      c) Book                      d) Drink
3. aNNa  
a) Sister                      b) Mother                      c) Father                      d) Brother
4. Ketta  
a) Good                      b) Bad                      c) New                      d) Big
5. Huduga  
a) Son                      b) Boy                      c) Daughter                      d) Sister

Match the following using the Table given below :

|    |         |      |        |
|----|---------|------|--------|
| a) | Fruit   | i)   | Vaidya |
| b) | Tree    | ii)  | Desha  |
| c) | Flower  | iii) | Mara   |
| d) | Doctor  | iv)  | Hannu  |
| e) | Country | v)   | Hoovu  |

6. Fruit  
a) = i                      b) = ii                      c) = iii                      d) = iv
7. Tree  
a) = i                      b) = ii                      c) = iii                      d) = iv
8. Flower  
a) = i                      b) = iii                      c) = iv                      d) = v
9. Doctor  
a) = i                      b) = ii                      c) = iii                      d) = v
10. Country  
a) = i                      b) = ii                      c) = iv                      d) = v

Translate the following English sentence into Kannada sentence :

11. Who are they?  
a) Neenu yaaru                      b) Avaru yaaru                      c) Neenu yelli                      d) Avaru Enu
12. What is her Name?  
a) Avalu yaru                      b) Avana hesarenu  
c) Avala hesarenu                      d) Ninna hesarenu
13. This is my Nation  
a) Idu Avara desha                      b) Idu Nanna desha  
c) Adu Ninna desha                      d) Idu Ninna desha
14. What is your name?  
a) Avala Hesaru Enu?                      b) Avana Hesaru Enu?  
c) Ninna Hesaru Enu?                      d) Nanna Hesaru Enu?
15. This is my pen.  
a) Idu nanna pustaka                      b) Idu Avara vastu  
c) Idu Avara Lekhani                      d) Idu Nanna Lekhani

Fill in the blank for the given English words into Kannada meaning :

16. Small  
a) Udda                      b) agala                      c) Dappa                      d) Chikka
17. Salty  
a) huLi                      b) Uppu                      c) Khara                      d) Sihi

18. Simple  
a) Khatina b) Tappu c) Sarala d) Khasagi
19. Daily  
a) Nale b) Nenne c) Nadiddu d) Dina Nitya
20. Dignity  
a) Samasye b) Ghanathe c) Sari d) Olleya

**Write the Kannada Vocabulary for the following English words :**

21. Child \_\_\_\_\_  
a) navu b) Avaru c) Magu d) vidyarthi
22. Plant \_\_\_\_\_  
a) Mara b) Gida c) Hoova d) Rembe
23. Story \_\_\_\_\_  
a) Kate b) Angadi c) Mane d) Kavana
24. Where \_\_\_\_\_  
a) Enu b) Eke c) yaaru d) Elli
25. Bad \_\_\_\_\_  
a) hosa b) Naataka c) Ketta d) Olleya

**Translate the following Kannada words into English :**

26. Maga  
a) Father b) Sister c) Daughter d) Son
27. Mane  
a) Temple b) House c) Church d) Hospital
28. Kempu BaNNa  
a) Black colour b) Red colour c) White colour d) Green colour

**Translate the following English words into Kannada :**

29. We  
a) Naavu b) Namma c) Namage d) Nimage
30. This  
a) Adu b) Alli c) Idu d) Both (a) and (b)
31. He  
a) Avanu b) Avalu c) Naanu d) Neenu

Match the following using Table given below :

|    |          |      |             |
|----|----------|------|-------------|
| a) | White    | i)   | Hudugi      |
| b) | Girl     | ii)  | Sihi        |
| c) | Daughter | iii) | Granthalaya |
| d) | Sweet    | iv)  | Bili        |
| e) | Library  | v)   | Magalu      |

32. White  
a) = i                      b) = ii                      c) = iii                      d) = iv
33. Girl  
a) = i                      b) = ii                      c) = iii                      d) = v
34. Daughter  
a) = i                      b) = ii                      c) = iv                      d) = v
35. Sweet  
a) = i                      b) = ii                      c) = iii                      d) = iv
36. Library  
a) = i                      b) = ii                      c) = iii                      d) = iv

Write appropriate words for the following :

37. Mother  
a) Huduga                      b) Hudugi                      c) Tayi                      d) Tande
38. Festival  
a) Maduve                      b) Habba                      c) Mane                      d) Raste
39. Bitter  
a) Sihi                      b) Khara                      c) Uppu                      d) Kahi
40. Holiday  
a) Mara                      b) Raje                      c) Nagu                      d) Ooru
41. Town  
a) Halli                      b) Angadi                      c) Nagara                      d) Rajya

Translate the following Kannada words as per the given model [Q.No. 42 and 43]  
Example : Kacheri – Kacheriyalli.

42. Mane  
a) Manadalli                      b) Maneyinda                      c) Maneyalli                      d) Mane
43. Angadi  
a) Angadiyalli                      b) Angadihattira                      c) Angadiyage                      d) Angadiyolage

44. How would you write "This is my Book" in Kannada?  
a) Adu Nanna pustaka                      b) idu nanna pustaka  
c) Idu Avaru pustaka                      d) idu nannadu
45. How would you write "what is this" in Kannada?  
a) Idu elli                      b) Idu yarige                      c) Idu yavudu                      d) Idu Avaradu
46. How would you write Fruit in Kannada  
a) Hoovu                      b) Ele                      c) HaNNu                      d) BaNNa
47. How would you write "why sir" in Kannada?  
a) Elli sir                      b) Yaake Saar                      c) Hogi Saar                      d) Banni Saar
48. What is the meaning of "Student" in Kannada?  
a) Huduga                      b) Vidyarthi                      c) Jana                      d) Shale
49. How would you write "don't" in Kannada?  
a) Beku                      b) Beke                      c) Beda                      d) Yaake
50. How would you write "Marriage" in Kannada?  
a) Habba                      b) Maduve                      c) Sabe                      d) Prarthane

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**Question Paper Version : D**

**First/Second Semester B.E./B.Tech Degree Examination,  
Dec.2024/Jan.2025  
Balake Kannada**

Time: 1 hr.]

[Max. Marks: 50

## INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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**Translate the following Kannada words as per the given model [Q.No. 1 and 2]  
Example : Kacheri – Kacheriyalli.**

1. Mane  
a) Manadalli      b) Maneyinda      c) Maneyalli      d) Mane
2. Angadi  
a) Angadiyalli      b) Angadihattira      c) Angadiyage      d) Angadiyolage
3. How would you write “This is my Book” in Kannada?  
a) Adu Nanna pustaka      b) idu nanna pustaka  
c) Idu Avaru pustaka      d) idu nannadu
4. How would you write “what is this” in Kannada?  
a) Idu elli      b) Idu yarige      c) Idu yavudu      d) Idu Avaradu
5. How would you write Fruit in Kannada  
a) Hoovu      b) Ele      c) HaNNu      d) BaNNa
6. How would you write “why sir” in Kannada?  
a) Elli sir      b) Yaake Saar      c) Hogi Saar      d) Banni Saar
7. What is the meaning of “Student” in Kannada?  
a) Huduga      b) Vidyarthi      c) Jana      d) Shale

8. How would you write “don’t” in Kannada?  
 a) Beku                      b) Beke                      c) Beda                      d) Yaake
9. How would you write “Marriage” in Kannada?  
 a) Habba                      b) Maduve                      c) Sabe                      d) Prarthane

**Write the English word for the following :**

10. Chikka  
 a) Big                      b) Huge                      c) Small                      d) House
11. PaaTa  
 a) Story                      b) Lesson                      c) Book                      d) Drink
12. aNNa  
 a) Sister                      b) Mother                      c) Father                      d) Brother
13. Ketta  
 a) Good                      b) Bad                      c) New                      d) Big
14. Huduga  
 a) Son                      b) Boy                      c) Daughter                      d) Sister

**Match the following using the Table given below :**

|    |         |      |        |
|----|---------|------|--------|
| a) | Fruit   | i)   | Vaidya |
| b) | Tree    | ii)  | Desha  |
| c) | Flower  | iii) | Mara   |
| d) | Doctor  | iv)  | Hannu  |
| e) | Country | v)   | Hoovu  |

15. Fruit  
 a) = i                      b) = ii                      c) = iii                      d) = iv
16. Tree  
 a) = i                      b) = ii                      c) = iii                      d) = iv
17. Flower  
 a) = i                      b) = iii                      c) = iv                      d) = v
18. Doctor  
 a) = i                      b) = ii                      c) = iii                      d) = v
19. Country  
 a) = i                      b) = ii                      c) = iv                      d) = v

**Translate the following English sentence into Kannada sentence :**

20. Who are they?  
 a) Neenu yaaru                      b) Avaru yaaru                      c) Neenu yelli                      d) Avaru Enu

21. What is her Name?

- a) Avalu yaru  
c) Avala hesarenu

- b) Avana hesarenu  
d) Ninna hesarenu

22. This is my Nation

- a) Idu Avara desha  
c) Adu Ninna desha

- b) Idu Nanna desha  
d) Idu Ninna desha

23. What is your name?

- a) Avala Hesaru Enu?  
c) Ninna Hesaru Enu?

- b) Avana Hesaru Enu?  
d) Nanna Hesaru Enu?

24. This is my pen.

- a) Idu nanna pustaka  
c) Idu Avara Lekhani

- b) Idu Avara vastu  
d) Idu Nanna Lekhani

**Fill in the blank for the given English words into Kannada meaning :**

25. Small

- a) Udda b) agala c) Dappa d) Chikka

26. Salty

- a) huLi b) Uppu c) Khara d) Sihi

27. Simple

- a) Khatina b) Tappu c) Sarala d) Khasagi

28. Daily

- a) Nale b) Nenne c) Nadiddu d) Dina Nitya

29. Dignity

- a) Samasye b) Ghanathe c) Sari d) Olleya

**Write the Kannada Vocabulary for the following English words :**

30. Child \_\_\_\_\_

- a) navu b) Avaru c) Magu d) vidyarthi

31. Plant \_\_\_\_\_

- a) Mara b) Gida c) Hoova d) Rembe

32. Story \_\_\_\_\_

- a) Kate b) Angadi c) Mane d) Kavana

33. Where \_\_\_\_\_

- a) Enu b) Eke c) yaaru d) Elli

34. Bad \_\_\_\_\_

- a) hosa b) Naataka c) Ketta d) Olleya

**Translate the following Kannada words into English :**

35. Maga  
a) Father                      b) Sister                      c) Daughter                      d) Son
36. Mane  
a) Temple                      b) House                      c) Church                      d) Hospital
37. Kempu BaNNa  
a) Black colour                      b) Red colour                      c) White colour                      d) Green colour

**Translate the following English words into Kannada :**

38. We  
a) Naavu                      b) Namma                      c) Namage                      d) Nimage
39. This  
a) Adu                      b) Alli                      c) Idu                      d) Both (a) and (b)
40. He  
a) Avanu                      b) Avalu                      c) Naanu                      d) Neenu

**Match the following using Table given below :**

|    |          |      |             |
|----|----------|------|-------------|
| a) | White    | i)   | Hudugi      |
| b) | Girl     | ii)  | Sihi        |
| c) | Daughter | iii) | Granthalaya |
| d) | Sweet    | iv)  | Bili        |
| e) | Library  | v)   | Magalu      |

41. White  
a) = i                      b) = ii                      c) = iii                      d) = iv
42. Girl  
a) = i                      b) = ii                      c) = iii                      d) = v
43. Daughter  
a) = i                      b) = ii                      c) = iv                      d) = v
44. Sweet  
a) = i                      b) = ii                      c) = iii                      d) = iv
45. Library  
a) = i                      b) = ii                      c) = iii                      d) = iv

**Write appropriate words for the following :**

46. Mother  
a) Huduga                      b) Hudugi                      c) Tayi                      d) Tande

47. Festival  
a) Maduve b) Habba c) Mane d) Raste
48. Bitter  
a) Sihi b) Khara c) Uppu d) Kahi
49. Holiday  
a) Mara b) Raje c) Nagu d) Ooru
50. Town  
a) Halli b) Angadi c) Nagara d) Rajya

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First/Second Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025

## ಸಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

## ಸೂಚನೆಗಳು

1. ಎಲ್ಲ ಜಿಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ಜೇಡರ ದಾಸಿಮಯ್ಯನವರ ಅಂಕಿತನಾಮ  
a) ಚನ್ನ b) ಶಿವಪ್ಪ c) ಗುಹೇಶ್ವರ d) ರಾಮನಾಥ.
2. ಆಲಿಸುವ ಅಥವಾ ಗ್ರಹಿಸುವ ಕೌಶಲ ಈ ವಾಕ್ಯದಲ್ಲಿರುವ ಷಷ್ಟಿ ವಿಭಕ್ತಿ ಯಾವುದು?  
a) ಆ b) ಎಲ್ಲಿ c) ಗೆ d) ದೆಸೆಯಿಂದ
3. ವಿನೋಬಾ ಭಾವೆಯವರು ಕನ್ನಡ ಲಿಪಿಯನ್ನು ಏನೆಂದು ಕರೆದಿದ್ದಾರೆ?  
a) ರಾಜರಾಣಿ b) ಲಿಪಿಗಳ ರಾಣಿ  
c) ಕನ್ನಡಭಾಷೆ d) ಕುಸುಮರಾಣಿ
4. ಥಾಮಸ್ ಮನ್ರೋವರ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿ ಕಲೆಕ್ಟರ್ ಆಗಿ ಕಾರ್ಯ ನಿರ್ವಹಿಸಿದ್ದರು?  
a) ಬಾಗಲಕೋಟೆ b) ಬೆಳಗಾವಿ c) ಬೀದರ್ d) ಬಳ್ಳಾರಿ.
5. ಕಳೆದ ಪದದ ಸಮನಾರ್ಥಕ ಪದ  
a) ರಾಗಿ b) ಜೋಳ c) ಭತ್ತ d) ಗೋಧಿ

6. ಹಂಪ ನಾಗರಾಜಯ್ಯನವರ ಹುಟ್ಟಿದ ಸ್ಥಳ  
a) ಹಂಪ ಸಂದ್ರ b) ಮಂಡ್ಯ c) ಬಳ್ಳಾರಿ d) ಮೈಸೂರು.
7. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಕೃತಿಯನ್ನು ರಚಿಸಿದ ಕವಿ ಯಾರು?  
a) ಪಂಪ b) ರನ್ನ c) ಜನ್ನ d) ಅಂಡಯ್ಯ
8. ತಂತ್ರಜ್ಞಾನದ ಬೆಳವಣಿಗೆಯ ಫಲವಾಗಿ ಕನ್ನಡ ಭಾಷೆಗೆ ಸಾಕಷ್ಟು \_\_\_\_\_ ಪದಗಳು ಸೇರಿಕೊಂಡಿವೆ.  
a) ತೆಲಗು b) ತಮಿಳು c) ಇಂಗ್ಲೀಷ್ d) ಮಲಯಾಳಿ
9. ಶರಣರ ಚಳುವಳಿಗೆ ಪ್ರೇರಕ ಶಕ್ತಿಯಾಗಿ ನಿಂತ ಮಹಾ ಮಾನವತಾವಾದಿ.  
a) ಅಕ್ಕಮಹಾದೇವಿ b) ಆಯ್ದಕ್ಕಿಮಾರಯ್ಯ  
c) ಬಸವಣ್ಣ d) ಆಯ್ದಕ್ಕಿ ಲಕ್ಕಮ್ಮ
10. ಭಕ್ತಿ ಪದದ ತದ್ಭವ ರೂಪ \_\_\_\_\_  
a) ಪ್ರಕೃತಿ b) ವಿಕೃತಿ c) ಸಂಸ್ಕೃತಿ d) ಭಕುತಿ
11. ವಿಷ್ಣುವ ಪದದ ಅರ್ಥ.  
a) ಕ್ರಾಂತಿ b) ಸ್ವರ್ಗ c) ಕವಿತೆ d) ಕಾರ್ಯ
12. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ 'ಆಚಾರ' ಎಂದರೆ  
a) ಅವಿಗೆ ಅಥವಾ ಮುಚ್ಚಳ b) ಮಣ್ಣು  
c) ನೀರು d) ಮನ.
13. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಕಬ್ಬಿಣ ಮತ್ತು ಉಕ್ಕಿನ ಕಾರ್ಖಾನೆ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿದೆ?  
a) ಮಂಡ್ಯ b) ಮಂಗಳೂರು  
c) ಮೈಸೂರು d) ಶಿವಮೊಗ್ಗ
14. ಗಾಂಧೀಜಿಯವರು ಯಾರನ್ನು "ಅನ್ನಬ್ರಹ್ಮ" ಎಂದು ಕರೆದಿದ್ದಾರೆ?  
a) ಕುವೆಂಪು  
b) ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯ  
c) ಡಿ.ವಿ.ಜಿ  
d) ಅಕ್ಕಿಹೆಬ್ಬಾಳು ನರಸಿಂಹಮೂರ್ತಿ
15. ಯುಗಾದಿ ಕಥೆಯ ಲೇಖಕರು ಯಾರು?  
a) ಡಿ.ವಿ.ಜಿ b) ವಸುಧೇಂದ್ರ c) ಅ.ನ.ಕೃ d) ವಿ.ಕೃ.ಗೋಕಾಕ್

16. ಕಾಸಿಂಸಾಬರ ಮಗಳ ಹೆಸರೇನು?  
 a) ಚಾಂದಿನಿ                      b) ಹಂಸ                      c) ರೇಖಾ                      d) ರೂಪ.
17. ಹಸಿವಾದರೆ ಉರೂಳಿಗೆ ಭಿಕ್ಷಾನ್ನಗಳುಂಟು, ತೃಷೆಯಾದರೆ ಕೆರೆ ಬಾವಿ ಹಳ್ಳಗಳುಂಟು, ಶಯನಕ್ಕೆ ಹಾಳುದೇಗುಲಗಳುಂಟು, ಈ ವಚನವನ್ನು ರಚಿಸಿದವರು.  
 a) ಅಕ್ಕಮಾಹಾದೇವಿ  
 b) ಬಸವಣ್ಣ  
 c) ರಾಮನಾಥ  
 d) ಅಲ್ಲಮ ಪ್ರಭು
18. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘವನ್ನು ಯಾವ ಕಾರಣಕ್ಕಾಗಿ ಸ್ಥಾಪಿಸಲಾಯಿತು.  
 a) ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿಗಾಗಿ  
 b) ದೇಶಕ್ಕಾಗಿ  
 c) ಭಾಷೆಗಾಗಿ  
 d) ಕರ್ನಾಟಕ ಏಕೀಕರಣಕ್ಕಾಗಿ
19. ತಲ್ಲಣಿಸದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆಯ ಕೀರ್ತನಕಾರರು ಯಾರು?  
 a) ಪುರಂದರದಾಸರು  
 b) ಶ್ರೀರಂಗರು  
 c) ಕನಕದಾಸರು  
 d) ವೆಂಕಟಸುಬ್ಬಯ್ಯ
20. ವೀರರ ಜೀವನ ಸ್ಮಾರಕಗಳನ್ನು ಏನೆಂದು ಕರೆಯುವರು?  
 a) ಸಿಡಿತಿಲೆ                      b) ವೀರಗಲ್ಲು                      c) ಮಾಸ್ತಿಗಲ್ಲು                      d) ಬರಹದಕಲ್ಲು
21. ಹಂಪಿನಾಗರಾಜಯ್ಯನವರು ಹುಟ್ಟಿದ ವರ್ಷ.  
 a) 1936                      b) 1946                      c) 1947                      d) 1935
22. ಕನ್ನಡ ಬರಹಗಳಿಗೆ \_\_\_\_\_ ವರ್ಷಗಳ ಇತಿಹಾಸವಿದೆ.  
 a) 1956                      b) 1884                      c) 1956                      d) 1500
23. ಹುಲ್ಲಾಗು ಬೆಟ್ಟದಡಿ ಮನೆಗೆ \_\_\_\_\_ ಆಗು.  
 a) ಸಂಪಿಗೆ                      b) ಮಲ್ಲಿಗೆ                      c) ಇಂಪಿಗೆ                      d) ತಂಪಿಗೆ
24. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ ಕುಂಬಾರನಿಗೆ ಚಿನ್ನ ಎಂದರೆ.  
 a) ಬಂಗಾರ                      b) ಮಣ್ಣು                      c) ತೋಟ                      d) ಕಾರ್ಯ

25. ಪಥ ಪದದ ಅರ್ಥ.
- a) ರಕ್ತ                      b) ಸ್ವರ್ಗ ಜೀವನ                      c) ದಾರಿ                      d) ಜ್ಞಾನಿ
26. ಜನಪದ ಅಕಾಡೆಮಿಯ ಅಧ್ಯಕ್ಷರು \_\_\_\_\_
- a) ಹುಚ್ಚಪ್ಪ ಮಾಸ್ತರ  
b) ಶ್ರೀ,ಎಸ್,ಕೆ.ಕರೀಂಖಾನ್  
c) ಹಿ.ಚಿ.ಬೋರಲಿಂಗಯ್ಯ  
d) ಶ್ರೀನಿವಾಸಮೂರ್ತಿ
27. ಕವಿರಾಜಮಾರ್ಗದಲ್ಲಿ \_\_\_\_\_ ಅವರನ್ನು ಕುರಿತು ವರ್ಣಿಸಲಾಗಿದೆ?
- a) ತಮಿಳರು                      b) ಕೇರಳಿಗರು                      c) ಕನ್ನಡಿಗರು                      d) ಮರಾಠಿಗರು
28. ಕನ್ನಡ ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳಿವೆ?
- a) 10                      b) 20                      c) 4                      d) 8.
29. ವಚನ ಎಂದರೆ \_\_\_\_\_.
- a) ಮಾತು                      b) ಸಮಾಜ                      c) ಭಾಷೆ                      d) ದೇಶ
30. ಸುಂಕ ಪದದ ಅರ್ಥ.
- a) ಕಾಸು                      b) ಆಣಿ                      c) ತೆರಿಗೆ                      d) ನೋವು.
31. ಕಳ್ಳಗಂಜಿ \_\_\_\_\_ ಹೊಕ್ಕಡೆ ಹುಲಿ ತಿಂಬುದ ಮಾಬುದೇ?
- a) ಮನೆ                      b) ಕಾಡು                      c) ಸ್ಮಶಾನ                      d) ಹುತ್ತ.
32. ಕರಿ ಘನ \_\_\_\_\_ ಕಿರಿದೆನ್ನಬಹುದೆ ಬಾರದಯ್ಯ.
- a) ಜ್ಯೋತಿ                      b) ನೆನಪು                      c) ಮರೆವು                      d) ಅಂಕುಶ.
33. ತೊರೆಯೊಳು \_\_\_\_\_ ದುರಿತ ಪೋಗುವುದೆ?
- a) ಮಿಂದರೆ                      b) ಹಾಡಿದರೆ                      c) ಕುಣಿದರೆ                      d) ಮರೆತರೆ.
34. ಪವಳದ ಲತೆಗೆ \_\_\_\_\_ ಇಟ್ಟವರು ಯಾರು ?
- a) ಕಪ್ಪು                      b) ಹಸಿರು                      c) ಚಿತ್ರ                      d) ಕೆಂಪು.
35. ಬ್ರಹ್ಮಾಡವನ್ನೆಲ್ಲ ಯಾರು ತುಂಬಿಕೊಂಡಿದ್ದಾರೆಂದು ಶರೀಫರು ಹೇಳಿದ್ದಾರೆ?
- a) ಕುಂಬಾರ                      b) ಬಡಿಗ                      c) ಕುಂಬಾರಕಿ                      d) ಗಾಣತಿ

36. ಎಷ್ಟು ಕಾಸಿಗೆ ಕುಂಬಾರಕಿ ಒಂದು ಕುಡಕಿಯನ್ನು ಮಾರುವಳು?  
a) 3 b) 6 c) 9 d) 10
37. ನಗುವು ಸಹಜದ ಧರ್ಮ ನಗಿಸುವುದು \_\_\_\_  
a) ಪರಧರ್ಮ b) ಅಧರ್ಮ c) ಧರ್ಮ d) ಮನುಷ್ಯ ಧರ್ಮ
38. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಅವರ ತವರೂರು ಯಾವುದು?  
a) ರಾಣೆಬೆನ್ನೂರು b) ಮಂಡ್ಯ c) ಮೈಸೂರು d) ಮುದ್ದೇನಹಳ್ಳಿ
39. ಗ್ರಂಥಕರ್ತರ ಸಮ್ಮೇಳನವು \_\_\_\_ ರಲ್ಲಿ ನಡೆಯಿತು.  
a) 1977 b) 1907 c) 1970 d) 1974.
40. ಭಾರತ ಸರ್ಕಾರವು ಕನ್ನಡ ಭಾಷೆಯನ್ನು \_\_\_\_ ಎಂದು ಘೋಷಿಸಿದೆ.  
a) ಶಾಸ್ತ್ರೀಯ ಭಾಷೆ  
b) ಕನ್ನಡಭಾಷೆ  
c) ತ್ರಿಭಾಷೆ  
d) ಲಿಪಿ.
41. ವಸುಧೇಂದ್ರರವರು ಬರೆದ ಕಾದಂಬರಿ ಇದಾಗಿದೆ.  
a) ಮನೀಷೆ b) ಕೃತಿ c) ಕರ್ತೃ d) ಸಂಕಲನ
42. ಪ್ರಹ್ಲಾದ \_\_\_\_ ಇಂಜಿನಿಯರ್.  
a) ಸಾಫ್ಟ್‌ವೇರ್ b) ಇಂಜಿನಿಯರ್ c) ಸಿವಿಲ್ ಇಂಜಿನಿಯರ್ d) ಮೆಕಾನಿಕ್
43. ವಿಜಯನಗರ ಸಾಮ್ರಾಜ್ಯದ ರಾಜಧಾನಿ  
a) ಹಂಪೆ b) ಬೆಲೂರು c) ಹಳೇಬೀಡು d) ಕೋಲಾರ
44. ನಮ್ಮ ರಾಜ್ಯದಲ್ಲಿ \_\_\_\_ ಸೂತ್ರ ಬಳಕೆಯಲ್ಲಿದೆ.  
a) ಏಕಭಾಷಾ b) ದ್ವಿಭಾಷಾ c) ತ್ರಿಭಾಷಾ d) ಭಾಷಾ
45. ಧರ್ಮಸಹಿಷ್ಣುತೆ ಬಗ್ಗೆ ಹೇಳುವ ಶಾಸನ \_\_\_\_  
a) ಹಳೇಬೀಡು b) ಬೆಲೂರು c) ಹಂಪಿ d) ಹಾಸನ
46. ಬುಕ್ಕರಾಯನ ಕಾಲ ಸರಿಸುಮಾರು \_\_\_\_ ವರ್ಷ.  
a) 1368 b) 1468 c) 1568 d) 1668.

47. ಕುರುಡು ಕಾಂಚಾಣ ಪದ್ಯದ ಕವಿ.  
 a) ಕುವೆಂಪು  
 b) ದ.ರಾ.ಬೇಂದ್ರೆ  
 c) ಡಿ.ವಿ.ಜಿ  
 d) ಗೋಕಾಕ್
48. ಯುಗಾದಿ ಕಥೆಯಲ್ಲಿ ಯಾರನ್ನು 'ಖಡಕ್' ಮಾಸ್ತರ ಎಂದು ಕರೆಯಲಾಗಿದೆ.  
 a) ಗೋಪಣ್ಣ  
 b) ಪ್ರಹ್ಲಾದ  
 c) ಇಸ್ಮಾಯಿಲ್  
 d) ಕಾಸಿಂಸಾಬ್
49. ಯದ್ಧದಲ್ಲಿ ಗೆದ್ದು ಬಂದರೆ ಕೀರ್ತಿ, ಸತ್ತರೆ \_\_\_\_\_  
 a) ನರಕ  
 b) ಹಣ  
 c) ಪ್ರಶಸ್ತಿ  
 d) ಸ್ವರ್ಗ
50. ಹಲವು ಕಾಲ ಕಲ್ಲು ನೀರೊಳಗಿದ್ದರೇನು? ಬಲುನೆನೆದು \_\_\_\_\_ ಆಗುವುದೆ.  
 a) ಅಮೃತಶಿಲೆ  
 b) ಹವಳ  
 c) ಕಪ್ಪೆ  
 d) ನಿಧಿ.

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First/Second Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025

## ಸಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

## ಸೂಚನೆಗಳು

1. ಎಲ್ಲಾ ಜಿಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ವಸುಧೇಂದ್ರರವರು ಬರೆದ ಕಾದಂಬರಿ ಇದಾಗಿದೆ.  
a) ಮನೀಷೆ b) ಕೃತಿ c) ಕರ್ತೃ d) ಸಂಕಲನ
2. ಪ್ರಹ್ಲಾದ \_\_\_\_\_ ಇಂಜಿನಿಯರ್.  
a) ಸಾಫ್ಟ್‌ವೇರ್ b) ಇಂಜಿನಿಯರ್ c) ಸಿವಿಲ್ ಇಂಜಿನಿಯರ್ d) ಮೆಕಾನಿಕ್
3. ವಿಜಯನಗರ ಸಾಮ್ರಾಜ್ಯದ ರಾಜಧಾನಿ  
a) ಹಂಪಿ b) ಬೆಲೂರು c) ಹಳೇಬೀಡು d) ಕೋಲಾರ
4. ನಮ್ಮ ರಾಜ್ಯದಲ್ಲಿ \_\_\_\_\_ ಸೂತ್ರ ಬಳಕೆಯಲ್ಲಿದೆ.  
a) ಏಕಭಾಷಾ b) ದ್ವಿಭಾಷಾ c) ತ್ರಿಭಾಷಾ d) ಭಾಷಾ
5. ಧರ್ಮಸಹಿಷ್ಣುತೆ ಬಗ್ಗೆ ಹೇಳುವ ಶಾಸನ \_\_\_\_\_  
a) ಹಳೇಬೀಡು b) ಬೆಲೂರು c) ಹಂಪಿ d) ಹಾಸನ

6. ಬುಕ್ಕರಾಯನ ಕಾಲ ಸರಿಸುಮಾರು \_\_\_\_ ವರ್ಷ.
- a) 1368                      b) 1468                      c) 1568                      d) 1668.
7. ಕುರುಡು ಕಾಂಚಾಣ ಪದ್ಯದ ಕವಿ.
- a) ಕುವೆಂಪು                      b) ದ.ರಾ.ಬೇಂದ್ರೆ  
c) ಡಿ.ವಿ.ಜಿ                      d) ಗೋಕಾಕ್
8. ಯುಗಾದಿ ಕಥೆಯಲ್ಲಿ ಯಾರನ್ನು 'ಖಡಕ್' ಮಾಸ್ತರ ಎಂದು ಕರೆಯಲಾಗಿದೆ.
- a) ಗೋಪಣ್ಣ                      b) ಪ್ರಹ್ಲಾದ  
c) ಇಸ್ಮಾಯಿಲ್                      d) ಕಾಸಿಂಸಾಬ್
9. ಯದ್ಧದಲ್ಲಿ ಗೆದ್ದು ಬಂದರೆ ಕೀರ್ತಿ, ಸತ್ತರೆ \_\_\_\_
- a) ನರಕ                      b) ಹಣ                      c) ಪ್ರಶಸ್ತಿ                      d) ಸ್ವರ್ಗ
10. ಹಲವು ಕಾಲ ಕಲ್ಲು ನೀರೊಳಗಿದ್ದರೇನು? ಬಲುನೆನೆದು \_\_\_\_ ಆಗುವುದೆ.
- a) ಅಮೃತಶಿಲೆ                      b) ಹವಳ                      c) ಕಪ್ಪೆ                      d) ನಿಧಿ.
11. ಕಳ್ಳಗಂಜಿ \_\_\_\_ ಹೊಕ್ಕಡೆ ಹುಲಿ ತಿಂಬುದ ಮಾಬುದೇ?
- a) ಮನೆ                      b) ಕಾಡು                      c) ಸ್ಮಶಾನ                      d) ಹುತ್ತ.
12. ಕರಿ ಘನ \_\_\_\_ ಕಿರಿದೆನ್ನಬಹುದೆ ಬಾರದಯ್ಯ.
- a) ಜ್ಯೋತಿ                      b) ನೆನಪು                      c) ಮರೆಪು                      d) ಅಂಕುಶ.
13. ತೊರೆಯೊಳು \_\_\_\_ ದುರಿತ ಪೋಗುವುದೆ?
- a) ಮಿಂದರೆ                      b) ಹಾಡಿದರೆ                      c) ಕುಣಿದರೆ                      d) ಮರೆತರೆ.
14. ಪವಳದ ಲತೆಗೆ \_\_\_\_ ಇಟ್ಟವರು ಯಾರು ?
- a) ಕಪ್ಪು                      b) ಹಸಿರು                      c) ಚಿತ್ರ                      d) ಕೆಂಪು.
15. ಬ್ರಹ್ಮಾಡವನ್ನೆಲ್ಲ ಯಾರು ತುಂಬಿಕೊಂಡಿದ್ದಾರೆಂದು ಶರೀಫರು ಹೇಳಿದ್ದಾರೆ?
- a) ಕುಂಬಾರ                      b) ಬಡಗಿ                      c) ಕುಂಬಾರಕಿ                      d) ಗಾಣತಿ
16. ಎಷ್ಟು ಕಾಸಿಗೆ ಕುಂಬಾರಕಿ ಒಂದು ಕುಡಕಿಯನ್ನು ಮಾರುವಳು?
- a) 3                      b) 6                      c) 9                      d) 10
17. ನಗುವು ಸಹಜದ ಧರ್ಮ ನಗಿಸುವುದು \_\_\_\_
- a) ಪರಧರ್ಮ                      b) ಅಧರ್ಮ                      c) ಧರ್ಮ                      d) ಮನುಷ್ಯ ಧರ್ಮ

18. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಅವರ ತವರೂರು ಯಾವುದು?  
 a) ರಾಣಿಬೆನ್ನೂರು      b) ಮಂಡ್ಯ      c) ಮೈಸೂರು      d) ಮುದ್ದೇನಹಳ್ಳಿ
19. ಗ್ರಂಥಕರ್ತರ ಸಮ್ಮೇಳನವು \_\_\_\_ ರಲ್ಲಿ ನಡೆಯಿತು.  
 a) 1977      b) 1907      c) 1970      d) 1974.
20. ಭಾರತ ಸರ್ಕಾರವು ಕನ್ನಡ ಭಾಷೆಯನ್ನು \_\_\_\_\_ ಎಂದು ಘೋಷಿಸಿದೆ.  
 a) ಶಾಸ್ತ್ರೀಯ ಭಾಷೆ  
 b) ಕನ್ನಡಭಾಷೆ  
 c) ತ್ರಿಭಾಷೆ  
 d) ಲಿಪಿ.
21. ವಿಫಲವಾದ ಪದದ ಅರ್ಥ.  
 a) ಕ್ರಾಂತಿ      b) ಸ್ವರ್ಗ      c) ಕವಿತೆ      d) ಕಾರ್ಯ
22. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ 'ಆಚಾರ' ಎಂದರೆ  
 a) ಆದಿಗೇ ಅಥವಾ ಮುಚ್ಚಳ      b) ಮಣ್ಣು  
 c) ನೀರು      d) ಮನ.
23. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಕಬ್ಬಿಣ ಮತ್ತು ಉಕ್ಕಿನ ಕಾರ್ಖಾನೆ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿದೆ?  
 a) ಮಂಡ್ಯ      b) ಮಂಗಳೂರು  
 c) ಮೈಸೂರು      d) ಶಿವಮೊಗ್ಗ
24. ಗಾಂಧೀಜಿಯವರು ಯಾರನ್ನು "ಅನ್ವಬ್ರಹ್ಮ" ಎಂದು ಕರೆದಿದ್ದಾರೆ?  
 a) ಕುವೆಂಪು  
 b) ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯ  
 c) ಡಿ.ವಿ.ಜಿ  
 d) ಅಕ್ಕಿಹೆಬ್ಬಾಳು ನರಸಿಂಹಮೂರ್ತಿ
25. ಯುಗಾದಿ ಕಥೆಯ ಲೇಖಕರು ಯಾರು?  
 a) ಡಿ.ವಿ.ಜಿ      b) ವಸುಧೇಂದ್ರ      c) ಅ.ನ.ಕೃ      d) ವಿ.ಕೃ.ಗೋಕಾಕ್
26. ಕಾಸಿಂಸಾಬರ ಮಗಳ ಹೆಸರೇನು?  
 a) ಚಾಂದಿನಿ      b) ಹಂಸ      c) ರೇಖಾ      d) ರೂಪ.

27. ಹಸಿವಾದರೆ ಉರೂಗಳಿಗೆ ಭಿಕ್ಷಾನ್ನಗಳುಂಟು, ತೃಷೆಯಾದರೆ ಕೆರೆ ಬಾವಿ ಹಳ್ಳಗಳುಂಟು, ಶಯನಕ್ಕೆ ಹಾಳುದೇಗುಲಗಳುಂಟು, ಈ ವಚನವನ್ನು ರಚಿಸಿದವರು.
- ಅಕ್ಕಮಾಹಾದೇವಿ
  - ಬಸವಣ್ಣ
  - ರಾಮನಾಥ
  - ಅಲ್ಲಮ ಪ್ರಭು
28. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘವನ್ನು ಯಾವ ಕಾರಣಕ್ಕಾಗಿ ಸ್ಥಾಪಿಸಲಾಯಿತು.
- ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿಗಾಗಿ
  - ದೇಶಕ್ಕಾಗಿ
  - ಭಾಷೆಗಾಗಿ
  - ಕರ್ನಾಟಕ ಏಕೀಕರಣಕ್ಕಾಗಿ
29. ತಲ್ಲಣಿಸದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆಯ ಕೀರ್ತನಕಾರರು ಯಾರು?
- ಪುರಂದರದಾಸರು
  - ಶ್ರೀರಂಗರು
  - ಕನಕದಾಸರು
  - ವೆಂಕಟಸುಬ್ಬಯ್ಯ
30. ವೀರರ ಜೀವನ ಸ್ಮಾರಕಗಳನ್ನು ಏನೆಂದು ಕರೆಯುವರು?
- ಸಿಡಿತಲೆ
  - ವೀರಗಲ್ಲು
  - ಮಾಸ್ತಿಗಲ್ಲು
  - ಬರಹದಕಲ್ಲು
31. ಜೇಡರ ದಾಸಿಮಯ್ಯನವರ ಅಂಕಿತನಾಮ
- ಚನ್ನ
  - ಶಿವಪ್ಪ
  - ಗುಹೇಶ್ವರ
  - ರಾಮನಾಥ.
32. ಆಲಿಸುವ ಅಥವಾ ಗ್ರಹಿಸುವ ಕೌಶಲ ಈ ವಾಕ್ಯದಲ್ಲಿರುವ ಷಷ್ಠಿ ವಿಭಕ್ತಿ ಯಾವುದು?
- ಆ
  - ಎಲ್ಲಿ
  - ಗೆ
  - ದೇಸೆಯಿಂದ
33. ವಿನೋಬಾ ಭಾವೆಯವರು ಕನ್ನಡ ಲಿಪಿಯನ್ನು ಏನೆಂದು ಕರೆದಿದ್ದಾರೆ?
- ರಾಜರಾಣಿ
  - ಲಿಪಿಗಳ ರಾಣಿ
  - ಕನ್ನಡಭಾಷೆ
  - ಕುಸುಮರಾಣಿ
34. ಥಾಮಸ್ ಮನ್ರೋವರ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿ ಕಲೆಕ್ಟರ್ ಆಗಿ ಕಾರ್ಯ ನಿರ್ವಹಿಸಿದ್ದರು?
- ಬಾಗಲಕೋಟೆ
  - ಬೆಳಗಾವಿ
  - ಬೀದರ್
  - ಬಳ್ಳಾರಿ.

35. ಕಳೆದ ಪದದ ಸಮನಾರ್ಥಕ ಪದ  
a) ರಾಗಿ b) ಜೋಳ c) ಭತ್ತ d) ಗೋಧಿ
36. ಹಂಪ ನಾಗರಾಜಯ್ಯನವರ ಹುಟ್ಟಿದ ಸ್ಥಳ  
a) ಹಂಪೆ ಸಂದ್ರ b) ಮಂಡ್ಯ c) ಬಳ್ಳಾರಿ d) ಮೈಸೂರು.
37. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಕೃತಿಯನ್ನು ರಚಿಸಿದ ಕವಿ ಯಾರು?  
a) ಪಂಪ b) ರನ್ನ c) ಜನ್ನ d) ಅಂಡಯ್ಯ
38. ತಂತ್ರಜ್ಞಾನದ ಬೆಳವಣಿಗೆಯ ಫಲವಾಗಿ ಕನ್ನಡ ಭಾಷೆಗೆ ಸಾಕಷ್ಟು \_\_\_\_\_ ಪದಗಳು ಸೇರಿಕೊಂಡಿವೆ.  
a) ತೆಲಗು b) ತಮಿಳು c) ಇಂಗ್ಲೀಷ್ d) ಮಲಯಾಳಿ
39. ಶರಣರ ಚಳುವಳಿಗೆ ಪ್ರೇರಕ ಶಕ್ತಿಯಾಗಿ ನಿಂತ ಮಹಾ ಮಾನವತಾವಾದಿ.  
a) ಅಕ್ಕಮಹಾದೇವಿ b) ಆಯ್ದಕ್ಕಿಮಾರಯ್ಯ  
c) ಬಸವಣ್ಣ d) ಆಯ್ದಕ್ಕಿ ಲಕ್ಕಮ್ಮ
40. ಭಕ್ತಿ ಪದದ ತದ್ಭವ ರೂಪ \_\_\_\_\_  
a) ಪ್ರಕೃತಿ b) ವಿಕೃತಿ c) ಸಂಸ್ಕೃತಿ d) ಭಕುತಿ
41. ಹಂಪನಾಗರಾಜಯ್ಯನವರು ಹುಟ್ಟಿದ ವರ್ಷ.  
a) 1936 b) 1946 c) 1947 d) 1935
42. ಕನ್ನಡ ಬರಹಗಳಿಗೆ \_\_\_\_\_ ವರ್ಷಗಳ ಇತಿಹಾಸವಿದೆ.  
a) 1956 b) 1884 c) 1956 d) 1500
43. ಹುಲ್ಲಾಗು ಬೆಟ್ಟದಡಿ ಮನೆಗೆ \_\_\_\_\_ ಆಗು.  
a) ಸಂಪಿಗೆ b) ಮಲ್ಲಿಗೆ c) ಇಂಪಿಗೆ d) ತಂಪಿಗೆ
44. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ ಕುಂಬಾರನಿಗೆ ಚಿನ್ನ ಎಂದರೆ.  
a) ಬಂಗಾರ b) ಮಣ್ಣು c) ತೋಟ d) ಕಾರ್ಯ
45. ಪಥ ಪದದ ಅರ್ಥ.  
a) ರಕ್ತ b) ಸ್ವರ್ಗ ಜೀವನ c) ದಾರಿ d) ಜ್ಞಾನಿ

46. ಜನಪದ ಅಕಾಡೆಮಿಯ ಅಧ್ಯಕ್ಷರು \_\_\_\_\_  
 a) ಹುಚ್ಚಪ್ಪ ಮಾಸ್ತರ  
 b) ಶ್ರೀ,ಎಸ್,ಕೆ.ಕರೀಂಖಾನ್  
 c) ಹಿ.ಚಿ.ಬೋರಲಿಂಗಯ್ಯ  
 d) ಶ್ರೀನಿವಾಸಮೂರ್ತಿ
47. ಕವಿರಾಜಮಾರ್ಗದಲ್ಲಿ \_\_\_\_\_ ಅವರನ್ನು ಕುರಿತು ವರ್ಣಿಸಲಾಗಿದೆ?  
 a) ತಮಿಳರು                      b) ಕೇರಳಿಗರು                      c) ಕನ್ನಡಿಗರು                      d) ಮರಾಠಿಗರು
48. ಕನ್ನಡ ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳಿವೆ?  
 a) 10                      b) 20                      c) 4                      d) 8.
49. ವಚನ ಎಂದರೆ \_\_\_\_\_.  
 a) ಮಾತು                      b) ಸಮಾಜ                      c) ಭಾಷೆ                      d) ದೇಶ
50. ಸುಂಕ ಪದದ ಅರ್ಥ.  
 a) ಕಾಸು                      b) ಆಣೆ                      c) ತೆರಿಗೆ                      d) ನೋವು.

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First/Second Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025

## ಸಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

## ಸೂಚನೆಗಳು

1. ಎಲ್ಲಾ ಜಿಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ಹಂಪನಾಗರಾಜಯ್ಯನವರು ಹುಟ್ಟಿದ ವರ್ಷ.  
a) 1936                      b) 1946                      c) 1947                      d) 1935
2. ಕನ್ನಡ ಬರಹಗಳಿಗೆ \_\_\_\_\_ ವರ್ಷಗಳ ಇತಿಹಾಸವಿದೆ.  
a) 1956                      b) 1884                      c) 1956                      d) 1500
3. ಹುಲ್ಲಾಗು ಬೆಟ್ಟದಡಿ ಮನೆಗೆ \_\_\_\_\_ ಆಗು.  
a) ಸಂಪಿಗೆ                      b) ಮಲ್ಲಿಗೆ                      c) ಇಂಪಿಗೆ                      d) ತಂಪಿಗೆ
4. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ ಕುಂಬಾರನಿಗೆ ಚಿನ್ನ ಎಂದರೆ.  
a) ಬಂಗಾರ                      b) ಮಣ್ಣು                      c) ತೋಟ                      d) ಕಾರ್ಯ
5. ಪಥ ಪದದ ಅರ್ಥ.  
a) ರಕ್ತ                      b) ಸ್ವರ್ಗ ಜೀವನ                      c) ದಾರಿ                      d) ಜ್ಞಾನಿ

6. ಜನಪದ ಅಕಾಡೆಮಿಯ ಅಧ್ಯಕ್ಷರು \_\_\_\_\_  
 a) ಹುಚ್ಚಪ್ಪ ಮಾಸ್ತರ  
 b) ಶ್ರೀ.ಎಸ್.ಕೆ.ಕರೀಂಖಾನ್  
 c) ಹಿ.ಚಿ.ಬೋರಲಿಂಗಯ್ಯ  
 d) ಶ್ರೀನಿವಾಸಮೂರ್ತಿ
7. ಕವಿರಾಜಮಾರ್ಗದಲ್ಲಿ \_\_\_\_\_ ಅವರನ್ನು ಕುರಿತು ವರ್ಣಿಸಲಾಗಿದೆ?  
 a) ತಮಿಳರು                      b) ಕೇರಳಿಗರು                      c) ಕನ್ನಡಿಗರು                      d) ಮರಾಠಿಗರು
8. ಕನ್ನಡ ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳಿವೆ?  
 a) 10                      b) 20                      c) 4                      d) 8.
9. ವಚನ ಎಂದರೆ \_\_\_\_\_.  
 a) ಮಾತು                      b) ಸಮಾಜ                      c) ಭಾಷೆ                      d) ದೇಶ
10. ಸುಂಕ ಪದದ ಅರ್ಥ.  
 a) ಕಾಸು                      b) ಆಣೆ                      c) ತೆರಿಗೆ                      d) ನೋವು.
11. ವಸುಧೇಂದ್ರವರು ಬರೆದ ಕಾದಂಬರಿ ಇದಾಗಿದೆ.  
 a) ಮನೀಷೆ                      b) ಕೃತಿ                      c) ಕರ್ತೃ                      d) ಸಂಕಲನ
12. ಪ್ರಹ್ಲಾದ \_\_\_\_\_ ಇಂಜಿನಿಯರ್.  
 a) ಸಾಫ್ಟ್‌ವೇರ್                      b) ಇಂಜಿನಿಯರ್                      c) ಸಿವಿಲ್ ಇಂಜಿನಿಯರ್                      d) ಮೆಕಾನಿಕ್
13. ವಿಜಯನಗರ ಸಾಮ್ರಾಜ್ಯದ ರಾಜಧಾನಿ  
 a) ಹಂಪೆ                      b) ಬೆಲೂರು                      c) ಹಳೇಬೀಡು                      d) ಕೋಲಾರ
14. ನಮ್ಮ ರಾಜ್ಯದಲ್ಲಿ \_\_\_\_\_ ಸೂತ್ರ ಬಳಕೆಯಲ್ಲಿದೆ.  
 a) ಏಕಭಾಷಾ                      b) ದ್ವಿಭಾಷಾ                      c) ತ್ರಿಭಾಷಾ                      d) ಭಾಷಾ
15. ಧರ್ಮಸಹಿಷ್ಣುತೆ ಬಗ್ಗೆ ಹೇಳುವ ಶಾಸನ \_\_\_\_\_  
 a) ಹಳೇಬೀಡು                      b) ಬೆಲೂರು                      c) ಹಂಪಿ                      d) ಹಾಸನ
16. ಬುಕ್ಕರಾಯನ ಕಾಲ ಸರಿಸುಮಾರು \_\_\_\_\_ ವರ್ಷ.  
 a) 1368                      b) 1468                      c) 1568                      d) 1668.

17. ಕುರುಡು ಕಾಂಚಾಣ ಪದ್ಯದ ಕವಿ.  
 a) ಕುವೆಂಪು b) ದ.ರಾ.ಬೇಂದ್ರೆ  
 c) ಡಿ.ವಿ.ಜಿ d) ಗೋಕಾಕ್
18. ಯುಗಾದಿ ಕಥೆಯಲ್ಲಿ ಯಾರನ್ನು 'ಖಡಕ್' ಮಾಸ್ತರ ಎಂದು ಕರೆಯಲಾಗಿದೆ.  
 a) ಗೋಪಣ್ಣ b) ಪ್ರಹ್ಲಾದ  
 c) ಇಸ್ಮಾಯಿಲ್ d) ಕಾಸಿಂಸಾಬ್
19. ಯದ್ಧದಲ್ಲಿ ಗೆದ್ದು ಬಂದರೆ ಕೀರ್ತಿ, ಸತ್ತರೆ \_\_\_\_\_  
 a) ನರಕ b) ಹಣ c) ಪ್ರಶಸ್ತಿ d) ಸ್ವರ್ಗ
20. ಹಲವು ಕಾಲ ಕಲ್ಲು ನೀರೊಳಗಿದ್ದರೇನು? ಬಲುನೆನೆದು \_\_\_\_\_ ಆಗುವುದೆ.  
 a) ಅಮೃತಶಿಲೆ b) ಹವಳ c) ಕಪ್ಪೆ d) ನಿಧಿ.
21. ಕಳ್ಳಗಂಜಿ \_\_\_\_\_ ಹೊಕ್ಕಡೆ ಹುಲಿ ತಿಂಬುದ ಮಾಬುದೇ?  
 a) ಮನೆ b) ಕಾಡು c) ಸ್ಮಶಾನ d) ಹುತ್ತ.
22. ಕರಿ ಘನ \_\_\_\_\_ ಕಿರಿದೆನ್ನಬಹುದೆ ಬಾರದಯ್ಯ.  
 a) ಜ್ಯೋತಿ b) ನೆನಪು c) ಮರೆವು d) ಅಂಕುಶ.
23. ತೊರೆಯೊಳು \_\_\_\_\_ ದುರಿತ ಪೋಗುವುದೆ?  
 a) ಮಿಂದರೆ b) ಹಾಡಿದರೆ c) ಕುಣಿದರೆ d) ಮರೆತರೆ.
24. ಪವಳದ ಲತೆಗೆ \_\_\_\_\_ ಇಟ್ಟವರು ಯಾರು ?  
 a) ಕಪ್ಪು b) ಹಸಿರು c) ಚಿತ್ರ d) ಕೆಂಪು.
25. ಬ್ರಹ್ಮಾಡವನ್ನೆಲ್ಲ ಯಾರು ತುಂಬಿಕೊಂಡಿದ್ದಾರೆಂದು ಶರೀಫರು ಹೇಳಿದ್ದಾರೆ?  
 a) ಕುಂಬಾರ b) ಬಡಗಿ c) ಕುಂಬಾರಕಿ d) ಗಾಣತಿ
26. ಎಷ್ಟು ಕಾಸಿಗೆ ಕುಂಬಾರಕಿ ಒಂದು ಕುಡಕಿಯನ್ನು ಮಾರುವಳು?  
 a) 3 b) 6 c) 9 d) 10
27. ನಗುವು ಸಹಜದ ಧರ್ಮ ನಗಿಸುವುದು \_\_\_\_\_  
 a) ಪರಧರ್ಮ b) ಅಧರ್ಮ c) ಧರ್ಮ d) ಮನುಷ್ಯ ಧರ್ಮ

28. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಅವರ ತವರೂರು ಯಾವುದು?  
 a) ರಾಣಿಬೆನ್ನೂರು      b) ಮಂಡ್ಯ      c) ಮೈಸೂರು      d) ಮುದ್ದೇನಹಳ್ಳಿ
29. ಗ್ರಂಥಕರ್ತರ ಸಮ್ಮೇಳನವು \_\_\_\_ ರಲ್ಲಿ ನಡೆಯಿತು.  
 a) 1977      b) 1907      c) 1970      d) 1974.
30. ಭಾರತ ಸರ್ಕಾರವು ಕನ್ನಡ ಭಾಷೆಯನ್ನು \_\_\_\_\_ ಎಂದು ಘೋಷಿಸಿದೆ.  
 a) ಶಾಸ್ತ್ರೀಯ ಭಾಷೆ  
 b) ಕನ್ನಡಭಾಷೆ  
 c) ತ್ರಿಭಾಷೆ  
 d) ಲಿಪಿ.
31. ವಿಫಲವಾದ ಪದದ ಅರ್ಥ.  
 a) ಕ್ರಾಂತಿ      b) ಸ್ವರ್ಗ      c) ಕವಿತೆ      d) ಕಾರ್ಯ
32. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ 'ಆಚಾರ' ಎಂದರೆ  
 a) ಆವಿಗೆ ಅಥವಾ ಮುಚ್ಚಳ      b) ಮಣ್ಣು  
 c) ನೀರು      d) ಮನ.
33. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಕಬ್ಬಿಣ ಮತ್ತು ಉಕ್ಕಿನ ಕಾರ್ಖಾನೆ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿದೆ?  
 a) ಮಂಡ್ಯ      b) ಮಂಗಳೂರು  
 c) ಮೈಸೂರು      d) ಶಿವಮೊಗ್ಗ
34. ಗಾಂಧೀಜಿಯವರು ಯಾರನ್ನು "ಅನ್ವಬ್ರಹ್ಮ" ಎಂದು ಕರೆದಿದ್ದಾರೆ?  
 a) ಕುವೆಂಪು  
 b) ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯ  
 c) ಡಿ.ವಿ.ಜಿ  
 d) ಅಕ್ಕಿಹೆಬ್ಬಾಳು ನರಸಿಂಹಮೂರ್ತಿ
35. ಯುಗಾದಿ ಕಥೆಯ ಲೇಖಕರು ಯಾರು?  
 a) ಡಿ.ವಿ.ಜಿ      b) ವಸುಧೇಂದ್ರ      c) ಅ.ನ.ಕೃ      d) ವಿ.ಕೃ.ಗೋಕಾಕ್
36. ಕಾಸಿಂಸಾಬರ ಮಗಳ ಹೆಸರೇನು?  
 a) ಚಾಂದಿನಿ      b) ಹಂಸ      c) ರೇಖಾ      d) ರೂಪ.

37. ಹಸಿವಾದರೆ ಉರೋಳಿಗೆ ಭಿಕ್ಷಾನ್ನಗಳುಂಟು, ತೃಷೆಯಾದರೆ ಕೆರೆ ಬಾವಿ ಹಳ್ಳಗಳುಂಟು, ಶಯನಕ್ಕೆ ಹಾಳುದೇಗುಲಗಳುಂಟು, ಈ ವಚನವನ್ನು ರಚಿಸಿದವರು.  
 a) ಅಕ್ಕಮಹಾದೇವಿ  
 b) ಬಸವಣ್ಣ  
 c) ರಾಮನಾಥ  
 d) ಅಲ್ಲಮ ಪ್ರಭು
38. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘವನ್ನು ಯಾವ ಕಾರಣಕ್ಕಾಗಿ ಸ್ಥಾಪಿಸಲಾಯಿತು.  
 a) ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿಗಾಗಿ  
 b) ದೇಶಕ್ಕಾಗಿ  
 c) ಭಾಷೆಗಾಗಿ  
 d) ಕರ್ನಾಟಕ ಏಕೀಕರಣಕ್ಕಾಗಿ
39. ತಲ್ಲಣಿಸದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆಯ ಕೀರ್ತನಕಾರರು ಯಾರು?  
 a) ಪುರಂದರದಾಸರು  
 b) ಶ್ರೀರಂಗರು  
 c) ಕನಕದಾಸರು  
 d) ವೆಂಕಟಸುಬ್ಬಯ್ಯ
40. ವೀರರ ಜೀವನ ಸ್ಮಾರಕಗಳನ್ನು ಏನೆಂದು ಕರೆಯುವರು?  
 a) ಸಿಡಿತಲೆ  
 b) ವೀರಗಲ್ಲು  
 c) ಮಾಸ್ತಿಗಲ್ಲು  
 d) ಬರಹದಕಲ್ಲು
41. ಜೇಡರ ದಾಸಿಮಯ್ಯನವರ ಅಂಕಿತನಾಮ  
 a) ಚನ್ನ  
 b) ಶಿವಪ್ಪ  
 c) ಗುಹೇಶ್ವರ  
 d) ರಾಮನಾಥ.
42. ಆಲಿಸುವ ಅಥವಾ ಗ್ರಹಿಸುವ ಕೌಶಲ ಈ ವಾಕ್ಯದಲ್ಲಿರುವ ಷಷ್ಠಿ ವಿಭಕ್ತಿ ಯಾವುದು?  
 a) ಆ  
 b) ಎಲ್ಲಿ  
 c) ಗೆ  
 d) ದೆಸೆಯಿಂದ
43. ವಿನೋಬಾ ಭಾವೆಯವರು ಕನ್ನಡ ಲಿಪಿಯನ್ನು ಏನೆಂದು ಕರೆದಿದ್ದಾರೆ?  
 a) ರಾಜರಾಣಿ  
 b) ಲಿಪಿಗಳ ರಾಣಿ  
 c) ಕನ್ನಡಭಾಷೆ  
 d) ಕುಸುಮರಾಣಿ

44. ಥಾಮಸ್ ಮನ್ರೋರವರ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿ ಕಲೆಕ್ಟರ್ ಆಗಿ ಕಾರ್ಯ ನಿರ್ವಹಿಸಿದ್ದರು?  
a) ಬಾಗಲಕೋಟೆ b) ಬೆಳಗಾವಿ c) ಬೀದರ್ d) ಬಳ್ಳಾರಿ.
45. ಕಳೆದ ಪದದ ಸಮನಾರ್ಥಕ ಪದ  
a) ರಾಗಿ b) ಜೋಳ c) ಭತ್ತ d) ಗೋಧಿ
46. ಹಂಪ ನಾಗರಾಜಯ್ಯನವರ ಹುಟ್ಟಿದ ಸ್ಥಳ  
a) ಹಂಪೆ ಸಂದ್ರ b) ಮಂಡ್ಯ c) ಬಳ್ಳಾರಿ d) ಮೈಸೂರು.
47. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಕೃತಿಯನ್ನು ರಚಿಸಿದ ಕವಿ ಯಾರು?  
a) ಪಂಪ b) ರನ್ನ c) ಜನ್ನ d) ಅಂಡಯ್ಯ
48. ತಂತ್ರಜ್ಞಾನದ ಬೆಳವಣಿಗೆಯ ಫಲವಾಗಿ ಕನ್ನಡ ಭಾಷೆಗೆ ಸಾಕಷ್ಟು \_\_\_\_\_ ಪದಗಳು ಸೇರಿಕೊಂಡಿವೆ.  
a) ತೆಲಗು b) ತಮಿಳು c) ಇಂಗ್ಲೀಷ್ d) ಮಲಯಾಳಿ
49. ಶರಣರ ಚಳುವಳಿಗೆ ಪ್ರೇರಕ ಶಕ್ತಿಯಾಗಿ ನಿಂತ ಮಹಾ ಮಾನವತಾವಾದಿ.  
a) ಅಕ್ಕಮಹಾದೇವಿ b) ಆಯ್ದಕ್ಕಿಮಾರಯ್ಯ  
c) ಬಸವಣ್ಣ d) ಆಯ್ದಕ್ಕಿ ಲಕ್ಕಮ್ಮ
50. ಭಕ್ತಿ ಪದದ ತದ್ಭವ ರೂಪ \_\_\_\_\_  
a) ಪ್ರಕೃತಿ b) ವಿಕೃತಿ c) ಸಂಸ್ಕೃತಿ d) ಭಕುತಿ

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First/Second Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025

## ಸಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

## ಸೂಚನೆಗಳು

1. ಎಲ್ಲ ಜಿಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ಕಳ್ಳಗಂಜಿ \_\_\_\_\_ ಹೊಕ್ಕಡೆ ಹುಲಿ ತಿಂಬುದ ಮಾಬುದೇ?  
a) ಮನೆ b) ಕಾಡು c) ಸ್ಮಶಾನ d) ಹುತ್ತ.
2. ಕರಿ ಘನ \_\_\_\_\_ ಕಿರಿದೆನ್ನಬಹುದೆ ಬಾರದಯ್ಯ.  
a) ಜ್ಯೋತಿ b) ನೆನಪು c) ಮರೆವು d) ಅಂಕುಶ.
3. ತೊರೆಯೊಳು \_\_\_\_\_ ದುರಿತ ಪೋಗುವುದೆ?  
a) ಮಿಂದರೆ b) ಹಾಡಿದರೆ c) ಕುಣಿದರೆ d) ಮರೆತರೆ.
4. ಪವಳದ ಲತೆಗೆ \_\_\_\_\_ ಇಟ್ಟವರು ಯಾರು ?  
a) ಕಪ್ಪು b) ಹಸಿರು c) ಚಿತ್ರ d) ಕೆಂಪು.
5. ಬ್ರಹ್ಮಾಡವನ್ನೆಲ್ಲ ಯಾರು ತುಂಬಿಕೊಂಡಿದ್ದಾರೆಂದು ಶರೀಫರು ಹೇಳಿದ್ದಾರೆ?  
a) ಕುಂಬಾರ b) ಬಡಗ c) ಕುಂಬಾರಕಿ d) ಗಾಣತಿ

6. ಎಷ್ಟು ಕಾಸಿಗೆ ಕುಂಬಾರಕಿ ಒಂದು ಕುಡಕಿಯನ್ನು ಮಾರುವಳು?  
a) 3 b) 6 c) 9 d) 10
7. ನಗುವು ಸಹಜದ ಧರ್ಮ ನಗಿಸುವುದು \_\_\_\_  
a) ಪರಧರ್ಮ b) ಅಧರ್ಮ c) ಧರ್ಮ d) ಮನುಷ್ಯ ಧರ್ಮ
8. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಅವರ ತವರೂರು ಯಾವುದು?  
a) ರಾಣಿಬೆನ್ನೂರು b) ಮಂಡ್ಯ c) ಮೈಸೂರು d) ಮುದ್ದೇನಹಳ್ಳಿ
9. ಗ್ರಂಥಕರ್ತರ ಸಮ್ಮೇಳನವು \_\_\_\_ ರಲ್ಲಿ ನಡೆಯಿತು.  
a) 1977 b) 1907 c) 1970 d) 1974.
10. ಭಾರತ ಸರ್ಕಾರವು ಕನ್ನಡ ಭಾಷೆಯನ್ನು \_\_\_\_ ಎಂದು ಘೋಷಿಸಿದೆ.  
a) ಶಾಸ್ತ್ರೀಯ ಭಾಷೆ  
b) ಕನ್ನಡಭಾಷೆ  
c) ತ್ರಿಭಾಷೆ  
d) ಲಿಪಿ.
11. ಜೇಡರ ದಾಸಿಮಯ್ಯನವರ ಅಂಕಿತನಾಮ  
a) ಚನ್ನ b) ಶಿವಪ್ಪ c) ಗುಹೇಶ್ವರ d) ರಾಮನಾಥ.
12. ಆಲಿಸುವ ಅಥವಾ ಗ್ರಹಿಸುವ ಕೌಶಲ ಈ ವಾಕ್ಯದಲ್ಲಿರುವ ಷಷ್ಠಿ ವಿಭಕ್ತಿ ಯಾವುದು?  
a) ಆ b) ಎಲ್ಲಿ c) ಗೆ d) ದೆಸೆಯಿಂದ
13. ವಿನೋಬಾ ಭಾವೆಯವರು ಕನ್ನಡ ಲಿಪಿಯನ್ನು ಏನೆಂದು ಕರೆದಿದ್ದಾರೆ?  
a) ರಾಜರಾಣಿ b) ಲಿಪಿಗಳ ರಾಣಿ  
c) ಕನ್ನಡಭಾಷೆ d) ಕುಸುಮರಾಣಿ
14. ಥಾಮಸ್ ಮನ್ರೋವರ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿ ಕಲೆಕ್ಟರ್ ಆಗಿ ಕಾರ್ಯ ನಿರ್ವಹಿಸಿದ್ದರು?  
a) ಬಾಗಲಕೋಟೆ b) ಬೆಳಗಾವಿ c) ಬೀದರ್ d) ಬಳ್ಳಾರಿ.
15. ಕಳೆದ ಪದದ ಸಮನಾರ್ಥಕ ಪದ  
a) ರಾಗಿ b) ಜೋಳ c) ಭತ್ತ d) ಗೋಧಿ
16. ಹಂಪ ನಾಗರಾಜಯ್ಯನವರ ಹುಟ್ಟಿದ ಸ್ಥಳ  
a) ಹಂಪ ಸಂದ್ರ b) ಮಂಡ್ಯ c) ಬಳ್ಳಾರಿ d) ಮೈಸೂರು.

17. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಕೃತಿಯನ್ನು ರಚಿಸಿದ ಕವಿ ಯಾರು?  
a) ಪಂಪ b) ರನ್ನ c) ಜನ್ನ d) ಆಂಡಯ್ಯ
18. ತಂತ್ರಜ್ಞಾನದ ಬೆಳವಣಿಗೆಯ ಫಲವಾಗಿ ಕನ್ನಡ ಭಾಷೆಗೆ ಸಾಕಷ್ಟು \_\_\_\_\_ ಪದಗಳು ಸೇರಿಕೊಂಡಿವೆ.  
a) ತೆಲಗು b) ತಮಿಳು c) ಇಂಗ್ಲೀಷ್ d) ಮಲಯಾಳಿ
19. ಶರಣರ ಚಳುವಳಿಗೆ ಪ್ರೇರಕ ಶಕ್ತಿಯಾಗಿ ನಿಂತ ಮಹಾ ಮಾನವತಾವಾದಿ.  
a) ಅಕ್ಕಮಹಾದೇವಿ b) ಆಯ್ದಕ್ಕಿಮಾರಯ್ಯ  
c) ಬಸವಣ್ಣ d) ಆಯ್ದಕ್ಕಿ ಲಕ್ಕಮ್ಮ
20. ಭಕ್ತಿ ಪದದ ತದ್ಭವ ರೂಪ \_\_\_\_\_  
a) ಪ್ರಕೃತಿ b) ವಿಕೃತಿ c) ಸಂಸ್ಕೃತಿ d) ಭಕುತಿ
21. ವಸುಧೇಂದ್ರರವರು ಬರೆದ ಕಾದಂಬರಿ ಇದಾಗಿದೆ.  
a) ಮನೀಷೆ b) ಕೃತಿ c) ಕರ್ತೃ d) ಸಂಕಲನ
22. ಪ್ರಹ್ಲಾದ \_\_\_\_\_ ಇಂಜಿನಿಯರ್.  
a) ಸಾಫ್ಟ್‌ವೇರ್ b) ಇಂಜಿನಿಯರ್ c) ಸಿವಿಲ್ ಇಂಜಿನಿಯರ್ d) ಮೆಕಾನಿಕ್
23. ವಿಜಯನಗರ ಸಾಮ್ರಾಜ್ಯದ ರಾಜಧಾನಿ  
a) ಹಂಪೆ b) ಬೇಲೂರು c) ಹಳೇಬೀಡು d) ಕೋಲಾರ
24. ನಮ್ಮ ರಾಜ್ಯದಲ್ಲಿ \_\_\_\_\_ ಸೂತ್ರ ಬಳಕೆಯಲ್ಲಿದೆ.  
a) ಏಕಭಾಷಾ b) ದ್ವಿಭಾಷಾ c) ತ್ರಿಭಾಷಾ d) ಭಾಷಾ
25. ಧರ್ಮಸಹಿಷ್ಣುತೆ ಬಗ್ಗೆ ಹೇಳುವ ಶಾಸನ \_\_\_\_\_  
a) ಹಳೇಬೀಡು b) ಬೇಲೂರು c) ಹಂಪಿ d) ಹಾಸನ
26. ಬುಕ್ಕರಾಯನ ಕಾಲ ಸರಿಸುಮಾರು \_\_\_\_\_ ವರ್ಷ.  
a) 1368 b) 1468 c) 1568 d) 1668.
27. ಕುರುಡು ಕಾಂಚಾಣ ಪದ್ಯದ ಕವಿ.  
a) ಕುವೆಂಪು b) ದ.ರಾ.ಬೇಂದ್ರೆ  
c) ಡಿ.ವಿ.ಜಿ d) ಗೋಕಾಕ್

28. ಯುಗಾದಿ ಕಥೆಯಲ್ಲಿ ಯಾರನ್ನು 'ಖಡಕ್' ಮಾಸ್ತರ ಎಂದು ಕರೆಯಲಾಗಿದೆ.  
 a) ಗೋಪಣ್ಣ b) ಪ್ರಹ್ಲಾದ  
 c) ಇಸ್ಮಾಯಿಲ್ d) ಕಾಸಿಂಸಾಬ್
29. ಯದ್ಧದಲ್ಲಿ ಗೆದ್ದು ಬಂದರೆ ಕೀರ್ತಿ, ಸತ್ತರೆ \_\_\_\_  
 a) ನರಕ b) ಹಣ c) ಪ್ರಶಸ್ತಿ d) ಸ್ವರ್ಗ
30. ಹಲವು ಕಾಲ ಕಲ್ಲು ನೀರೊಳಗಿದ್ದರೇನು? ಬಲುನೆನೆದು \_\_\_\_ ಆಗುವುದೆ.  
 a) ಅಮೃತಶಿಲೆ b) ಹವಳ c) ಕಪ್ಪೆ d) ನಿಧಿ.
31. ಹಂಪನಾಗರಾಜಯ್ಯನವರು ಹುಟ್ಟಿದ ವರ್ಷ.  
 a) 1936 b) 1946 c) 1947 d) 1935
32. ಕನ್ನಡ ಬರಹಗಳಿಗೆ \_\_\_\_ ವರ್ಷಗಳ ಇತಿಹಾಸವಿದೆ.  
 a) 1956 b) 1884 c) 1956 d) 1500
33. ಹುಲ್ಲಾಗು ಬೆಟ್ಟದಡಿ ಮನೆಗೆ \_\_\_\_ ಆಗು.  
 a) ಸಂಪಿಗೆ b) ಮಲ್ಲಿಗೆ c) ಇಂಪಿಗೆ d) ತಂಪಿಗೆ
34. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ ಕುಂಬಾರನಿಗೆ ಚಿನ್ನ ಎಂದರೆ.  
 a) ಬಂಗಾರ b) ಮಣ್ಣು c) ತೋಟ d) ಕಾರ್ಯ
35. ಪಥ ಪದದ ಅರ್ಥ.  
 a) ರಕ್ತ b) ಸ್ವರ್ಗ ಜೀವನ c) ದಾರಿ d) ಜ್ಞಾನಿ
36. ಜನಪದ ಅಕಾಡೆಮಿಯ ಅಧ್ಯಕ್ಷರು \_\_\_\_  
 a) ಹುಚ್ಚಪ್ಪ ಮಾಸ್ತರ  
 b) ಶ್ರೀ,ಎಸ್,ಕೆ.ಕರೀಂಖಾನ್  
 c) ಹಿ.ಚಿ.ಬೋರಲಿಂಗಯ್ಯ  
 d) ಶ್ರೀನಿವಾಸಮೂರ್ತಿ
37. ಕವಿರಾಜಮಾರ್ಗದಲ್ಲಿ \_\_\_\_ ಅವರನ್ನು ಕುರಿತು ವರ್ಣಿಸಲಾಗಿದೆ?  
 a) ತಮಿಳರು b) ಕೇರಳಿಗರು c) ಕನ್ನಡಿಗರು d) ಮರಾಠಿಗರು
38. ಕನ್ನಡ ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳಿವೆ?  
 a) 10 b) 20 c) 4 d) 8.

39. ವಚನ ಎಂದರೆ \_\_\_\_.
- a) ಮಾತು                      b) ಸಮಾಜ                      c) ಭಾಷೆ                      d) ದೇಶ
40. ಸುಂಕ ಪದದ ಅರ್ಥ.
- a) ಕಾಸು                      b) ಆಣೆ                      c) ತೆರಿಗೆ                      d) ನೋವು.
41. ವಿಘ್ನ ಪದದ ಅರ್ಥ.
- a) ಕ್ರಾಂತಿ                      b) ಸ್ವರ್ಗ                      c) ಕವಿತೆ                      d) ಕಾರ್ಯ
42. ಶಿಶುನಾಳ ಷರೀಫರ ಪ್ರಕಾರ 'ಆಚಾರ' ಎಂದರೆ
- a) ಆವಿಗೆ ಅಥವಾ ಮುಚ್ಚಳ                      b) ಮಣ್ಣು  
c) ನೀರು                      d) ಮನ.
43. ವಿಶ್ವೇಶ್ವರಯ್ಯ ಕಬ್ಬಿಣ ಮತ್ತು ಉಕ್ಕಿನ ಕಾರ್ಖಾನೆ ಯಾವ ಜಿಲ್ಲೆಯಲ್ಲಿದೆ?
- a) ಮಂಡ್ಯ                      b) ಮಂಗಳೂರು  
c) ಮೈಸೂರು                      d) ಶಿವಮೊಗ್ಗ
44. ಗಾಂಧೀಜಿಯವರು ಯಾರನ್ನು "ಅನ್ನಬ್ರಹ್ಮ" ಎಂದು ಕರೆದಿದ್ದಾರೆ?
- a) ಕುವೆಂಪು  
b) ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯ  
c) ಡಿ.ವಿ.ಜಿ  
d) ಅಕ್ಕಿಹೆಬ್ಬಾಳು ನರಸಿಂಹಮೂರ್ತಿ
45. ಯುಗಾದಿ ಕಥೆಯ ಲೇಖಕರು ಯಾರು?
- a) ಡಿ.ವಿ.ಜಿ                      b) ವಸುಧೇಂದ್ರ                      c) ಅ.ನ.ಕೃ                      d) ವಿ.ಕೃ.ಗೋಕಾಕ್
46. ಕಾಸಿಂಸಾಬರ ಮಗಳ ಹೆಸರೇನು?
- a) ಚಾಂದಿನಿ                      b) ಹಂಸ                      c) ರೇಖಾ                      d) ರೂಪ.
47. ಹಸಿವಾದರೆ ಉರೂಳಿಗೆ ಭಿಕ್ಷಾನ್ನಗಳುಂಟು, ತೃಷೆಯಾದರೆ ಕೆರೆ ಬಾವಿ ಹಳ್ಳಗಳುಂಟು, ಶಯನಕ್ಕೆ ಹಾಳುದೇಗುಲಗಳುಂಟು, ಈ ವಚನವನ್ನು ರಚಿಸಿದವರು.
- a) ಅಕ್ಕಮಹಾದೇವಿ  
b) ಬಸವಣ್ಣ  
c) ರಾಮನಾಥ  
d) ಅಲ್ಲಮ ಪ್ರಭು.

48. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘವನ್ನು ಯಾವ ಕಾರಣಕ್ಕಾಗಿ ಸ್ಥಾಪಿಸಲಾಯಿತು.

- a) ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿಗಾಗಿ
- b) ದೇಶಕ್ಕಾಗಿ
- c) ಭಾಷೆಗಾಗಿ
- d) ಕರ್ನಾಟಕ ಏಕೀಕರಣಕ್ಕಾಗಿ

49. ತಲ್ಲಣಿಸದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆಯ ಕೀರ್ತನಕಾರರು ಯಾರು?

- a) ಪುರಂದರದಾಸರು
- b) ಶ್ರೀರಂಗರು
- c) ಕನಕದಾಸರು
- d) ವೆಂಕಟಸುಬ್ಬಯ್ಯ

50. ವೀರರ ಜೀವನ ಸ್ಮಾರಕಗಳನ್ನು ಏನೆಂದು ಕರೆಯುವರು?

- a) ಸಿಡಿತಲೆ
- b) ವೀರಗಲ್ಲು
- c) ಮಾಸ್ತಿಗಲ್ಲು
- d) ಬರಹದಕಲ್ಲು

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BMATC101

## First Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Mathematics–I for Civil Engineering Stream

Time: 3 hrs.

Max. Marks:100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.  
3. VTU formula Handbook is permuted.

|                   |    | Module – 1                                                                                                                                                                          | M | L  | C   |
|-------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 1                 | a. | With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$ .                                                                                                                 | 6 | L2 | CO1 |
|                   | b. | Find the angle between the curves $r = 4\sec^2(\theta/2)$ and $r = 9\operatorname{cosec}^2(\theta/2)$ .                                                                             | 7 | L2 | CO1 |
|                   | c. | Find the radius of curvature of the curve.<br>$\sqrt{x} + \sqrt{y} = \sqrt{a}$ at $(\frac{a}{4}, \frac{a}{4})$ .                                                                    | 7 | L3 | CO1 |
| <b>OR</b>         |    |                                                                                                                                                                                     |   |    |     |
| 2                 | a. | Find the pedal equation of the curve $r^m = a^m(\cos m\theta + \sin m\theta)$ .                                                                                                     | 6 | L2 | CO1 |
|                   | b. | Derive the radius of curvature in Cartesian form.                                                                                                                                   | 7 | L2 | CO1 |
|                   | c. | Using modern mathematical tool write a program to plot the curve<br>$r = 2(1 + \cos \theta)$ .                                                                                      | 7 | L3 | CO5 |
| <b>Module – 2</b> |    |                                                                                                                                                                                     |   |    |     |
| 3                 | a. | Using Maclaurin's series expand $e^{\sin x}$ in powers of x upto the term containing $x^5$ .                                                                                        | 6 | L2 | CO2 |
|                   | b. | If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$ . | 7 | L3 | CO2 |
|                   | c. | Find the extreme values of the function :<br>$f(x, y) = x^3 + y^3 - 3x - 12y + 20$ .                                                                                                | 7 | L3 | CO2 |
| <b>OR</b>         |    |                                                                                                                                                                                     |   |    |     |
| 4                 | a. | Evaluate : i) $\lim_{x \rightarrow 0} \left( \frac{\tan x}{x} \right)^{\frac{1}{x^2}}$ ii) $\lim_{x \rightarrow 0} (\sin x)^{\tan x}$ .                                             | 6 | L2 | CO2 |
|                   | b. | If: $u = x + 3y^2 - z^3$ ; $v = 4x^2yz$ ; $w = 2z^2 - xy$<br>find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ at $(1, -1, 0)$ .                                                   | 7 | L3 | CO2 |
|                   | c. | Using modern mathematical tool write a program to evaluate $\lim_{x \rightarrow 0} (\sin x)^{\tan x}$ .                                                                             | 7 | L3 | CO5 |
| <b>1 of 3</b>     |    |                                                                                                                                                                                     |   |    |     |

Module – 3

|   |    |                                                                        |   |    |     |
|---|----|------------------------------------------------------------------------|---|----|-----|
| 5 | a. | Solve :<br>$x \frac{dy}{dx} + y \log y = xye^x$ .                      | 6 | L2 | C03 |
|   | b. | Find the orthogonal trajectories of the family of curves $y^2 = 4ax$ . | 7 | L3 | C03 |
|   | c. | Solve : $yp^2 + (x - y)p - x = 0$ .                                    | 7 | L3 | C03 |

OR

|   |    |                                                                                                                                                                                                                         |   |    |     |
|---|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 6 | a. | Solve $(4xy + 3y^2 - x)dx + x(x + 2y)dy = 0$ .                                                                                                                                                                          | 6 | L2 | C03 |
|   | b. | A bottle of mineral water at a room temperature of 72°F is kept in a refrigerator where the temperature is 44°F. After half an hour water cooled to 61°F. What is the temperature of the water in another half an hour? | 7 | L3 | C03 |
|   | c. | Find the general and singular solution of $xp^2 - py + kp + a = 0$ .                                                                                                                                                    | 7 | L3 | C03 |

Module – 4

|   |    |                                                                                                                               |   |    |     |
|---|----|-------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 7 | a. | Solve $(D^4 + 18D^2 + 81)y = 0$ .                                                                                             | 6 | L2 | C03 |
|   | b. | Solve $D^3y + 8y = \sin(2x)$ .                                                                                                | 7 | L2 | C03 |
|   | c. | Solve Legendre's linear differential equation :<br>$(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4\cos[\log(1+x)]$ . | 7 | L2 | C03 |

OR

|   |    |                                                                                      |   |    |     |
|---|----|--------------------------------------------------------------------------------------|---|----|-----|
| 8 | a. | Solve $D^2y + 3Dy + 2y = 12x^2$ .                                                    | 6 | L2 | C03 |
|   | b. | Solve by method of variation of parameters $\frac{d^2y}{dx^2} + y = \sec x \tan x$ . | 7 | L2 | C03 |
|   | c. | Solve $x^2 \frac{d^2y}{dx^2} + 5x \frac{dy}{dx} + 4y = x^4$ .                        | 7 | L2 | C03 |

Module – 5

|    |    |                                                                                                                                                                                                                                     |   |    |     |
|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 9  | a. | Find the rank of the matrix $A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$ .                                                                                                                 | 6 | L2 | C04 |
|    | b. | Using Gauss-elimination method solve :<br>$x + y + z = 9$<br>$x - 2y + 3z = 8$<br>$2x + y - z = 3$ .                                                                                                                                | 7 | L3 | C04 |
|    | c. | Using Rayleigh's power method find the dominant eigen value and the corresponding eigen vector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ , taking initial vector as $[1, 1, 1]^T$ . | 7 | L3 | C04 |
| OR |    |                                                                                                                                                                                                                                     |   |    |     |
| 10 | a. | Apply Gauss – Jordan method to solve the equations :<br>$x + y + z = 9$<br>$2x - 3y + 4z = 13$<br>$3x + 4y + 5z = 40$ .                                                                                                             | 6 | L2 | C04 |
|    | b. | Test for consistency and solve :<br>$x + y + z = 6$<br>$x - y + 2z = 5$<br>$3x + y + z = 8$ .                                                                                                                                       | 7 | L2 | C04 |
|    | c. | Write a program to solve the system of equations using Gauss – Seidel method by taking initial approximations (0, 0, 0). Carry out 3 iterations.<br>$x + y + 54z = 110$<br>$27x + 6y - z = 85$<br>$6x + 15y + 2z = 72$ .            | 7 | L3 | C05 |

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BMATC201

## Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Mathematics – II for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                       | M | L  | C   |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dy dx$ .                                                                                                             | 7 | L2 | CO1 |
|            | b. | Evaluate $\int_0^a \int_y^a \frac{x}{x^2 + y^2} dx dy$ by changing the order of integration.                                                                          | 7 | L3 | CO1 |
|            | c. | Derive the relation $\beta(m, n) = \frac{\gamma(m) \cdot \gamma(n)}{\gamma(m + n)}$ .                                                                                 | 6 | L2 | CO1 |
| OR         |    |                                                                                                                                                                       |   |    |     |
| Q.2        | a. | Evaluate $\int_{-a}^a \int_0^{\sqrt{a^2 - x^2}} \sqrt{x^2 + y^2} dy dx$ by changing into polar coordinates.                                                           | 7 | L3 | CO1 |
|            | b. | Using double integration find the area of a plane in the form of a quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ .                                  | 7 | L2 | CO1 |
|            | c. | Using modern mathematical program to evaluate the integral, $\int_0^3 \int_0^{3-x} \int_0^{3-x-y} xyz dz dy dx$ .                                                     | 6 | L3 | CO5 |
| Module – 2 |    |                                                                                                                                                                       |   |    |     |
| Q.3        | a. | If $\vec{F} = \nabla(xy^3z^2)$ , find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ at the point (1, -1, 1).                                                       | 7 | L2 | CO2 |
|            | b. | Define an irrotational vector. Find the constants a, b, c such that $\vec{F} = (x + y + az) \hat{i} + (bx + 2y - z) \hat{j} + (x + cy + 2z) \hat{k}$ is irrotational. | 7 | L2 | CO2 |
|            | c. | Find the directional derivative of $\phi = x^2yz + 4xz^2$ at the point (1, -2, -1) in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$ .                   | 6 | L2 | CO2 |

| OR         |       |                                                                                                                                                                                                                                                                                                                                                                                                                        |       |       |       |      |      |      |            |       |       |       |       |       |   |    |     |
|------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|------|------|------|------------|-------|-------|-------|-------|-------|---|----|-----|
| Q.4        | a.    | Using Green's theorem, evaluate $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ , where C is the boundary of the region enclosed by $y = \sqrt{x}$ and $y = x^2$ .                                                                                                                                                                                                                                                              | 7     | L3    | CO2   |      |      |      |            |       |       |       |       |       |   |    |     |
|            | b.    | If $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$ , evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve represented by, $x = t, y = t^2, z = t^3 - 1 \leq t \leq 1$                                                                                                                                                                                                                                           | 7     | L2    | CO2   |      |      |      |            |       |       |       |       |       |   |    |     |
|            | c.    | Write the modern mathematical tool program to find the divergence of the vector field, $\vec{F} = x^2y\hat{i} + y^2z\hat{j} + z^2xy\hat{k}$ .                                                                                                                                                                                                                                                                          | 6     | L3    | CO5   |      |      |      |            |       |       |       |       |       |   |    |     |
| Module – 3 |       |                                                                                                                                                                                                                                                                                                                                                                                                                        |       |       |       |      |      |      |            |       |       |       |       |       |   |    |     |
| Q.5        | a.    | Form the partial differential equation by eliminating the arbitrary function from the relation, $\ell x + my + nz = f(x^2 + y^2 + z^2)$                                                                                                                                                                                                                                                                                | 7     | L2    | CO3   |      |      |      |            |       |       |       |       |       |   |    |     |
|            | b.    | Solve $\frac{\partial^2 z}{\partial y^2} = Z$ , given that $y = 0, z = e^x$ and $\frac{\partial z}{\partial y} = e^{-x}$ .                                                                                                                                                                                                                                                                                             | 7     | L3    | CO3   |      |      |      |            |       |       |       |       |       |   |    |     |
|            | c.    | Solve $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$ using Lagrange's multipliers.                                                                                                                                                                                                                                                                                                                                     | 6     | L2    | CO3   |      |      |      |            |       |       |       |       |       |   |    |     |
| OR         |       |                                                                                                                                                                                                                                                                                                                                                                                                                        |       |       |       |      |      |      |            |       |       |       |       |       |   |    |     |
| Q.6        | a.    | Form the partial differential equation by eliminating the arbitrary constants from $(x - a)^2 + (y - b)^2 + z^2 = C^2$ .                                                                                                                                                                                                                                                                                               | 7     | L2    | CO3   |      |      |      |            |       |       |       |       |       |   |    |     |
|            | b.    | Solve $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$ for which $\frac{\partial z}{\partial y} = -2 \sin y$ , when $x = 0$ and $z = 0$ if y is an odd multiple of $\frac{\pi}{2}$ .                                                                                                                                                                                                                       | 7     | L3    | CO3   |      |      |      |            |       |       |       |       |       |   |    |     |
|            | c.    | Derive one dimensional wave equation.                                                                                                                                                                                                                                                                                                                                                                                  | 6     | L2    | CO3   |      |      |      |            |       |       |       |       |       |   |    |     |
| Module – 4 |       |                                                                                                                                                                                                                                                                                                                                                                                                                        |       |       |       |      |      |      |            |       |       |       |       |       |   |    |     |
| Q.7        | a.    | By Newton's-Raphson method find the root of $x \sin x + \cos x = 0$ , which is near to $x = \pi$ .                                                                                                                                                                                                                                                                                                                     | 7     | L2    | CO4   |      |      |      |            |       |       |       |       |       |   |    |     |
|            | b.    | The population of a town is given by the following table :<br><table border="1"> <tr> <td>Year</td><td>1951</td><td>1961</td><td>1971</td><td>1981</td><td>1991</td></tr> <tr> <td>Population</td><td>19.96</td><td>39.65</td><td>58.81</td><td>72.21</td><td>94.61</td></tr> </table> Using Forward and Backward Newton's interpolation formula, calculate the increase in population between the years 1955 to 1985. | Year  | 1951  | 1961  | 1971 | 1981 | 1991 | Population | 19.96 | 39.65 | 58.81 | 72.21 | 94.61 | 7 | L2 | CO4 |
| Year       | 1951  | 1961                                                                                                                                                                                                                                                                                                                                                                                                                   | 1971  | 1981  | 1991  |      |      |      |            |       |       |       |       |       |   |    |     |
| Population | 19.96 | 39.65                                                                                                                                                                                                                                                                                                                                                                                                                  | 58.81 | 72.21 | 94.61 |      |      |      |            |       |       |       |       |       |   |    |     |
|            | c.    | Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{\cos \theta} d\theta$ by taking 7 ordinates using Simpson's $\left(\frac{1}{3}\right)^{rd}$ rule.                                                                                                                                                                                                                                                                               | 6     | L3    | CO4   |      |      |      |            |       |       |       |       |       |   |    |     |

| OR         |    |                                                                                                                                                                                                                                             |    |     |     |   |   |      |    |   |    |     |   |    |     |
|------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|-----|---|---|------|----|---|----|-----|---|----|-----|
| Q.8        | a. | Compute the real root of the equation $x \log_{10} x - 1.2 = 0$ by the Regula Falsi method taking Four decimal places.                                                                                                                      | 7  | L2  | CO4 |   |   |      |    |   |    |     |   |    |     |
|            | b. | Compute the value of y when x = 4, using Lagrange's interpolation formula given,<br><table border="1"><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x)</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table> | x  | 0   | 2   | 3 | 6 | f(x) | -4 | 2 | 14 | 158 | 7 | L2 | CO4 |
| x          | 0  | 2                                                                                                                                                                                                                                           | 3  | 6   |     |   |   |      |    |   |    |     |   |    |     |
| f(x)       | -4 | 2                                                                                                                                                                                                                                           | 14 | 158 |     |   |   |      |    |   |    |     |   |    |     |
|            | c. | Evaluate $\int_0^{0.3} (1 - 8x^3)^{\frac{1}{2}} dx$ by using Simpson's $\left(\frac{1}{3}\right)^{\text{rd}}$ rule, by taking 3 equal parts.                                                                                                | 6  | L3  | CO4 |   |   |      |    |   |    |     |   |    |     |
| Module – 5 |    |                                                                                                                                                                                                                                             |    |     |     |   |   |      |    |   |    |     |   |    |     |
| Q.9        | a. | Solve by using modified Euler's method. $\frac{dy}{dx} = x - y^2$ , $y(0) = 1$ taking $h = 0.1$ , find $y(0.2)$ .                                                                                                                           | 7  | L2  | CO4 |   |   |      |    |   |    |     |   |    |     |
|            | b. | Applying Milne's predictor and corrector method, find $y(0.8)$ from $\frac{dy}{dx} = x - y^2$ and given $y(0) = 0$ , $y(0.2) = 0.02$ , $y(0.4) = 0.0795$ , $y(0.6) = 0.1762$ .                                                              | 7  | L2  | CO4 |   |   |      |    |   |    |     |   |    |     |
|            | c. | Using Runge-Kutta method of order 4, find y at $x = 0.1$ , given that $\frac{dy}{dx} = 3e^x + 2y$ , $y(0) = 0$ and $h = 0.1$ .                                                                                                              | 6  | L3  | CO4 |   |   |      |    |   |    |     |   |    |     |
| OR         |    |                                                                                                                                                                                                                                             |    |     |     |   |   |      |    |   |    |     |   |    |     |
| Q.10       | a. | Use Taylor's series method to find $y(0.1)$ from $\frac{dy}{dx} = x - y^2$ and $y = 1$ at $x = 0$ upto 4 <sup>th</sup> degree.                                                                                                              | 7  | L2  | CO4 |   |   |      |    |   |    |     |   |    |     |
|            | b. | Using the Runge-Kutta method of order 4, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $h = 0.2$ , find $y(0.2)$ .                                                                                                 | 7  | L2  | CO4 |   |   |      |    |   |    |     |   |    |     |
|            | c. | Using mathematical tools, write a code to find the solution of $\frac{dy}{dx} = 1 + \frac{y}{x}$ at $y(2)$ taking $h = 0.2$ given that $y(1) = 2$ by Runge-Kutta method of order 4.                                                         | 6  | L3  | CO5 |   |   |      |    |   |    |     |   |    |     |

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**First Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025**  
**Mathematics - I for EEE Stream**

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
 2. M : Marks , L: Bloom's level , C: Course outcomes.  
 3. VTU Formula Handbook is permitted.*

| Module – 1 |    |                                                                                                                                                                                          | M | L  | C   |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | With usual notations , prove that $\tan \phi = r \cdot \frac{d\theta}{dr}$ .                                                                                                             | 6 | L2 | CO1 |
|            | b. | Show that the curves $r = a(1 + \cos\theta)$ and $r = b(1 - \cos\theta)$ intersects orthogonally.                                                                                        | 7 | L2 | CO1 |
|            | c. | Find the radius of curvature for the curve $x^3 + y^3 = 3axy$ at $(\frac{3a}{2}, \frac{3a}{2})$ .                                                                                        | 7 | L3 | CO1 |
| OR         |    |                                                                                                                                                                                          |   |    |     |
| Q.2        | a. | Find the angle of intersection between curves $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$ .                                                                                    | 7 | L2 | CO1 |
|            | b. | Find the pedal equation of the curve $r^m = a^m (\cos m\theta + \sin m\theta)$ .                                                                                                         | 8 | L2 | CO1 |
|            | c. | Using modern mathematical tool, write a program to plot the curve $r = 2 \cos 2\theta $ .                                                                                                | 5 | L3 | CO1 |
| Module – 2 |    |                                                                                                                                                                                          |   |    |     |
| Q.3        | a. | Expand $\sqrt{1+\sin 2x}$ using Maclaurin's series expansion upto terms containing $x^6$ .                                                                                               | 6 | L2 | CO1 |
|            | b. | If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$ . | 7 | L2 | CO1 |
|            | c. | Show that the function $f(x,y) = x^3 + y^3 - 3xy + 1$ is minimum at the point (1, 1).                                                                                                    | 7 | L3 | CO1 |
| OR         |    |                                                                                                                                                                                          |   |    |     |
| Q.4        | a. | If $u = \frac{xy}{z}$ , $v = \frac{yz}{x}$ and $w = \frac{xz}{y}$ then show that $J\left(\frac{u,v,w}{x,y,z}\right) = 4$ .                                                               | 7 | L2 | CO1 |
|            | b. | Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$ .                                                                                                          | 8 | L3 | CO1 |
|            | c. | Using modern tool write a program to evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$ .                                                                             | 5 | L3 | CO5 |

| Module – 3 |    |                                                                                                                                                                                                                                                   |   |    |     |
|------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.5        | a. | Solve $x \frac{dy}{dx} + y = x^3 y^6$ .                                                                                                                                                                                                           | 6 | L2 | CO2 |
|            | b. | Find the orthogonal trajectories of a family of curves $\frac{2a}{r} = 1 - \cos\theta$ .                                                                                                                                                          | 7 | L3 | CO2 |
|            | c. | Solve $xy(p^2) - (x^2 + y^2)p + xy = 0$ .                                                                                                                                                                                                         | 7 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                   |   |    |     |
| Q.6        | a. | Solve $(x^2 + y^2 + x)dx + xy dy = 0$ .                                                                                                                                                                                                           | 6 | L2 | CO2 |
|            | b. | A series circuit with resistance R, inductance L and electromotive force E is governed by the differential equation $L \frac{di}{dt} + Ri = E$ , where L and R are constants and initially the current i is zero. Find the current at any time t. | 7 | L3 | CO2 |
|            | c. | Solve $(px - y)(py + x) = a^2 p$ by reducing into Clairaut's form using the substitution $X = x^2$ and $Y = y^2$ .                                                                                                                                | 7 | L2 | CO2 |
| Module – 4 |    |                                                                                                                                                                                                                                                   |   |    |     |
| Q.7        | a. | Evaluate $\int_{-1}^{+1} \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$ .                                                                                                                                                                        | 6 | L2 | CO3 |
|            | b. | Change the order of integration and evaluate $\int_0^1 \int_x^{\sqrt{x}} xy dy dx$ .                                                                                                                                                              | 7 | L2 | CO3 |
|            | c. | Prove that $\beta(m, n) = \frac{\sqrt{m} \cdot \sqrt{n}}{\sqrt{m+n}}$ .                                                                                                                                                                           | 7 | L2 | CO3 |
| OR         |    |                                                                                                                                                                                                                                                   |   |    |     |
| Q.8        | a. | Evaluate $\int_0^\infty \int_0^\infty \frac{1}{e}(x^2 + y^2) dx dy$ by changing into polar coordinates.                                                                                                                                           | 6 | L2 | CO3 |
|            | b. | Evaluate $\int_0^{\pi/2} \sqrt{\cot\theta} d\theta$ by expressing in terms of gamma functions.                                                                                                                                                    | 7 | L2 | CO3 |
|            | c. | Using double integration find the area between the curves $y^2 = 4ax$ and $x^2 = 4ay$ .                                                                                                                                                           | 7 | L3 | CO3 |

| Module – 5  |           |                                                                                                                                                                                                                                         |          |                         |
|-------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------|
| <b>Q.9</b>  | <b>a.</b> | Find the rank of the matrix<br>$A = \begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$                                                                                                | <b>6</b> | <b>L2</b><br><b>CO4</b> |
|             | <b>b.</b> | Investigate the values of $\lambda$ and $\mu$ such that the system of equations $x + y + z = 6$ , $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ may have<br>i) Unique solution    ii) Infinite solution    iii) No solution.        | <b>7</b> | <b>L3</b><br><b>CO4</b> |
|             | <b>c.</b> | Using Rayleigh's power method find the largest eigen value and the corresponding eigen vector of the matrix<br>$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by taking $[1, 1, 1]^T$ as initial eigen vector. | <b>7</b> | <b>L3</b><br><b>CO4</b> |
| OR          |           |                                                                                                                                                                                                                                         |          |                         |
| <b>Q.10</b> | <b>a.</b> | Solve by using Gauss – Jordan method $x + y + z = 9$ , $x - 2y + 3z = 8$ and $2x + y - z = 3$ .                                                                                                                                         | <b>7</b> | <b>L2</b><br><b>CO4</b> |
|             | <b>b.</b> | Solve by using Gauss – Siedel method<br>$20x + y - 2z = 17$ , $3x + 20y - z = -18$ and $2x - 3y + 20z = 25$ .                                                                                                                           | <b>8</b> | <b>L2</b><br><b>CO4</b> |
|             | <b>c.</b> | Using modern mathematical model, write a program to test the consistency of the equations $x + 2y - z = 1$ , $2x + y + 4z = 2$ and $3x + 3y + 4z = 1$ .                                                                                 | <b>5</b> | <b>L3</b><br><b>CO5</b> |

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BMATE201

**Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025**  
**Mathematics – II for EEE stream**

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*  
*2. VTU Formula Hand Book is permitted.*  
*3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                         | M | L  | C   |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | If $\vec{F} = \nabla(xy^3z^2)$ find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ at the points (1, -1, 1).                                                                          | 7 | L2 | CO1 |
|            | b. | Show that $\vec{F} = (y + z)\mathbf{i} + (z + x)\mathbf{j} + (x + y)\mathbf{k}$ is irrotational. Also find a scalar function $\phi$ such that $\vec{F} = \nabla\phi$ .                  | 7 | L3 | CO1 |
|            | c. | Using Green's theorem, evaluate $\oint_C (xy + y^2)dx + x^2dy$ over the region bounded by the curves $y = x$ and $y = x^2$ .                                                            | 6 | L3 | CO1 |
| OR         |    |                                                                                                                                                                                         |   |    |     |
| Q.2        | a. | Find the directional derivatives of $\phi = x^2yz + 4xz^2$ at (1, -2, -1) along $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ .                                                              | 7 | L2 | CO1 |
|            | b. | If $\vec{F} = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$ , evaluate $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve represented by $x = t, y = t^2, z = t^3, -1 \leq t \leq 1$ .  | 7 | L2 | CO1 |
|            | c. | Using modern mathematical tools, write the code to find gradient of $\phi = x^2y + 2xz - 4$ .                                                                                           | 6 | L3 | CO5 |
| Module – 2 |    |                                                                                                                                                                                         |   |    |     |
| Q.3        | a. | Define Vector space, Subspace and Linear dependent.                                                                                                                                     | 7 | L2 | CO2 |
|            | b. | Find the dimension and basis of the subspace spanned by the vectors, (2, 4, 2), (1, -1, 0), (1, 2, 1) and (0, 3, 1) in $V_3R$ .                                                         | 7 | L2 | CO2 |
|            | c. | Prove that the following functions are linear transformation, $T : R^3 \rightarrow R^2$ defined by $T(x, y, z) = (y, -x, -z)$ .                                                         | 6 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                         |   |    |     |
| Q.4        | a. | Determine whether the vectors $V_1 = (1, 2, 3), V_2 = (3, 1, 7)$ and $V_3 = (2, 5, 8)$ are linearly dependent or linearly independent.                                                  | 7 | L2 | CO2 |
|            | b. | Let $T : V \rightarrow W$ be a linear transformation defined by, $T(x, y, z) = (x+y, x-y, 2x+z)$ . Find the range, null space, rank, nullity and hence verify the rank nullity theorem. | 7 | L2 | CO2 |

|                   |      |                                                                                                                                                                                                                                                                                                                                                                                                |      |      |      |    |    |     |   |      |      |      |      |      |   |    |     |
|-------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|----|----|-----|---|------|------|------|------|------|---|----|-----|
|                   | c.   | Using the modern mathematical tool, write a code to verify whether the following vectors (2, 1, 5, 4) and (3, 4, 7, 8) are orthogonal.                                                                                                                                                                                                                                                         | 6    | L3   | CO5  |    |    |     |   |      |      |      |      |      |   |    |     |
| <b>Module – 3</b> |      |                                                                                                                                                                                                                                                                                                                                                                                                |      |      |      |    |    |     |   |      |      |      |      |      |   |    |     |
| Q.5               | a.   | Find the Laplace transform of, $2^t + \frac{\cos 2t - \cos 3t}{t} + t \sin t$ .                                                                                                                                                                                                                                                                                                                | 7    | L2   | CO3  |    |    |     |   |      |      |      |      |      |   |    |     |
|                   | b.   | Given $f(t) = \begin{cases} E, & 0 < t < \frac{a}{2} \\ -E, & \frac{a}{2} < t < a \end{cases}$ where $f(t+a) = f(t)$ , show that $L[f(t)] = \frac{E}{S} \tanh\left(\frac{aS}{4}\right)$ .                                                                                                                                                                                                      | 7    | L3   | CO3  |    |    |     |   |      |      |      |      |      |   |    |     |
|                   | c.   | Express $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ \sin 2t, & \pi \leq t < 2\pi \\ \sin 3t, & t \geq 2\pi \end{cases}$ in terms of the Heaviside unit step function and hence find $L[f(t)]$ .                                                                                                                                                                                          | 6    | L3   | CO3  |    |    |     |   |      |      |      |      |      |   |    |     |
| <b>OR</b>         |      |                                                                                                                                                                                                                                                                                                                                                                                                |      |      |      |    |    |     |   |      |      |      |      |      |   |    |     |
| Q.6               | a.   | Find the inverse Laplace transform of, $\frac{1}{s(s+1)(s+2)(s+3)}$ .                                                                                                                                                                                                                                                                                                                          | 7    | L2   | CO3  |    |    |     |   |      |      |      |      |      |   |    |     |
|                   | b.   | Find $L^{-1}\left[\frac{S}{(S^2 + a^2)^2}\right]$ using convolution theorem.                                                                                                                                                                                                                                                                                                                   | 7    | L3   | CO3  |    |    |     |   |      |      |      |      |      |   |    |     |
|                   | c.   | Employ Laplace transform to solve the equation : $y'' + 5y' + 6y = 5e^{2t}$ , $y(0) = 2$ , $y'(0) = 1$ .                                                                                                                                                                                                                                                                                       | 6    | L3   | CO3  |    |    |     |   |      |      |      |      |      |   |    |     |
| <b>Module – 4</b> |      |                                                                                                                                                                                                                                                                                                                                                                                                |      |      |      |    |    |     |   |      |      |      |      |      |   |    |     |
| Q.7               | a.   | Using Newton-Raphson method, find the root that lies near $x = 4.5$ of the equation $\tan x = x$ . Correct to four decimal places.                                                                                                                                                                                                                                                             | 7    | L2   | CO4  |    |    |     |   |      |      |      |      |      |   |    |     |
|                   | b.   | The area of a circle (A) corresponding to diameter (D) is given below,<br><table border="1" data-bbox="331 1534 858 1608"> <tr> <td>D</td><td>80</td><td>85</td><td>90</td><td>95</td><td>100</td></tr> <tr> <td>A</td><td>5026</td><td>5674</td><td>6362</td><td>7088</td><td>7854</td></tr> </table> Find the area corresponding to diameter 105 using an appropriate interpolation formula. | D    | 80   | 85   | 90 | 95 | 100 | A | 5026 | 5674 | 6362 | 7088 | 7854 | 7 | L3 | CO4 |
| D                 | 80   | 85                                                                                                                                                                                                                                                                                                                                                                                             | 90   | 95   | 100  |    |    |     |   |      |      |      |      |      |   |    |     |
| A                 | 5026 | 5674                                                                                                                                                                                                                                                                                                                                                                                           | 6362 | 7088 | 7854 |    |    |     |   |      |      |      |      |      |   |    |     |
|                   | c.   | Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $\frac{1}{3}$ rule taking four equal parts.                                                                                                                                                                                                                                                                                            | 6    | L3   | CO4  |    |    |     |   |      |      |      |      |      |   |    |     |
| <b>OR</b>         |      |                                                                                                                                                                                                                                                                                                                                                                                                |      |      |      |    |    |     |   |      |      |      |      |      |   |    |     |
| Q.8               | a.   | Find the real root of $x \log_{10} x - 1.2 = 0$ by the method of False position. Carry out three iterations.                                                                                                                                                                                                                                                                                   | 7    | L2   | CO4  |    |    |     |   |      |      |      |      |      |   |    |     |

|                   |           |                                                                                                                                                                                                               |          |           |            |   |    |   |    |    |    |    |          |           |            |
|-------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|------------|---|----|---|----|----|----|----|----------|-----------|------------|
|                   | <b>b.</b> | Use Lagrange's interpolation formula to find y at x = 10 given,<br><table><tr><td>x</td><td>5</td><td>6</td><td>9</td><td>11</td></tr><tr><td>y</td><td>12</td><td>13</td><td>14</td><td>16</td></tr></table> | x        | 5         | 6          | 9 | 11 | y | 12 | 13 | 14 | 16 | <b>7</b> | <b>L2</b> | <b>CO4</b> |
| x                 | 5         | 6                                                                                                                                                                                                             | 9        | 11        |            |   |    |   |    |    |    |    |          |           |            |
| y                 | 12        | 13                                                                                                                                                                                                            | 14       | 16        |            |   |    |   |    |    |    |    |          |           |            |
|                   | <b>c.</b> | Evaluate $\int_0^1 \frac{dx}{1+x}$ taking seven ordinates by applying Simpson's $\frac{3}{8}$ rule.                                                                                                           | <b>6</b> | <b>L3</b> | <b>CO4</b> |   |    |   |    |    |    |    |          |           |            |
| <b>Module – 5</b> |           |                                                                                                                                                                                                               |          |           |            |   |    |   |    |    |    |    |          |           |            |
| <b>Q.9</b>        | <b>a.</b> | Use Taylor's series method to find y at x = 0.1 considering upto the third degree given that $\frac{dy}{dx} = x^2 + y^2$ and y(0) = 1.                                                                        | <b>7</b> | <b>L2</b> | <b>CO4</b> |   |    |   |    |    |    |    |          |           |            |
|                   | <b>b.</b> | Given $\frac{dy}{dx} = 3x + \frac{y}{2}$ , y(0) = 1. Compute y(0.2) by taking h = 0.2, using Runge-Kutta method of fourth order.                                                                              | <b>7</b> | <b>L2</b> | <b>CO4</b> |   |    |   |    |    |    |    |          |           |            |
|                   | <b>c.</b> | Given that $\frac{dy}{dx} = x - y^2$ and the data y(0) = 0, y(0.2) = 0.02, y(0.4) = 0.0795, y(0.6) = 0.1762. Compute y at x = 0.8 by using Milne's method.                                                    | <b>6</b> | <b>L2</b> | <b>CO4</b> |   |    |   |    |    |    |    |          |           |            |
| <b>OR</b>         |           |                                                                                                                                                                                                               |          |           |            |   |    |   |    |    |    |    |          |           |            |
| <b>Q.10</b>       | <b>a.</b> | Using modified Euler's formula, compute y at x = 0.2 given that $\frac{dy}{dx} = x + y$ , y(0) = 1 and h = 0.2                                                                                                | <b>7</b> | <b>L3</b> | <b>CO4</b> |   |    |   |    |    |    |    |          |           |            |
|                   | <b>b.</b> | Use fourth order Runge-Kutta method to compute y(1.1) given that $\frac{dy}{dx} = xy^{\frac{1}{3}}$ , y(1) = 1 and h = 0.1                                                                                    | <b>7</b> | <b>L2</b> | <b>CO4</b> |   |    |   |    |    |    |    |          |           |            |
|                   | <b>c.</b> | Using modern mathematical tools, write the code to find the solution of $\frac{dy}{dx} = 1 + \frac{y}{x}$ at y(2) taking h = 0.2 and y(1) = 2 by Runge-Kutta method of order four.                            | <b>6</b> | <b>L3</b> | <b>CO5</b> |   |    |   |    |    |    |    |          |           |            |

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## First Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Mathematics–I for ME Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                | M  | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Prove with usual notations, $\tan \phi = r \cdot \frac{d\theta}{dr}$                                                                                                           | 06 | L2 | CO1 |
|            | b. | Find the angle between the curves,<br>$r = a(1 + \cos \theta)$ and $r = b(1 - \cos \theta)$                                                                                    | 07 | L2 | CO1 |
|            | c. | Find the pedal equation to the curves<br>$r^n = a^n \cos n\theta$                                                                                                              | 07 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                |    |    |     |
| Q.2        | a. | Prove that for the radius of curvature in Cartesian form<br>$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$                                                                             | 07 | L2 | CO1 |
|            | b. | Find the radius of curvature for the curve $x^3 + y^3 = 3axy$ at the point<br>$\left(\frac{3a}{2}, \frac{3a}{2}\right)$ on it.                                                 | 08 | L2 | CO1 |
|            | c. | Using modern mathematical tools write the code to find the radius of curvature<br>$r = 4(1 + \cos t)$ at $t = \pi/2$                                                           | 05 | L3 | CO5 |
| Module – 2 |    |                                                                                                                                                                                |    |    |     |
| Q.3        | a. | Expand $e^{\sin x}$ by Maclaurin's series upto the terms containing $x^4$ .                                                                                                    | 06 | L2 | CO2 |
|            | b. | Evaluate : $\lim_{x \rightarrow 0} \left( \frac{a^x + b^x + c^x + d^x}{4} \right)^{1/x}$                                                                                       | 07 | L2 | CO2 |
|            | c. | If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then prove that<br>$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x + y + z}$ | 07 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                |    |    |     |
| Q.4        | a. | If $u = f(x - y, y - z, z - x)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$                                  | 07 | L2 | CO2 |
|            | b. | If $u = \frac{yz}{z}, v = \frac{zx}{y}, w = \frac{xy}{z}$ , show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$                                                        | 08 | L2 | CO2 |
|            | c. | Using modern mathematical tools, write the code to solve :<br>$y'' - 5y' + 6y = \cos 4x$                                                                                       | 05 | L3 | CO5 |
| Module – 3 |    |                                                                                                                                                                                |    |    |     |
| Q.5        | a. | Solve : $xy(1 + xy^2) \frac{dy}{dx} = 1$                                                                                                                                       | 06 | L2 | CO3 |
|            | b. | Solve : $(x^2 + y^2 + x)dx + xy dy = 0$                                                                                                                                        | 07 | L2 | CO3 |
|            | c. | Find the orthogonal trajectories of the family of curves<br>$\frac{x^2}{a^2} + \frac{y^2}{b^2 + \lambda} = 1$ where $\lambda$ is the parameter.                                | 07 | L2 | CO3 |

| OR         |    |                                                                                                                                                                                                                               |    |    |     |
|------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.6        | a. | Solve : $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$                                                                                                                                                           | 06 | L2 | CO3 |
|            | b. | Solve the equation $(px - y)(py + x) = 2$ by reducing in to Clairaut's form, taking the substitution. $X = x^2$ , $Y = y^2$ .                                                                                                 | 07 | L2 | CO3 |
|            | c. | If the temperature of the air is 30°C and a metal ball cools from 100°C to 70°C in 15 minutes, find how long will it take for the metal ball to reach a temperature of 40°C.                                                  | 07 | L3 | CO3 |
| Module – 4 |    |                                                                                                                                                                                                                               |    |    |     |
| Q.7        | a. | Solve : $(4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$                                                                                                                                                                               | 06 | L2 | CO3 |
|            | b. | Solve : $\frac{d^2y}{dx^2} - 4y = \cosh(2x - 1) + 3^x$                                                                                                                                                                        | 07 | L2 | CO3 |
|            | c. | Solve : $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 4\cos^2 x$                                                                                                                                                                 | 07 | L2 | CO3 |
| OR         |    |                                                                                                                                                                                                                               |    |    |     |
| Q.8        | a. | Solve : $\frac{d^2y}{dx^2} + y = \tan x$ by the method of variation of parameters.                                                                                                                                            | 06 | L2 | CO3 |
|            | b. | Solve : $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 4y = (1 + x)^2$ , using Cauchy's equation.                                                                                                                                | 07 | L3 | CO3 |
|            | c. | Solve the Legendre's linear equation<br>$(1 + x)^2 \frac{d^2y}{dx^2} + (1 + x) \frac{dy}{dx} + y = 2 \sin \log(1 + x)$                                                                                                        | 07 | L3 | CO3 |
| Module – 5 |    |                                                                                                                                                                                                                               |    |    |     |
| Q.9        | a. | Find the rank of a matrix by elementary row transformation<br>$A = \begin{bmatrix} 4 & 0 & 2 & 1 \\ 2 & 1 & 3 & 4 \\ 2 & 3 & 4 & 7 \\ 2 & 3 & 1 & 4 \end{bmatrix}$                                                            | 06 | L2 | CO4 |
|            | b. | Investigate the values of $\lambda$ and $\mu$ such that the system of equations:<br>$x + y + z = 6$ ; $x + 2y + 3z = 10$ ; $x + 2y + \lambda z = \mu$                                                                         | 07 | L2 | CO4 |
|            | c. | Solve the following system of equations by Gauss-Jordan method.<br>$x + y + z = 9$ ; $x - 2y + 3z = 8$ ; $2x + y - z = 3$                                                                                                     | 07 | L2 | CO4 |
| OR         |    |                                                                                                                                                                                                                               |    |    |     |
| Q.10       | a. | Using Rayleigh's power method, find numerically the largest eigen value and the corresponding eigen vectors of the matrix<br>$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$                      | 07 | L2 | CO4 |
|            | b. | Solve the following system of equations by Gauss-Seidel method.<br>$10x + y + z = 12$ ; $x + 10y + z = 12$ ; $x + y + 10z = 12$                                                                                               | 08 | L2 | CO4 |
|            | c. | Using modern mathematical tools, write the code to check whether the following system of homogeneous linear equation has non-trivial solution:<br>$x_1 + 2x_2 - x_3 = 0$ ; $2x_1 + x_2 + 4x_3 = 0$ ; $3x_1 + 3x_2 + 4x_3 = 0$ | 05 | L3 | CO5 |

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BMATM201

**Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025**  
**Mathematics – II for Mechanical Engineering Stream**

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*  
*2. VTU Formula Hand Book is permitted.*  
*3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                        | M | L  | C   |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Evaluate $\int_{-1}^1 \int_0^2 \int_{x-2}^{x+2} (x + y + z) \, dy \, dx \, dz$ .                                                                                       | 7 | L3 | CO1 |
|            | b. | Evaluate $\int_0^1 \int_{\sqrt{y}}^1 dx \, dy$ by changing the order of integration.                                                                                   | 7 | L3 | CO1 |
|            | c. | With usual notation show that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m + n)}$ .                                                                               | 6 | L2 | CO1 |
| OR         |    |                                                                                                                                                                        |   |    |     |
| Q.2        | a. | Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} x^3 y \, dx \, dy$ .                                                                                                          | 7 | L3 | CO1 |
|            | b. | Evaluate $\iint xy(x + y) \, dy \, dx$ taken over the area between $y = x^2$ and $y = x$ .                                                                             | 7 | L2 | CO1 |
|            | c. | Write a modern mathematical program to evaluate the integral $\int_0^3 \int_0^{3-x} \int_0^{3-x-y} (xyz) \, dz \, dy \, dx$ .                                          | 6 | L3 | CO5 |
| Module – 2 |    |                                                                                                                                                                        |   |    |     |
| Q.3        | a. | Find the angle between the surfaces $xy^2z = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point (1, -2, 1).                                                              | 7 | L2 | CO2 |
|            | b. | If $\vec{F} = \nabla(xy^3z^2)$ find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ at the point (1, -1, 1).                                                          | 7 | L2 | CO2 |
|            | c. | Define solenoidal vector. Find the values of a, b, c such that $\vec{F} = (x + y + az)\mathbf{i} + (bx + 2y - z)\mathbf{j} + (x + cy + 2z)\mathbf{k}$ is irrotational. | 6 | L2 | CO2 |
| OR         |    |                                                                                                                                                                        |   |    |     |
| Q.4        | a. | Apply Green's theorem to evaluate $\int (3x - 8y^2) \, dx + (4y - 6xy) \, dy$ , where C is the boundary of the region bounded by $x = 0, y = 0, x + y = 1$ .           | 7 | L3 | CO2 |

|            |     |                                                                                                                                                                                                                                                                                                        |     |     |     |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|----|----|------|------|-----|-----|-----|-----|-----|-----|-----|----|-----|
|            | b.  | Use Stoke's theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$ , where $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken round the rectangle bounded by $x = 0$ , $x = a$ , $y = 0$ and $y = b$ .                                                                                                     | 7   | L3  | CO2 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | c.  | Write the modern mathematical tool program to find the divergence of the vector field $\vec{F} = x^2yz\vec{i} + y^2zx\vec{j} + z^2xy\vec{k}$ .                                                                                                                                                         | 6   | L3  | CO5 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| Module – 3 |     |                                                                                                                                                                                                                                                                                                        |     |     |     |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| Q.5        | a.  | Form the PDE by eliminating the arbitrary function from the relation $f(x + y + z, x^2 + y^2 - z^2) = 0$ .                                                                                                                                                                                             | 7   | L2  | CO3 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | b.  | Solve $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y}$ , subject to the conditions $\frac{\partial z}{\partial x} = \log_e x$ when $y = 1$ and $z = 0$ when $x = 1$ .                                                                                                                        | 7   | L3  | CO3 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | c.  | Derive one-dimensional heat equation.                                                                                                                                                                                                                                                                  | 6   | L2  | CO3 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| OR         |     |                                                                                                                                                                                                                                                                                                        |     |     |     |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| Q.6        | a.  | Form the PDE by eliminating the arbitrary function from the relation $z = e^y f(x + y)$ .                                                                                                                                                                                                              | 7   | L2  | CO3 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | b.  | Solve $\frac{\partial^2 z}{\partial x^2} + z = 0$ given that when $x = 0$ , $z = e^y$ and $\frac{\partial z}{\partial x} = 1$ .                                                                                                                                                                        | 7   | L3  | CO3 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | c.  | Solve $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$ using Lagrange's multiplier.                                                                                                                                                                                                                      | 6   | L3  | CO3 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| Module – 4 |     |                                                                                                                                                                                                                                                                                                        |     |     |     |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| Q.7        | a.  | Find a real root of $x^3 - 9x + 1 = 0$ in $(2, 3)$ by the Regula-Falsi method in three iterations.                                                                                                                                                                                                     | 7   | L3  | CO4 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | b.  | Use an appropriate interpolation formula to compute $f(42)$ using the following data:<br><table border="1"><tr><td>x</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td><td>90</td></tr><tr><td>f(x)</td><td>184</td><td>204</td><td>226</td><td>250</td><td>276</td><td>304</td></tr></table> | x   | 40  | 50  | 60  | 70 | 80 | 90   | f(x) | 184 | 204 | 226 | 250 | 276 | 304 | 7   | L3 | CO4 |
| x          | 40  | 50                                                                                                                                                                                                                                                                                                     | 60  | 70  | 80  | 90  |    |    |      |      |     |     |     |     |     |     |     |    |     |
| f(x)       | 184 | 204                                                                                                                                                                                                                                                                                                    | 226 | 250 | 276 | 304 |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | c.  | Using Simpson's $\frac{1}{3}$ <sup>rd</sup> rule, evaluate $\int_0^{0.6} e^{-x^2} dx$ by considering seven ordinates.                                                                                                                                                                                  | 6   | L3  | CO4 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| OR         |     |                                                                                                                                                                                                                                                                                                        |     |     |     |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| Q.8        | a.  | Find a real root of the equation $x \sin x + \cos x = 0$ near $x = \pi$ correct to four decimal places using Newtons-Raphson method.                                                                                                                                                                   | 7   | L3  | CO4 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
|            | b.  | Using Lagrange's interpolation formula. Find $f(5)$ from the following data :<br><table border="1"><tr><td>x</td><td>1</td><td>3</td><td>4</td><td>6</td><td>9</td></tr><tr><td>f(x)</td><td>3</td><td>9</td><td>30</td><td>132</td><td>156</td></tr></table>                                          | x   | 1   | 3   | 4   | 6  | 9  | f(x) | 3    | 9   | 30  | 132 | 156 | 7   | L3  | CO4 |    |     |
| x          | 1   | 3                                                                                                                                                                                                                                                                                                      | 4   | 6   | 9   |     |    |    |      |      |     |     |     |     |     |     |     |    |     |
| f(x)       | 3   | 9                                                                                                                                                                                                                                                                                                      | 30  | 132 | 156 |     |    |    |      |      |     |     |     |     |     |     |     |    |     |

|                   |    |                                                                                                                                                                                   |   |    |     |
|-------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|                   | c. | Use Simpson's $\frac{3}{8}$ rule to obtain the approximate value of $\int_0^{0.3} (1-8x^3)^{\frac{1}{2}} dx$ , by considering 3 equal intervals.                                  | 6 | L3 | CO4 |
| <b>Module – 5</b> |    |                                                                                                                                                                                   |   |    |     |
| <b>Q.9</b>        | a. | Use Taylor's method to find $x = 0.1$ considering terms up to the third degree given that $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$ .                                            | 7 | L2 | CO4 |
|                   | b. | Using Runge-Kutta method of order 4, find $y$ at $x = 0.1$ given that $\frac{dy}{dx} = 3e^x + 2y$ , $y(0) = 1$ .                                                                  | 7 | L3 | CO4 |
|                   | c. | Given that $\frac{dy}{dx} = x - y^2$ and the data $y(0) = 0$ , $y(0.2) = 0.02$ , $y(0.4) = 0.0795$ , $y(0.6) = 0.1762$ . Compute $y$ at $x = 0.8$ by applying Milne's method.     | 6 | L3 | CO4 |
| <b>OR</b>         |    |                                                                                                                                                                                   |   |    |     |
| <b>Q.10</b>       | a. | Using modified Euler's method, find $y$ at $x = 0.2$ given that $\frac{dy}{dx} = 3x + \frac{1}{2}y$ with $y(0) = 1$ taking $h = 0.1$ . Perform three iterations.                  | 7 | L3 | CO4 |
|                   | b. | Using Runge-Kutta method of fourth order, find $y(0.2)$ for the equation, $\frac{dy}{dx} = \frac{y-x}{y+x}$ , $y(0) = 1$ taking $h = 0.2$ .                                       | 7 | L3 | CO4 |
|                   | c. | Using modern mathematical tools write a program to solve $\frac{dy}{dx} = 1 + \left(\frac{y}{x}\right)$ at $y(2)$ taking $h = 0.2$ . Given that $y(1) = 2$ by Runge-Kutta method. | 6 | L3 | CO5 |

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## First Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

### Mathematics – I for CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.

| Module – 1 |    |                                                                                                                                                                                                   | M | L  | C   |
|------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.1        | a. | Find the angle between the curves, $r = \frac{a}{1 + \cos \theta}$ and $r = \frac{b}{1 - \cos \theta}$ .                                                                                          | 6 | L2 | CO1 |
|            | b. | Find the pedal equations of the curve $r^m = a^m \cos(m\theta)$ .                                                                                                                                 | 7 | L2 | CO1 |
|            | c. | Determine the radius of curvature of the curve $r^2 \sec(2\theta) = a^2$ .                                                                                                                        | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                                   |   |    |     |
| Q.2        | a. | With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$ .                                                                                                                               | 8 | L2 | CO1 |
|            | b. | Show that tangents to the cardioid $r = a(1 + \cos \theta)$ at the points $\theta = \frac{\pi}{3}$ and $\theta = \frac{2\pi}{3}$ are respectively parallel and perpendicular to the initial line. | 7 | L2 | CO1 |
|            | c. | Using modern mathematical tool write a programme/code to plot $r = 2 \cos 2\theta $ .                                                                                                             | 5 | L3 | CO5 |
| Module – 2 |    |                                                                                                                                                                                                   |   |    |     |
| Q.3        | a. | Expand $\sqrt{1 + \sin 2x}$ as Maclaurin's series up to fourth degree terms.                                                                                                                      | 6 | L2 | CO1 |
|            | b. | If $u = f(y - z, z - x, x - y)$ , prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ .                                                | 7 | L2 | CO1 |
|            | c. | Compute $J = \frac{\partial(x, y, z)}{\partial(\rho, \phi, z)}$ for $x = \rho \cos \phi$ , $y = \rho \sin \phi$ and $z = z$                                                                       | 7 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                                   |   |    |     |
| Q.4        | a. | If $u = e^{(ax+by)} f(ax - by)$ , prove that $b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2abu$ .                                                                         | 8 | L2 | CO1 |
|            | b. | Prove that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$ for $u = f\left(\frac{y - x}{xy}, \frac{z - x}{xz}\right)$             | 7 | L2 | CO1 |
|            | c. | Using modern mathematical tool write a programe/code to show that $u_{xx} + u_{yy} = 0$ , given that $u = e^x(x \cos y - y \sin y)$ .                                                             | 5 | L2 | CO5 |
| Module – 3 |    |                                                                                                                                                                                                   |   |    |     |
| Q.5        | a. | Solve $\left[y\left(1 + \frac{1}{x}\right) + \cos y\right]dx + [x + \log x - x \sin y]dy = 0$                                                                                                     | 6 | L2 | CO2 |
|            | b. | Show that the curve $y^2 = 4a(x + a)$ is self-orthogonal.                                                                                                                                         | 7 | L3 | CO2 |
|            | c. | A 12-volts battery connected to a series circuit in which the inductance is $\frac{1}{2}$ henry and resistance is 10 ohms. Find the current 'i' if the initial current is zero.                   | 7 | L3 | CO2 |

| OR         |    |                                                                                                                                                                                                       |   |    |     |
|------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| Q.6        | a. | Solve $x \frac{dy}{dx} + y = x^3 y^6$ .                                                                                                                                                               | 6 | L2 | CO2 |
|            | b. | Find orthogonal trajectories of the family $r^n \cos n\theta = a^n$ .                                                                                                                                 | 7 | L3 | CO2 |
|            | c. | Find the general solutions of the equations $(px - y)(py + x) = a^2 P$ by reducing into Clairaut's form by taking $u = x^2$ , $v = y^2$ .                                                             | 7 | L2 | CO2 |
| Module – 4 |    |                                                                                                                                                                                                       |   |    |     |
| Q.7        | a. | Find remainder when $(349 \times 74 \times 36)$ is divided by 3.                                                                                                                                      | 6 | L1 | CO3 |
|            | b. | Solve linear Diophantine equations $13x + 17y = 5$ .                                                                                                                                                  | 7 | L2 | CO3 |
|            | c. | Solve the system of linear congruence $x \equiv 2(\text{mod}3)$ , $x \equiv 3(\text{mod}5)$ and $x \equiv 2(\text{mod}7)$ , using remainder theorem.                                                  | 7 | L2 | CO3 |
| OR         |    |                                                                                                                                                                                                       |   |    |     |
| Q.8        | a. | Find the last digit in $7^{126}$ .                                                                                                                                                                    | 6 | L2 | CO3 |
|            | b. | Solve $2x + 6y \equiv 1(\text{mod}7)$<br>$4x + 3y \equiv 2(\text{mod}7)$                                                                                                                              | 7 | L2 | CO3 |
|            | c. | Find the remainder when $7^{121}$ is divisible by 13.                                                                                                                                                 | 7 | L2 | CO3 |
| Module – 5 |    |                                                                                                                                                                                                       |   |    |     |
| Q.9        | a. | Solve the system of equation by using Gauss-Jordan method.<br>$x + y + z = 9$ , $2x + y - z = 0$ , $2x + 5y + 7z = 52$ .                                                                              | 6 | L2 | CO4 |
|            | b. | For what values $\lambda$ and $\mu$ the system of equations, $x + y + z = 6$ , $x + 2y + 3z = 10$ , $x + 2y + \lambda z = \mu$ has (i) no solution (ii) a unique solution and (iii) Many solutions.   | 7 | L2 | CO4 |
|            | c. | Using power method, find the largest eigen value and corresponding vector of the matrix,<br>$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ .                             | 7 | L2 | CO4 |
| OR         |    |                                                                                                                                                                                                       |   |    |     |
| Q.10       | a. | Determine the rank of the matrix $A = \begin{bmatrix} 91 & 92 & 93 & 94 & 95 \\ 92 & 93 & 94 & 95 & 96 \\ 93 & 94 & 95 & 96 & 97 \\ 94 & 95 & 96 & 97 & 98 \\ 95 & 96 & 97 & 98 & 99 \end{bmatrix}$ . | 8 | L1 | CO4 |
|            | b. | Using the Gauss-Seidel iteration method, solve the equation $27x + 6y - z = 85$ , $6x + 15y + 2z = 72$ , $x + y + 54z = 110$ . Carry out four iterations.                                             | 7 | L2 | CO4 |
|            | c. | Using modern mathematical tool, write a program/code to find the largest eigen value of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ by power method.                      | 5 | L3 | CO5 |

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## Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Mathematics – II for CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                          | M  | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Evaluate $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dz dy dx$                                                                                | 07 | L2 | CO1 |
|            | b. | Prove that $\beta(m, n) = \frac{m \cdot n}{(m+n)}$                                                                                                       | 07 | L2 | CO1 |
|            | c. | Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx$ by changing to polar coordinates.                                                            | 06 | L3 | CO1 |
| OR         |    |                                                                                                                                                          |    |    |     |
| Q.2        | a. | Evaluate $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} xy dy dx$ by change the order of integration.                                                   | 07 | L2 | CO1 |
|            | b. | Show that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \pi$                                     | 07 | L2 | CO1 |
|            | c. | Write a program to find the volume of the tetrahedron bounded by the planes $x = 0, y = 0, z = 0, \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ .         | 06 | L3 | CO5 |
| Module – 2 |    |                                                                                                                                                          |    |    |     |
| Q.3        | a. | Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at the point $(2, -1, 2)$ along $2\hat{i} - 3\hat{j} + 6\hat{k}$ .                          | 07 | L2 | CO2 |
|            | b. | Find the value of the constants a, b, c such that $\vec{F} = (x + y + az)\hat{i} + (bx + 2y - z)\hat{j} + (x + cy + 2z)\hat{k}$ is irrotational.         | 07 | L2 | CO2 |
|            | c. | Show that the cylindrical co-ordinate system is orthogonal.                                                                                              | 06 | L3 | CO2 |
| OR         |    |                                                                                                                                                          |    |    |     |
| Q.4        | a. | Find the value of the constants 'a' such that the vector field, $\vec{F} = (axy - z^3)\hat{i} + (a - 2)x^2\hat{j} + (1 - a)xz^2\hat{k}$ is irrotational. | 07 | L2 | CO2 |
|            | b. | Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$ .                                            | 07 | L2 | CO2 |
|            | c. | Write a program to verify whether the following vectors $(2, 1, 5, 4)$ and $(3, 4, 7, 8)$ are orthogonal.                                                | 06 | L3 | CO5 |

| Module – 3      |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                  |         |         |         |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------|---------|---------|---------|-----------------|----|-----|----|-----|-----|-----|------|-----|----|-----|
| Q.5             | a.                                                                                                                                                                                           | Express the matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & -2 \end{bmatrix}$ in the vector spaces of $2 \times 2$ matrices as a linear combination of $B = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ , $C = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ , $D = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ | 07      | L2      | CO3     |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
|                 | b.                                                                                                                                                                                           | Determine whether the vectors $V_1 = (1, 2, 3)$ , $V_2 = (3, 1, 7)$ and $V_3 = (2, 5, 8)$ are linearly dependent or linearly independent.                                                                                                                                                                        | 07      | L2      | CO3     |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
|                 | c.                                                                                                                                                                                           | Verify the rank nullity theorem for the linear transformation $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$                                                                                                                                                                | 06      | L3      | CO3     |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| OR              |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                  |         |         |         |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| Q.6             | a.                                                                                                                                                                                           | Let W be the subspace of $R^5$ spanned by $x_1 = (1, 2, -1, 3, 4)$ , $x_2 = (2, 4, -2, 6, 8)$ , $x_3 = (1, 3, 2, 2, 6)$ , $x_4 = (1, 4, 5, 1, 8)$ and $x_5 = (2, 7, 3, 3, 9)$ . Find a subset of vectors which forms a basis of W.                                                                               | 07      | L2      | CO3     |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
|                 | b.                                                                                                                                                                                           | Consider the following polynomials in p(t) and inner product :<br>$f(t) = t + 2$ , $g(t) = 3t - 2$ $h(t) = t^3 - 2t - 3$ and $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$ .<br>(i) Find $\langle f, g \rangle$ and $\langle f, h \rangle$<br>(ii) Find $\  f \ $ and $\  g \ $                                  | 07      | L2      | CO3     |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
|                 | c.                                                                                                                                                                                           | If V is a vector space of polynomials over R. Find a basis and dimension of the subspaces W and V, spanned by the polynomials.<br>$x_1 = t^3 - 2t^2 + 4t + 1$ , $x_2 = 2t^3 - 3t^2 + 9t - 1$<br>$x_3 = t^3 + 6t - 5$ , $x_4 = 2t^3 - 5t^2 + 7t + 5$                                                              | 06      | L2      | CO3     |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| Module – 4      |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                  |         |         |         |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| Q.7             | a.                                                                                                                                                                                           | Find the real root of the equation $x \log_{10} x - 1.2 = 0$ by Regular Falsi method. Correct to four decimal places.                                                                                                                                                                                            | 07      | L2      | CO4     |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
|                 | b.                                                                                                                                                                                           | From the following table find the number of students who have obtained less than 45 marks. <table><tr><td>Marks</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td><td>70 – 80</td></tr><tr><td>No. of students</td><td>31</td><td>42</td><td>51</td><td>35</td><td>31</td></tr></table>        | Marks   | 30 – 40 | 40 – 50 | 50 – 60 | 60 – 70 | 70 – 80 | No. of students | 31 | 42  | 51 | 35  | 31  | 07  | L2   | CO4 |    |     |
|                 | Marks                                                                                                                                                                                        | 30 – 40                                                                                                                                                                                                                                                                                                          | 40 – 50 | 50 – 60 | 60 – 70 | 70 – 80 |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| No. of students | 31                                                                                                                                                                                           | 42                                                                                                                                                                                                                                                                                                               | 51      | 35      | 31      |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| c.              | Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $(1/3)^{rd}$ rule taking four equal strips.                                                                                          | 06                                                                                                                                                                                                                                                                                                               | L3      | CO4     |         |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| OR              |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                  |         |         |         |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| Q.8             | a.                                                                                                                                                                                           | Fit the polynomial for the following data using Newton's divided difference formula and hence find f(3). <table><tr><td>x</td><td>2</td><td>4</td><td>5</td><td>6</td><td>8</td><td>10</td></tr><tr><td>y</td><td>10</td><td>96</td><td>196</td><td>350</td><td>868</td><td>1746</td></tr></table>               | x       | 2       | 4       | 5       | 6       | 8       | 10              | y  | 10  | 96 | 196 | 350 | 868 | 1746 | 07  | L2 | CO4 |
|                 | x                                                                                                                                                                                            | 2                                                                                                                                                                                                                                                                                                                | 4       | 5       | 6       | 8       | 10      |         |                 |    |     |    |     |     |     |      |     |    |     |
| y               | 10                                                                                                                                                                                           | 96                                                                                                                                                                                                                                                                                                               | 196     | 350     | 868     | 1746    |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| b.              | Using Lagrange's interpolation formula find f(4). <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>y</td><td>- 4</td><td>2</td><td>14</td><td>158</td></tr></table> | x                                                                                                                                                                                                                                                                                                                | 0       | 2       | 3       | 6       | y       | - 4     | 2               | 14 | 158 | 07 | L2  | CO4 |     |      |     |    |     |
| x               | 0                                                                                                                                                                                            | 2                                                                                                                                                                                                                                                                                                                | 3       | 6       |         |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |
| y               | - 4                                                                                                                                                                                          | 2                                                                                                                                                                                                                                                                                                                | 14      | 158     |         |         |         |         |                 |    |     |    |     |     |     |      |     |    |     |

|                   |           |                                                                                                                                                                                                                                  |           |           |            |
|-------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|------------|
| <b>Q.8</b>        | <b>c.</b> | Use Simpson's (3/8) <sup>th</sup> rule to evaluate $\int_1^4 e^{1/x} dx$ by taking four ordinates.                                                                                                                               | <b>06</b> | <b>L3</b> | <b>CO4</b> |
| <b>Module – 5</b> |           |                                                                                                                                                                                                                                  |           |           |            |
| <b>Q.9</b>        | <b>a.</b> | Employ Taylor's series method to solve the initial value problem $\frac{dy}{dx} = x - y^2$ ; $y(0) = 1$ at the point $x = 0.1$ by considering upto 4 <sup>th</sup> degree terms.                                                 | <b>07</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b.</b> | Apply Milne's method to compute $y(1.4)$ for the differential equation $\frac{dy}{dx} = x^2 + \frac{y}{2}$ , given that $y(1) = 2$ , $y(1.1) = 2.2156$ , $y(1.3) = 2.4649$ and $y(1.3) = 2.7514$ correct to four decimal places. | <b>07</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>c.</b> | Use fourth order Runge Kutta method to find the value of $y$ at $x = 0.1$ , given that $\frac{dy}{dx} = 3e^x + 2y$ , $y(0) = 0$ and $h = 0.1$ .                                                                                  | <b>06</b> | <b>L2</b> | <b>CO4</b> |
| <b>OR</b>         |           |                                                                                                                                                                                                                                  |           |           |            |
| <b>Q.10</b>       | <b>a.</b> | Use Modified Euler's method to compute $y(0.1)$ , given that $\frac{dy}{dx} = x^2 + y$ ; $y(0) = 1$ by taking $h = 0.05$ .                                                                                                       | <b>07</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>b.</b> | If $\frac{dy}{dx} = 2e^x - y$ ; $y(0) = 2$ , $y(0.1) = 2.010$ , $y(0.2) = 2.040$ and $y(0.3) = 2.090$ . Find the value of $y$ at $x = 0.4$ correct to four decimal places by applying Milne's predictor and corrector method.    | <b>07</b> | <b>L2</b> | <b>CO4</b> |
|                   | <b>c.</b> | Write a program to solve : $\frac{dy}{dx} - 2y = 3e^x$ with $y(0) = 0$ using Taylor's series method at $x_1 = 0.1$ , $x_2 = 0.2$ and $x_3 = 0.3$ .                                                                               | <b>06</b> | <b>L3</b> | <b>CO5</b> |

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## First/Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

### Applied Physics for Civil Engg Stream

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. VTU Formula Hand Book is permitted.  
3. M : Marks , L: Bloom's level , C: Course outcomes.

| Module – 1 |    |                                                                                                                                                                                                          | M  | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Define Stiffness factor of a spring and hence derive expressions for the effective spring constant of springs in series and parallel combinations.                                                       | 07 | L2 | CO1 |
|            | b. | Explain various forces acting on a system under damped oscillations, setup differential equation and assuming the solution mention the variation of amplitude with respect to time.                      | 08 | L2 | CO1 |
|            | c. | Calculate the mach angle if the object is travelling with a speed of 660 m/s given the speed of sound in the medium is 330 m/s.                                                                          | 05 | L3 | CO1 |
| <b>OR</b>  |    |                                                                                                                                                                                                          |    |    |     |
| Q.2        | a. | Mention four mach regimes and explain the construction and working of Reddy shock tube with the help of neat sketch.                                                                                     | 09 | L3 | CO1 |
|            | b. | Explain the types of springs and their applications.                                                                                                                                                     | 06 | L2 | CO1 |
|            | c. | Evaluate the resonance frequency of a spring of force constant 2467 Nm carrying a mass of 100 gm.                                                                                                        | 05 | L3 | CO5 |
| Module – 2 |    |                                                                                                                                                                                                          |    |    |     |
| Q.3        | a. | Mention the types of engineering materials and with figure explain brittle and ductile fractures in engineering materials.                                                                               | 09 | L2 | CO1 |
|            | b. | Define a beam and explain the types of beams.                                                                                                                                                            | 06 | L2 | CO1 |
|            | c. | A metal cube of side 0.20 m is subjected to a shearing force of 4000 N. The top surface is displaced through 0.50 cm with respect to the bottom. Calculate the shear modulus of elasticity of the metal. | 05 | L3 | CO5 |
| <b>OR</b>  |    |                                                                                                                                                                                                          |    |    |     |
| Q.4        | a. | Describe stress hardening and stress softening.                                                                                                                                                          | 06 | L2 | CO1 |
|            | b. | Define Poisson's ratio, Young's modulus and rigidity modulus and derive the relation between them.                                                                                                       | 09 | L2 | CO1 |
|            | c. | Calculate the force required to produce an extension of 1mm in wire made of material with Young's modulus 100 GPa and of length 1 m and diameter 1 mm.                                                   | 05 | L3 | CO5 |

| Module – 3 |    |                                                                                                                                                                                                                                                                                                                   |    |    |     |
|------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.5        | a. | Give the qualitative explanation of radiometric quantities such as radiant energy, radiant power, radiant intensity, radiance, radiant existence etc along with respective equations.                                                                                                                             | 10 | L2 | CO2 |
|            | b. | Describe reflectance and transmittance.                                                                                                                                                                                                                                                                           | 05 | L2 | CO2 |
|            | c. | Explain the impact of noise in multi-storied buildings.                                                                                                                                                                                                                                                           | 05 | L2 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                                                                                   |    |    |     |
| Q.6        | a. | Define reverberation and reverberation time and hence derive Sabine's formula.                                                                                                                                                                                                                                    | 09 | L2 | CO2 |
|            | b. | Explain Cosine law and Inverse square law.                                                                                                                                                                                                                                                                        | 06 | L2 | CO2 |
|            | c. | The reverberation time is found to be 1.5 sec for an empty hall and it is found to be 1 sec when a curtain cloth of $20 \text{ m}^2$ is suspended at the center of the hall. If the dimensions of the hall are $10 \times 8 \times 6 \text{ m}^3$ , calculate the coefficient of absorption of the curtain cloth. | 05 | L3 | CO2 |
| Module – 4 |    |                                                                                                                                                                                                                                                                                                                   |    |    |     |
| Q.7        | a. | Enumerate the requisites of a laser system and describe the construction and working of semiconductor laser with a neat sketch and energy level diagram.                                                                                                                                                          | 09 | L2 | CO3 |
|            | b. | Derive an expression for numerical aperture in terms of refractive indices of core, cladding and the surrounding.                                                                                                                                                                                                 | 06 | L2 | CO3 |
|            | c. | Calculate the number of photons emitted per second for a LASER with power output 10 MW given the wavelength of fiber 690 nm.                                                                                                                                                                                      | 05 | L3 | CO5 |
| OR         |    |                                                                                                                                                                                                                                                                                                                   |    |    |     |
| Q.8        | a. | Define attenuation in fiber with the expression for attenuation coefficient and describe the various fiber losses.                                                                                                                                                                                                | 10 | L2 | CO3 |
|            | b. | Explain the construction and working of fiber optic temperature sensor.                                                                                                                                                                                                                                           | 05 | L2 | CO3 |
|            | c. | Calculate the numerical aperture and acceptance angle for an optical fiber of RI of core 1.5 and RI of cladding 1.45 placed in water of RI 1.33.                                                                                                                                                                  | 05 | L3 | CO5 |
| Module – 5 |    |                                                                                                                                                                                                                                                                                                                   |    |    |     |
| Q.9        | a. | Discuss the classification of earthquakes.                                                                                                                                                                                                                                                                        | 09 | L2 | CO4 |
|            | b. | Enumerate the causes and adverse effects of tsunami waves.                                                                                                                                                                                                                                                        | 06 | L2 | CO4 |
|            | c. | Calculate the intensity of earthquakes of magnitude 6.5 assuming the base intensity as $I_0$ .                                                                                                                                                                                                                    | 05 | L3 | CO4 |
| OR         |    |                                                                                                                                                                                                                                                                                                                   |    |    |     |
| Q.10       | a. | Define landslide and describe the causes for landslides.                                                                                                                                                                                                                                                          | 08 | L2 | CO4 |
|            | b. | Discuss the engineering structures to withstand earthquakes and Tsunami waves.                                                                                                                                                                                                                                    | 07 | L2 | CO4 |
|            | c. | The intensity of one earthquake is 100 times the intensity of the other. If the magnitude of the first earthquake is 8.9 estimate the magnitude of the other.                                                                                                                                                     | 05 | L3 | CO4 |

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## First/Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

### Applied Physics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

*Note:1. Answer any FIVE full questions, choosing ONE full question from each module.*

*2. VTU Formula Hand Book is permitted.*

*3. M : Marks, L: Bloom's level, C: Course outcomes.*

*4. Constants: Planck's constant ' $h$ ' =  $6.63 \times 10^{-34}$  Js, Boltzman constant ' $k$ ' =  $1.38 \times 10^{-23}$  JK<sup>-1</sup>*

*Mass of electron ' $m$ ' =  $9.11 \times 10^{-31}$  kg. Charge of electron ' $e$ ' =  $1.603 \times 10^{-19}$  C*

*Velocity of light in free space ' $C$ ' =  $3 \times 10^8$  m/s, Permittivity (free space) ' $\epsilon_0$ ' =  $8.854 \times 10^{-12}$  Fm<sup>-1</sup>*

| Module – 1 |    |                                                                                                                                                                                                       | M  | L  | C   |
|------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Using de-Broglie's hypothesis setup time independent Schrodinger wave equation for a particle in one-dimension.                                                                                       | 07 | L2 | CO1 |
|            | b. | State and explain Heisenberg's uncertainty principle. Using uncertainty principle, prove that electron does not exist inside the nucleus.                                                             | 09 | L2 | CO1 |
|            | c. | Find the de-Broglie wavelength of the electron in its ground state, if it is confined in an infinite potential well of width 10 nm.                                                                   | 04 | L3 | CO1 |
| OR         |    |                                                                                                                                                                                                       |    |    |     |
| Q.2        | a. | Explain briefly (i) Matter waves (ii) Group velocity (iii) Wave function                                                                                                                              | 06 | L2 | CO1 |
|            | b. | Find the eigen function and energy eigen values for particle in a one dimensional potential well of infinite height.                                                                                  | 10 | L2 | CO1 |
|            | c. | A particle of mass $9.1 \times 10^{-31}$ kg is confined in a region of width $8 \times 10^{-9}$ m. Calculate the minimum uncertainty in the measurement of its velocity.                              | 04 | L3 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                       |    |    |     |
| Q.3        | a. | What is Meissner effect? Discuss types of superconductors with examples based in critical field.                                                                                                      | 08 | L2 | CO2 |
|            | b. | What are polar dielectrics? Describe the different polarization mechanisms.                                                                                                                           | 08 | L2 | CO2 |
|            | c. | Calculate the probability of an electron occupying an energy level 0.02 eV below the Fermi level in a metal at thermal equilibrium (300 K).                                                           | 04 | L3 | CO2 |
| OR         |    |                                                                                                                                                                                                       |    |    |     |
| Q.4        | a. | What is Fermi factor? Discuss the dependence of Fermi factor on temperature and energy.                                                                                                               | 07 | L2 | CO2 |
|            | b. | What is internal field in dielectrics? Derive Clausius – Mossotti equation for an isotropic solid dielectric material with cubic symmetry.                                                            | 08 | L2 | CO2 |
|            | c. | Explain construction and working of a DC-SQUID.                                                                                                                                                       | 05 | L2 | CO2 |
| Module – 3 |    |                                                                                                                                                                                                       |    |    |     |
| Q.5        | a. | Explain the various phenomena when radiation interacts with matter and derive an expression for energy density of radiation under thermal equilibrium condition, in terms of Einstein's coefficients. | 10 | L2 | CO1 |
|            | b. | Discuss the different types of optical fibers with suitable diagrams.                                                                                                                                 | 06 | L2 | CO1 |
|            | c. | The ratio of population of two energy levels in a laser material at 57°C is $0.944 \times 10^{30}$ . Find the wavelength of the light emitted by spontaneous emissions.                               | 04 | L3 | CO1 |

| OR         |    |                                                                                                                                                                                                                                                                                                                                                      |    |    |     |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.6        | a. | Describe the construction and working of CO <sub>2</sub> laser with the help of necessary diagrams.                                                                                                                                                                                                                                                  | 10 | L2 | CO1 |
|            | b. | Derive an expression for numerical aperture in an optical fiber with a neat diagram.                                                                                                                                                                                                                                                                 | 06 | L2 | CO1 |
|            | c. | The attenuation in an optical fiber is 2.2 dB/km. What fraction of its initial light intensity remains after 2 km length?                                                                                                                                                                                                                            | 04 | L3 | CO1 |
| Module – 4 |    |                                                                                                                                                                                                                                                                                                                                                      |    |    |     |
| Q.7        | a. | State and derive Gauss's divergence theorem.                                                                                                                                                                                                                                                                                                         | 08 | L2 | CO3 |
|            | b. | Derive the electromagnetic wave equation using Maxwell's equations.                                                                                                                                                                                                                                                                                  | 08 | L2 | CO3 |
|            | c. | A vector field is given $\vec{F} = 2x^3\vec{i} + 5y^2\vec{j} + 9z\vec{k}$ . Find the divergence of $\vec{F}$ at (1, 2, 3).                                                                                                                                                                                                                           | 04 | L3 | CO3 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                      |    |    |     |
| Q.8        | a. | Write the four Maxwell's equations for time-varying conditions in vacuum and explain each notations used in it.                                                                                                                                                                                                                                      | 08 | L2 | CO3 |
|            | b. | What is displacement current density? Derive an expression for displacement current.                                                                                                                                                                                                                                                                 | 08 | L2 | CO3 |
|            | c. | Show that given vector field $\vec{A}$ is irrotational.<br>Given $\vec{A} = (3x^2 + y + 2z)\vec{i} + (x - 5y^3 - 2z)\vec{j} + (2x - 2y + 3z^2)\vec{k}$                                                                                                                                                                                               | 04 | L3 | CO3 |
| Module – 5 |    |                                                                                                                                                                                                                                                                                                                                                      |    |    |     |
| Q.9        | a. | State law of mass action in semiconductors. Derive an expression for electrical conductivity of an intrinsic semiconductor.                                                                                                                                                                                                                          | 08 | L2 | CO4 |
|            | b. | Describe four Probe method to determine the resistivity of a thin semiconductor sample.                                                                                                                                                                                                                                                              | 08 | L2 | CO4 |
|            | c. | In the experiment of dielectric constant of the material of capacitor, the intersecting time interval for charging and discharging ( $T_p$ ) is found to be 7 sec with a resistor of 110 k $\Omega$ . Find the area of the capacitor plate "A". Given that thickness of dielectric material and dielectric constant are 1.02mm and 4.2 respectively. | 04 | L3 | CO5 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                      |    |    |     |
| Q.10       | a. | What is hall effect? Derive an expression for Hall-coefficient with necessary diagrams.                                                                                                                                                                                                                                                              | 08 | L2 | CO4 |
|            | b. | What is direct band gap semiconductor? Explain the construction and working of a GaAs laser.                                                                                                                                                                                                                                                         | 08 | L2 | CO4 |
|            | c. | The electrical conductivity of a intrinsic semiconductor with carrier density $2.37 \times 10^{19}/\text{m}^3$ at 300 K is $2.12 \Omega^{-1}\text{m}^{-1}$ . Assuming electron mobility as $0.38 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$ , calculate mobility of holes.                                                                               | 04 | L3 | CO4 |

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# CBCS SCHEME

USN

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BPHYM102/202

**First/Second Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025**

## Physics for ME Stream

Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**

**2. M : Marks , L: Bloom's level , C: Course outcomes.**

**3. VTU formula Handbook is permuted.**

| Module – 1        |    |                                                                                                                                                                                                                                                           | M  | L  | C   |
|-------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 1                 | a. | Obtain a differential equation for a body undergoing forced oscillation and mention expression for amplitude and phase of oscillation. Discuss the three cases for variation of amplitude with frequency in forced oscillation.                           | 10 | L2 | CO1 |
|                   | b. | Define Mach number and Mach angle. Discuss the various types waves based as Mach number.                                                                                                                                                                  | 5  | L2 | CO1 |
|                   | c. | A spring undergoes an extension of 5 cm for a load of 50 gm. Find its frequency of oscillations if it is set for vertical oscillations with a load of 200 gm, attached to its bottom, Ignore the mass of the spring.                                      | 5  | L3 | CO5 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                           |    |    |     |
| 2                 | a. | Describe the construction and working of hand operated Reddy shock tube. Mention any three applications of shock waves.                                                                                                                                   | 10 | L2 | CO1 |
|                   | b. | Derive the expression for equivalent force constant for two springs connected in series combinations and mention the expression for period of oscillation.                                                                                                | 6  | L2 | CO1 |
|                   | c. | The distance between the two pressure sensors in shock tube is 100 mm. The time taken by a shock wave to travel this distance is 100 microsecond. If the velocity of sound under the same conditions is 340 m/s. Find the match number of the shock wave. | 4  | L3 | CO1 |
| <b>Module – 2</b> |    |                                                                                                                                                                                                                                                           |    |    |     |
| 3                 | a. | Derive the expression for bending moment in terms of moment of inertia and hence arrive at the expression for bending moment for a beam for circular and rectangular cross section.                                                                       | 10 | L2 | CO1 |
|                   | b. | With neat diagram explain the stress-strain curve for elastic materials.                                                                                                                                                                                  | 6  | L2 | CO1 |
|                   | c. | Calculate the extension produced in a wire of length 2 m and radius 0.013 cm due to force of 15 N applied along its length. (Given Yong's modulus of wire $Y = 2.1 \times 10^{11} \text{ N/m}^2$ ).                                                       | 4  | L3 | CO1 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                           |    |    |     |
| 4                 | a. | Define Young's modulus, bulk modulus and rigidity modulus. Derive relation between Y, n and $\sigma$ .                                                                                                                                                    | 10 | L2 | CO2 |
|                   | b. | Discuss any three types of beams.                                                                                                                                                                                                                         | 6  | L2 | CO2 |
|                   | c. | Calculate the Poisson's ratio for the material given that $Y = 12.25 \times 10^{10} \text{ N/mm}^2$ and $\eta = 4.55 \times 10^{10} \text{ N/mm}^2$ .                                                                                                     | 4  | L3 | CO2 |
| <b>Module – 3</b> |    |                                                                                                                                                                                                                                                           |    |    |     |
| 5                 | a. | What is thermocouple? Describe the seeback effect and the Peltier effect.                                                                                                                                                                                 | 8  | L2 | CO2 |
|                   | b. | Describe the construction of and working of thermoelectric cooler.                                                                                                                                                                                        | 7  | L2 | CO2 |
|                   | c. | EMF of a thermocouple is 1200 $\mu\text{V}$ , when working between $0^\circ\text{C}$ and $100^\circ\text{C}$ . Its neutral temperature is $300^\circ\text{C}$ . Find the values of a and b for it.                                                        | 5  | L3 | CO2 |
| <b>1 of 2</b>     |    |                                                                                                                                                                                                                                                           |    |    |     |

## OR

|   |    |                                                                                                                                                                                                          |   |    |     |
|---|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 6 | a. | State and explain Laws of thermoelectricity.                                                                                                                                                             | 8 | L2 | CO2 |
|   | b. | Explain the construction and working of thermo couples. Mention their advantages.                                                                                                                        | 7 | L2 | CO2 |
|   | c. | The e.m.f in lead-iron thermocouple, one junction of which is at 0°C, is given by $E = 1784t - 2.41t^2$ (in $\mu$ Volt) where t is temperature in °C. Find the neutral temperature, $\pi$ and $\sigma$ . | 5 | L3 | CO2 |

## Module – 4

|   |    |                                                                                                                                                            |   |    |     |
|---|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 7 | a. | Explain Joule Thomson effect. Derive $\Delta T$ ; $\Delta T = (P_1 - P_2)/C_p [2a/RT - b]$ .                                                               | 8 | L2 | CO3 |
|   | b. | Describe the process of liquefaction of oxygen by cascade process.                                                                                         | 8 | L2 | CO3 |
|   | c. | Calculate inversion temperature of gas. Given $a = 0.244 \text{ atm L}^2/\text{mol}^2$ , $b = 0.027 \text{ L/Mol}$ and $R = 0.0821 \text{ L atm /K/mol}$ . | 4 | L3 | CO3 |

## OR

|   |    |                                                                                                        |   |    |     |
|---|----|--------------------------------------------------------------------------------------------------------|---|----|-----|
| 8 | a. | Describe construction and working of porous plug experiment. What conclusions have been drawn from it? | 8 | L2 | CO3 |
|   | b. | Describe construction and working of platinum resistance thermometer.                                  | 8 | L2 | CO3 |
|   | c. | Mention the properties of Liquid Helium.                                                               | 4 | L3 | CO3 |

## Module – 5

|   |    |                                                                                                                                                                                                   |   |    |     |
|---|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 9 | a. | With neat diagram, explain the principle, construction and working of atomic force microscopy.                                                                                                    | 8 | L2 | CO4 |
|   | b. | Determine the construction and working of transmission electron microscope (TEM)                                                                                                                  | 8 | L2 | CO4 |
|   | c. | The spacing between principle planes of NaCl crystal is $2.82 \text{ \AA}$ . It is found that first order Bragg reflection occurs at an angle of $10^\circ$ . Calculate the wavelength of X-rays. | 4 | L3 | CO4 |

## OR

|    |    |                                                                                                                                                                               |   |    |     |
|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 10 | a. | With neat diagram, explain the principle, construction and working of scanning electron microscopy.                                                                           | 8 | L2 | CO4 |
|    | b. | Explain the construction and working of X-ray diffractometer and how the crystal size is determined using Scherrer equation.                                                  | 8 | L2 | CO4 |
|    | c. | Determine the crystallite size given the wavelength of x-ray $10 \text{ nm}$ , the peak width $0.5^\circ$ and peak position $25^\circ$ for a cubic crystal given $K = 0.94$ . | 4 | L3 | CO4 |

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# CBCS SCHEME

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BPHYS102/202

**First / Second Semester B.E./B.Tech. Degree Examination,  
Dec.2024/Jan.2025**

## Applied Physics for CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.  
3. VTU Handbook is permitted.*

| Module – 1        |    |                                                                                                                                                                                                                                                                                         |  | M | L  | C   |
|-------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|----|-----|
| Q.1               | a. | Obtain the expression for energy density equation using Einstein's coefficients at thermal equilibrium condition.                                                                                                                                                                       |  | 9 | L2 | CO1 |
|                   | b. | Discuss the types of optical fibers based on modes of Propagation and Refractive Index profile.                                                                                                                                                                                         |  | 6 | L2 | CO1 |
|                   | c. | Given the numerical aperture 0.30 and RI of core 1.49. Calculate the critical angle for the core – cladding interface.                                                                                                                                                                  |  | 5 | L3 | CO1 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                                                         |  |   |    |     |
| Q.2               | a. | Illustrate the construction and working of semiconductor laser with a neat sketch with energy level diagram.                                                                                                                                                                            |  | 8 | L2 | CO1 |
|                   | b. | Define Acceptance angle and Numerical aperture and hence derive an expression for Numerical aperture in terms of Refractive Indices of core, cladding and surrounding.                                                                                                                  |  | 7 | L2 | CO1 |
|                   | c. | In an optical fiber experiment the LASER light propagating through optical fibre cable of 1.5m, made a spot diameter of 8mm on the screen. The distance between the end of the optical fibre cable and the screen is 0.031m. Calculate angle of contact and N.A of given optical fibre. |  | 5 | L3 | CO5 |
| <b>Module – 2</b> |    |                                                                                                                                                                                                                                                                                         |  |   |    |     |
| Q.3               | a. | Derive an expression for de Broglie wavelength by analogy and hence discuss the significance of de Broglie waves.                                                                                                                                                                       |  | 6 | L2 | CO2 |
|                   | b. | Set – up Schrodinger time independent wave equation in one dimension.                                                                                                                                                                                                                   |  | 9 | L2 | CO2 |
|                   | c. | Calculate the energy of the first three energy states for an electron in one dimensional potential well of width 0.1nm.                                                                                                                                                                 |  | 5 | L3 | CO2 |
| <b>OR</b>         |    |                                                                                                                                                                                                                                                                                         |  |   |    |     |
| Q.4               | a. | State and explain Heisenberg's uncertainty principle and show that electron does not exist inside the nucleus using Heisenberg's uncertainty principle.                                                                                                                                 |  | 7 | L2 | CO2 |
|                   | b. | Explain Eigen values and Eigen functions and hence derive the Eigen function of a particle inside infinite potential well of width 'a' using the method of Normalization.                                                                                                               |  | 8 | L2 | CO2 |

|                   |    |                                                                                                                                                                                                                              |   |    |     |
|-------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
|                   | c. | The kinetic energy of an electron is equal to the energy of a photon with a wave length of 560 nm. Calculate the de Broglie wave length of the electron.                                                                     | 5 | L3 | CO2 |
| <b>Module – 3</b> |    |                                                                                                                                                                                                                              |   |    |     |
| Q.5               | a. | State the Pauli matrices and apply Pauli matrices on the states $ 0\rangle$ and $ 1\rangle$ .                                                                                                                                | 9 | L2 | CO2 |
|                   | b. | Discuss the CNOT gate and its operation on four different input states.                                                                                                                                                      | 6 | L2 | CO2 |
|                   | c. | Given $ \Psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$ and $ \phi\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$ , Prove that $\langle \Psi   \phi \rangle = \langle \phi   \Psi \rangle^*$ . | 5 | L3 | CO2 |
| <b>OR</b>         |    |                                                                                                                                                                                                                              |   |    |     |
| Q.6               | a. | Explain the representation of qubit using Bloch sphere.                                                                                                                                                                      | 7 | L2 | CO3 |
|                   | b. | Describe the working of controlled – Z gate mentioning its matrix representation and truth table.                                                                                                                            | 8 | L3 | CO3 |
|                   | c. | A linear operator 'X' operates such that $X 0\rangle =  1\rangle$ and $X 1\rangle =  0\rangle$ . Find the matrix representation of 'X'.                                                                                      | 5 | L3 | CO3 |
| <b>Module – 4</b> |    |                                                                                                                                                                                                                              |   |    |     |
| Q.7               | a. | Enumerate the assumptions of Quantum free electron theory of metals and mention the failures and classical free electron theory.                                                                                             | 7 | L2 | CO3 |
|                   | b. | Describe Meissner effect and hence classify superconductors into Soft and Hard super conductors using M – H graphs.                                                                                                          | 8 | L2 | CO3 |
|                   | c. | Calculate the probability of occupation of an energy level 0.2eV above Fermi level at temperature 27°C.                                                                                                                      | 5 | L3 | CO3 |
| <b>OR</b>         |    |                                                                                                                                                                                                                              |   |    |     |
| Q.8               | a. | Define Fermi factor and discuss the variation of Fermi factor with temperature and energy.                                                                                                                                   | 7 | L2 | CO3 |
|                   | b. | Explain the phenomenon of superconductivity and discuss qualitatively the BCS theory of super conductivity for negligible resistance of metal at temperature close to absolute zero.                                         | 8 | L2 | CO3 |
|                   | c. | A superconductivity Tin has a critical temperature of 3.7K at zero magnetic field and a critical field of 0.0306 Tesla at 0°K. Find the critical field at 2K.                                                                | 5 | L3 | CO3 |
| <b>Module – 5</b> |    |                                                                                                                                                                                                                              |   |    |     |
| Q.9               | a. | Elucidate the importance of size and scale , weight and strength in animations.                                                                                                                                              | 8 | L2 | CO4 |
|                   | b. | Discuss the salient features of normal distribution using bell curves.                                                                                                                                                       | 7 | L2 | CO4 |

|             |           |                                                                                                                                                                                                                                                                                  |          |           |            |
|-------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------|------------|
|             | <b>c.</b> | A slowing object in an animations has a first frame distance 0.5m and the first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.                                                                                                           | <b>5</b> | <b>L3</b> | <b>CO4</b> |
| <b>OR</b>   |           |                                                                                                                                                                                                                                                                                  |          |           |            |
| <b>Q.10</b> | <b>a.</b> | Describe Jumping and parts of Jump.                                                                                                                                                                                                                                              | <b>8</b> | <b>L2</b> | <b>CO4</b> |
|             | <b>b.</b> | Discuss modeling the probability for proton decay.                                                                                                                                                                                                                               | <b>7</b> | <b>L2</b> | <b>CO4</b> |
|             | <b>c.</b> | In a diffraction grating experiment the laser light undergoes second order diffraction for diffraction angle $1.48^\circ$ . The grating constant $d = 5.05 \times 10^{-5}\text{m}$ and the distance between the grating and screen is 0.60m. Find the wavelength of LASER light. | <b>5</b> | <b>L3</b> | <b>CO5</b> |

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# CBCS SCHEME

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BPLCK105B

## First Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Introduction to Python Programming

Time: 3 hrs.

Max. Marks:100

**Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.

| Module – 1 |    |                                                                                                                                            | M | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 1          | a. | Explain basic data types like int, float, double and string with an example.                                                               | 6 | L2 | CO1 |
|            | b. | Differentiate between local scope and global scope.                                                                                        | 6 | L2 | CO1 |
|            | c. | Develop a program to calculate factorial of a number. Program to compute binomial coefficient (Given N and R).                             | 8 | L3 | CO1 |
| OR         |    |                                                                                                                                            |   |    |     |
| 2          | a. | Define functions. Explain how to pass parameters through the function with return statement.                                               | 6 | L2 | CO1 |
|            | b. | What is exception? How exception are handled in python? Write a program to solve divide by zero exception.                                 | 6 | L2 | CO1 |
|            | c. | Develop a program to generate Fibonacci sequence of length (N). Read N from the console.                                                   | 8 | L3 | CO1 |
| Module – 2 |    |                                                                                                                                            |   |    |     |
| 3          | a. | Explain Augmented short hand assignment operators with an example.                                                                         | 7 | L2 | CO2 |
|            | b. | Explain different type of methods like append( ), Remove( ), sort( ), pop( ) in python programming list.                                   | 7 | L2 | CO2 |
|            | c. | Develop a program to find mean, variance and standard deviation.                                                                           | 6 | L3 | CO3 |
| OR         |    |                                                                                                                                            |   |    |     |
| 4          | a. | Explain set( ) and setdefault( ) method in dictionary.                                                                                     | 7 | L2 | CO2 |
|            | b. | Develop a python to print area of rectangle.                                                                                               | 6 | L3 | CO2 |
|            | c. | Define pretty printing. How does pretty print work in python with an example.                                                              | 7 | L2 | CO2 |
| Module – 3 |    |                                                                                                                                            |   |    |     |
| 5          | a. | Explain useful string functions like :<br>i) Capitalize<br>ii) Count<br>iii) Find<br>iv) Lower<br>v) Upper<br>vi) Replace with an example. | 8 | L2 | CO3 |
|            | b. | Develop a python code to determine whether given string is a palindrome or not a palindrome.                                               | 6 | L3 | CO3 |
|            | c. | Explain : i) isalpha ii) isalnum iii) isspace( ).                                                                                          | 6 | L2 | CO3 |

## OR

|   |    |                                                                                 |   |    |     |
|---|----|---------------------------------------------------------------------------------|---|----|-----|
| 6 | a. | Explain OS path module with an example.                                         | 6 | L3 | CO2 |
|   | b. | Explain the concept of file path. Also discuss absolute and relative file path. | 8 | L3 | CO3 |
|   | c. | Program to print of multi clipboard with appropriate message.                   | 6 | L3 | CO3 |

## Module – 4

|   |    |                                                                                                                                                                                         |   |    |     |
|---|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 7 | a. | Develop a program to backing up a given folder (folder in a current working directory) into a zip file by using relevant modules and suitable methods.                                  | 6 | L3 | CO4 |
|   | b. | List out the difference between shutil.copy() and shutil.copytree() method.                                                                                                             | 6 | L1 | CO4 |
|   | c. | Explain the following file operations in python with suitable example :<br>i) Copying files and folders<br>ii) Moving files and folders<br>iii) Permanently deleting files and folders. | 8 | L2 | CO4 |

## OR

|   |    |                                                                                                                                                                                                                                                                                          |   |    |     |
|---|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 8 | a. | Briefly explain assertion and raising a exception.                                                                                                                                                                                                                                       | 8 | L2 | CO4 |
|   | b. | List out the benefits of using logging module with an example.                                                                                                                                                                                                                           | 6 | L1 | CO4 |
|   | c. | Write a function named DivExp which takes two parameters a, b and returns a value C(c= a/b). Write suitable assertion for a 70 in function DivExp and raise an exception for when b = 0. Develop a suitable program which reads two values from the console and calls a function DivExp. | 6 | L3 | CO4 |

## Module – 5

|   |    |                                                                                                                                                                                                                                                                                                                        |   |    |     |
|---|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 9 | a. | Define a function which takes two objects representing complex numbers and returns a new complex number with a addition of two complex numbers. Define a suitable class 'complex' to represent the complex number. Develop a program to read N(N >= 2) complex numbers and compute the addition of 10 complex numbers. | 8 | L3 | CO5 |
|   | b. | Explain the concept of inheritance with an example.                                                                                                                                                                                                                                                                    | 6 | L2 | CO5 |
|   | c. | Explain the __str__ and the __init__ method with an example.                                                                                                                                                                                                                                                           | 6 | L2 | CO5 |

## OR

|    |    |                                                                                                                                                                                                                              |   |    |     |
|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|-----|
| 10 | a. | Define a class and object, construct the class called rectangle and initialize it with height = 100, width = 200, starting point as (x = 0, y = 0). Write a program to display the centre point co-ordinates of a rectangle. | 8 | L3 | CO5 |
|    | b. | Briefly explain the printing of objects with an example.                                                                                                                                                                     | 6 | L2 | CO5 |
|    | c. | Differentiate operator over loading and operator overriding in python.                                                                                                                                                       | 6 | L2 | CO5 |

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## Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

**Introduction to Python Programming**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                                                                                 | M  | L  | C   |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Describe the following flow control statements in python programming:<br>(i) if<br>(ii) else<br>(iii) while                                                                                                                                                     | 8  | L1 | CO1 |
|            | b. | Write a function to calculate factorial of a number.                                                                                                                                                                                                            | 8  | L3 | CO1 |
|            | c. | Define def statements with parameters.                                                                                                                                                                                                                          | 4  | L1 | CO1 |
| OR         |    |                                                                                                                                                                                                                                                                 |    |    |     |
| Q.2        | a. | Explain the syntax and control flow diagrams for break and continue statement.                                                                                                                                                                                  | 8  | L2 | CO1 |
|            | b. | Illustrate the dissection of python program.                                                                                                                                                                                                                    | 8  | L2 | CO1 |
|            | c. | Describe the return values and return statements.                                                                                                                                                                                                               | 4  | L1 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                                                                                 |    |    |     |
| Q.3        | a. | Read N numbers from the console and create a list. Develop a program to print, mean, variance and standard deviation with suitable messages.                                                                                                                    | 8  | L3 | CO2 |
|            | b. | Summarize the sequence data types of python programming.                                                                                                                                                                                                        | 8  | L2 | CO2 |
|            | c. | Compare and contrast dictionaries Vs lists.                                                                                                                                                                                                                     | 4  | L2 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                                 |    |    |     |
| Q.4        | a. | Develop a program to print 10 most frequently appearing words in a text file. [Hint : Use dictionary with distinct words and their frequency of occurrences. Sort the dictionary in reverse order of frequency and display dictionary slice of first 10 items]. | 8  | L3 | CO2 |
|            | b. | Explain the slots of tic – tac – toe board with its corresponding keys using data structures in python programming.                                                                                                                                             | 6  | L2 | CO2 |
|            | c. | Paraphrase the working with lists in python programming.                                                                                                                                                                                                        | 6  | L1 | CO2 |
| Module – 3 |    |                                                                                                                                                                                                                                                                 |    |    |     |
| Q.5        | a. | Describe the Python string handling methods with examples : Split ( ), endswith ( ), ljust ( ), center ( ), lstrip ( ).                                                                                                                                         | 10 | L2 | CO3 |
|            | b. | Summarize the process of input validation in python programming.                                                                                                                                                                                                | 10 | L2 | CO3 |

| OR         |    |                                                                                                                                                        |    |    |     |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.6        | a. | Explain Python string handling methods with examples :<br>join ( ), startswith ( ), rjust ( ), strip ( ),rstrip ( ).                                   | 10 | L2 | CO3 |
|            | b. | Demonstrate the process of copying and pasting strings with pyperclip module.                                                                          | 10 | L2 | CO3 |
| Module – 4 |    |                                                                                                                                                        |    |    |     |
| Q.7        | a. | Develop a program to backing up a given folder (Folder in a current working directory) into a zip file by using relevant modules and suitable methods. | 10 | L3 | CO3 |
|            | b. | Describe the file reading or writing process in python programming.                                                                                    | 10 | L2 | CO3 |
| OR         |    |                                                                                                                                                        |    |    |     |
| Q.8        | a. | Explain the process of compressing files with the zip file module.                                                                                     | 10 | L2 | CO3 |
|            | b. | Summarize the organization of files using shutil module.                                                                                               | 10 | L3 | CO3 |
| Module – 5 |    |                                                                                                                                                        |    |    |     |
| Q.9        | a. | Explain about class and objects.                                                                                                                       | 10 | L2 | CO4 |
|            | b. | Explain purefunction and modifier.                                                                                                                     | 10 | L2 | CO4 |
| OR         |    |                                                                                                                                                        |    |    |     |
| Q.10       | a. | Illustrate operator overloading and polymorphism in python with an example.                                                                            | 10 | L2 | CO4 |
|            | b. | Explain _init_ and _str_ method with examples.                                                                                                         | 10 | L2 | CO4 |

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## Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

**Basics of Java Programming**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                        | M  | L  | C   |
|------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Write a Java Program to find greatest of 3 numbers.                                                                                                    | 6  | L2 | CO1 |
|            | b. | With reason identify the following declarations are valid or invalid :<br>#AvgTemp, Count, a4, \$test, 2Count, high-temp, _Sum, if                     | 6  | L2 | CO1 |
|            | c. | Explain type casting and type conversion in Java with an example program.                                                                              | 8  | L2 | CO1 |
| OR         |    |                                                                                                                                                        |    |    |     |
| Q.2        | a. | Write a Java Program to swap data in 2 variables without using temporary variable.                                                                     | 4  | L2 | CO1 |
|            | b. | Define Array. With Syntax and Example Program. Explain declaration of single dimensional array in Java.                                                | 6  | L2 | CO1 |
|            | c. | Briefly explain the basic principle of Java.                                                                                                           | 10 | L2 | CO1 |
| Module – 2 |    |                                                                                                                                                        |    |    |     |
| Q.3        | a. | Briefly explain the following operators in Java with an example program :<br>>> << ~                                                                   | 6  | L2 | CO2 |
|            | b. | With example program, explain following statements :<br>(i) Break (ii) Continue                                                                        | 8  | L2 | CO2 |
|            | c. | Write a Java Program to find largest of 2 numbers using ternary operator.                                                                              | 6  | L3 | CO2 |
| OR         |    |                                                                                                                                                        |    |    |     |
| Q.4        | a. | Write a Java Program that prints all real solutions to the quadratic equation,<br>$ax^2 + bx + c = 0$ . Read in a, b, c and use the quadratic formula. | 10 | L3 | CO1 |
|            | b. | With syntax and example program, explain iterative statements in Java.                                                                                 | 10 | L2 | CO1 |
| Module – 3 |    |                                                                                                                                                        |    |    |     |
| Q.5        | a. | Explain following keywords in Java with examples :<br>(i) this (ii) final (iii) Static                                                                 | 10 | L3 | CO3 |
|            | b. | Define Constructor. Explain types of constructors with example programs.                                                                               | 10 | L3 | CO3 |

| OR         |    |                                                                                                                                                                                                                                                                                                            |    |    |     |
|------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.6        | a. | Explain Garbage collection in Java with an example.                                                                                                                                                                                                                                                        | 6  | L2 | CO3 |
|            | b. | Write a Java program to find factorial of a number using recursion.                                                                                                                                                                                                                                        | 6  | L3 | CO3 |
|            | c. | Define method overloading. Explain with an example program.                                                                                                                                                                                                                                                | 8  | L3 | CO3 |
| Module – 4 |    |                                                                                                                                                                                                                                                                                                            |    |    |     |
| Q.7        | a. | Why multiple inheritance is not supported in Java. Justify with an example.                                                                                                                                                                                                                                | 8  | L3 | CO3 |
|            | b. | Explain the “Super” keyword in Java.                                                                                                                                                                                                                                                                       | 4  | L2 | CO3 |
|            | c. | Define method overriding. Explain with an example program.                                                                                                                                                                                                                                                 | 8  | L2 | CO3 |
| OR         |    |                                                                                                                                                                                                                                                                                                            |    |    |     |
| Q.8        | a. | Explain types of inheritance in Java with example.                                                                                                                                                                                                                                                         | 10 | L2 | CO3 |
|            | b. | Design a Super Class called staff with details as staffId, Name, Phone, Salary. Extend this class by writing three subclasses namely Teaching (domain, publications), Technical (Skills) and Contract (Period). Write a Java Program to read and display at least 3 staff objects of all three categories. | 10 | L3 | CO3 |
| Module – 5 |    |                                                                                                                                                                                                                                                                                                            |    |    |     |
| Q.9        | a. | With a neat block diagram, explain types of Java Exceptions.                                                                                                                                                                                                                                               | 10 | L2 | CO4 |
|            | b. | Define interface. Differentiate between Class and interface.                                                                                                                                                                                                                                               | 5  | L2 | CO4 |
|            | c. | Write a note on packages in Java.                                                                                                                                                                                                                                                                          | 5  | L2 | CO4 |
| OR         |    |                                                                                                                                                                                                                                                                                                            |    |    |     |
| Q.10       | a. | Explain following types of Exceptions :<br>(i) ArithmeticException<br>(ii) ArrayIndexOutOfBoundsException<br>(iii) SQLExceptions<br>(iv) NullPointerExceptions                                                                                                                                             | 12 | L3 | CO4 |
|            | b. | Illustrate “throw” and “throws” in Java Exceptions with example.                                                                                                                                                                                                                                           | 8  | L3 | CO4 |

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BPOPS103/203

## First/Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Principles of Programming using C

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.  
2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                              | M  | L  | C   |
|------------|----|------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | List and explain any 5 characteristics of a computer.                                                                        | 10 | L2 | CO1 |
|            | b. | Draw the block diagram of a computer, explain all of its blocks such as input, output storage and CPU.                       | 10 | L2 | CO1 |
| <b>OR</b>  |    |                                                                                                                              |    |    |     |
| Q.2        | a. | Explain the following programming paradigms.<br>i) Procedural programming<br>ii) Object-oriented programming                 | 10 | L2 | CO1 |
|            | b. | Explain the structure of a C program with an example.                                                                        | 10 | L2 | CO1 |
| Module – 2 |    |                                                                                                                              |    |    |     |
| Q.3        | a. | Explain Arithmetic and Relational operators of C with an example.                                                            | 10 | L2 | CO2 |
|            | b. | Develop a complete C program to find the real roots of a quadratic equation by accepting the coefficients.                   | 10 | L3 | CO2 |
| <b>OR</b>  |    |                                                                                                                              |    |    |     |
| Q.4        | a. | Explain logical and assignment operators of C with an example.                                                               | 10 | L2 | CO2 |
|            | b. | Write a C program to find the factorial of a given integer n. Explain the computation process.                               | 10 | L3 | CO2 |
| Module – 3 |    |                                                                                                                              |    |    |     |
| Q.5        | a. | Write a C program to find the sum and average of n integers.                                                                 | 10 | L3 | CO3 |
|            | b. | Explain the concept of function declaration and function definition with an example.                                         | 10 | L2 | CO5 |
| <b>OR</b>  |    |                                                                                                                              |    |    |     |
| Q.6        | a. | Write a C program to add two m×n matrices.                                                                                   | 10 | L3 | CO3 |
|            | b. | Explain the following with an example.<br>i) Passing the entire array<br>ii) Passing the individual elements of an ID array. | 10 | L2 | CO5 |
| Module – 4 |    |                                                                                                                              |    |    |     |
| Q.7        | a. | Write a C program to find the length of a given string without using inbuilt function.                                       | 10 | L2 | CO3 |
|            | b. | What is a pointer? Show the use of two pointer operators & and *.                                                            | 10 | L2 | CO3 |

| OR         |    |                                                                                         |    |    |     |
|------------|----|-----------------------------------------------------------------------------------------|----|----|-----|
| Q.8        | a. | Write a C program to compare two given string S1 and S2 without using inbuilt function. | 10 | L3 | CO3 |
|            | b. | How do you declare and initialize a pointer in C? Show with an example.                 | 10 | L2 | CO3 |
| Module – 5 |    |                                                                                         |    |    |     |
| Q.9        | a. | What is recursion? Give one example.                                                    | 10 | L2 | CO5 |
|            | b. | Differentiate between arrays and structures in C.                                       | 10 | L3 | CO4 |
| OR         |    |                                                                                         |    |    |     |
| Q.10       | a. | Explain the process of opening and closing a file in C.                                 | 10 | L2 | CO5 |
|            | b. | Differentiate between structure and unions in C with example programs.                  | 10 | L3 | CO4 |

\* \* \* \* \*

**BSFHK158/258**

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**Question Paper Version : A**

**First/Second Semester B.E./B.Tech Degree Examination,  
Dec.2024/Jan.2025**

## Scientific Foundations of Health

Time: 1 hr.]

[Max. Marks: 50]

## INSTRUCTIONS TO THE CANDIDATES

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  3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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  5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.
- 
1. Globalization , growth of trade unions, public relations and positive atmosphere are some factors that enable \_\_\_\_\_
    - a) Good friendship
    - b) Isolation from society
    - c) Happiness
    - d) Communication growth
  2. Which of the following is NOT a characteristic of addiction?
    - a) Denial
    - b) Negative consequences
    - c) Loss of control
    - d) Habitual behavior
  3. Denial is person's
    - a) Experience of a blackout episode or memory loss.
    - b) Inability to perceive that a behavior is self – destructive
    - c) Failure to stop an addictive behavior
    - d) Change in usual pattern of drinking or drug use
  4. Temporary physical and psychological symptoms that occur when use of an addictive substance is discontinued is
    - a) Relapse
    - b) Compulsion
    - c) Withdrawal
    - d) Addiction
  5. Chemicals that relay messages (impulses) from one nerve cell to another or to other cells are \_\_\_\_\_
    - a) Enzymes
    - b) Neuro transmitters
    - c) Synapses
    - d) Chemoreceptors

6. What is the definitions of overweight?  
a) BMI > 25 kg/m<sup>2</sup>                      b) BMI = 25 kg/m<sup>2</sup>  
c) BMI 25 – 29.9 kg/m<sup>2</sup>                d) BMI < 25 kg/m<sup>2</sup>
7. Communication is a part of \_\_\_\_ skills.  
a) Soft                                      b) Hard                                      c) Short                                      d) Rough
8. Which of these factors is not required for communication growth?  
a) Globalisation                                      b) Negative Atmosphere  
c) Growth size of organizations                d) Public relations
9. Body language plays an important role in  
a) Communication                      b) Judgment                      c) Both (a) & (b)                      d) None of these
10. This causes tunnel vision.  
a) Smoking                      b) Alcohol                      c) Barbiturates                      d) Vitamins A
11. This causes maximum accumulation of fat in the liver.  
a) Meat & Egg                      b) Alcohol                      c) Saturated fat                      d) Starch
12. Using abbreviations in communication leads to which type of communication barrier.  
a) Language / Linguistic                      b) Physical  
c) Cultural                                      d) Organizational
13. Obesity occurs due to ; More consumption of  
a) Calorie                      b) Protein                      c) Vitamins                      d) None of these
14. What is intimately related?  
a) Body & health                                      b) Body & mind  
c) Disease & Health                                      d) Body & Spiritual values
15. What influences your health?  
a) Heredity                                      b) Environment  
c) Physical environment                      d) All of these
16. A lack of judgment regarding whether a particular substance or behavior will be healthy of damaging to oneself is known as  
a) Denial                      b) Obsession                      c) Loss of control                      d) Compulsion
17. Which of the following diseases is transmitted by an insert vector?  
a) Malaria                      b) HIV/AIDS                      c) CTB                      d) None of these
18. Disease of the heart, joint and nervous system is called  
a) Degenerative diseases                      b) Mental diseases  
c) Communicable diseases                      d) Deficiency diseases

19. How often one must have a routine body checkup?  
a) Once in 10 year    b) Twice a year    c) Once a year    d) Once in 5 year
20. How do puzzles help you?  
a) Physical exercise    b) They are a time pass  
c) They help exercise the brain    d) They help you learn numbers
21. Alcohol based hand rub is the preferred method of hand hygiene.  
a) False    b) True    c) Not sure    d) None of these
22. Is addiction a  
a) Mental disorder    b) Physical disorder    c) Both (a) & (b)    d) None of these
23. Important roles of life are  
a) Feeling happy    b) Enjoy life    c) Fighting diseases    d) All of these
24. Signs of mental illness are  
a) Abnormal thinking , perception and judgment  
b) Abnormal changes in feeling and memory  
c) Both (a) & (b)  
d) Abnormal changes in behavior towards others.
25. People suffering from mental diseases in India.  
a) 1 – 1.5 millions    b) 6 – 7 millions    c) 10 – 15 millions    d) 60 – 70 millions
26. Social health hazards includes.  
a) Stigma    b) Anti – social behaviour  
c) Higher crime records    d) All of these
27. Hyper obesity value in terms of BMI.  
a) > 25    b) 40    c) > 40    d) < 40
28. Goals of communication.  
a) To inform , to persuade    b) To inform, fear of offending  
c) To persuade, fear of offending    d) None of these
29. In which year WHO articulated the right to health in its constitution?  
a) 2000    b) 1946    c) 1952    d) 1987
30. Spiritual health refers  
a) Body composition    b) Mediation    c) Exercising    d) All of these
31. When findings are difficult to generalize to the world outside of the laboratory, we say the research is lacking in \_\_\_\_\_ validity.  
a) Ecological    b) Economical    c) Empirical    d) Experimental



44. What is the goal of social engineering?  
a) Sabotage a person's social media  
b) To build trust  
c) To gain vital personal information  
d) To catfish someone.
45. After drinking Alcohol, consumption of this leads to death.  
a) Morphine  
b) Opium  
c) Onion  
d) None of these
46. How is substance use disorder treated?  
a) Multiple types of treatment  
b) Alcohol drinking  
c) Recognition of problem  
d) Both (a) & (c)
47. What are proactive factors for addictions?  
a) Positive relationship  
b) Drug experimentation  
c) Community poverty  
d) All of these
48. Objectives of communication skill are  
a) Active listening skill  
b) Aware of own communication barrier  
c) Both (a) & (b)  
d) None of these
49. Post Correspondence Problem (PCP) was originally developed as  
a) A drug to induce vomiting  
b) An anesthetic used in veterinary practice  
c) An anesthetic with which person remains conscious but feels no pain  
d) An experimental psychiatric medication.
50. Which of the following statement about amphetamines is true?  
a) They are not addictive  
b) They are stimulants  
c) They have no known side effects  
d) They cause drop in heart rate and respiration rate.

\* \* \* \* \*

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# CBCS SCHEME

BSFHK158/258

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Question Paper Version : B

**First/Second Semester B.E./B.Tech Degree Examination,  
Dec.2024/Jan.2025**

## **Scientific Foundations of Health**

Time: 1 hr.]

[Max. Marks: 50

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- 
1. Wellness dimensions are  
a) 2                      b) 4                      c) 6                      d) 8
  2. What are the reasons for taking drug?  
a) To feel good              b) To do better              c) To feel better              d) All of these
  3. Planning refers to  
a) Communities in which they live in              b) Health services  
c) Financial stability              d) None of these
  4. What is the goal of social engineering?  
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13. Communication disease can spread when we use something already used by some one like a towel , cup , handkerchief etc. This statement refers to which of the following means through which the disease can be spread. i) Insects ii) Contact.  
a) Only i  
b) Only ii  
c) Both i & ii  
d) Neither i & ii.
14. Social exclusion refers to  
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**BSFHK158/258**

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**Question Paper Version : C**

**First/Second Semester B.E./B.Tech Degree Examination,  
Dec.2024/Jan.2025**

## Scientific Foundations of Health

Time: 1 hr.]

[Max. Marks: 50]

## INSTRUCTIONS TO THE CANDIDATES

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Ver - C - 1 of 5

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BSFHK158/258

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**Question Paper Version : D**

**First/Second Semester B.E./B.Tech Degree Examination,**

**Dec.2024/Jan.2025**

# Scientific Foundations of Health

Time: 1 hr.]

[Max. Marks: 50]

## INSTRUCTIONS TO THE CANDIDATES

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a) 2 Billion                                  b) Half Billion  
c) 1 Billion                                  d) Quarter of a billion
  - Communication disease can spread when we use something already used by some one like a towel , cup , handkerchief etc. This statement refers to which of the following means through which the disease can be spread. i) Insects      ii) Contact.  
a) Only i                  b) Only ii                  c) Both i & ii                  d) Neither i & ii.
  - Social exclusion refers to  
a) Poverty                  b) Old age                  c) Mental ill health                  d) All of these
  - Which disease spreads through open defecation?  
a) Cholera                  b) Malaria                  c) Dengue                  d) Elephantiasis
  - Emotional health refers to  
a) Problem solving                  b) Empathy towards others feeling  
c) Drinking lot of water                  d) Having good nutrition

7. Basic Instincts of human life  
a) Self preservance    b) Social    c) Sexual    d) All of these
8. Signs of compulsive buying include  
a) Purchasing only one item of your favorite color  
b) Showing new clothes to your friends or family  
c) Repeatedly buying more than you need or can afford  
d) Purchasing several gifts when they go on sale.
9. The most noticeable effect of THC, the psycho active substance in marijuana is  
a) Hyper active    b) Blood shot eyes    c) Slouching posture    d) Loss of appetite
10. Emotional well being includes.  
a) Smoking    b) Drinking alcohol  
c) Stress situation management    d) None of these
11. Globalization , growth of trade unions, public relations and positive atmosphere are some factors that enable \_\_\_\_  
a) Good friendship    b) Isolation from society  
c) Happiness    d) Communication growth
12. Which of the following is NOT a characteristic of addiction?  
a) Denial    b) Negative consequences  
c) Loss of control    d) Habitual behavior
13. Denial is person's  
a) Experience of a blackout episode or memory loss.  
b) Inability to perceive that a behavior is self – destructive  
c) Failure to stop an addictive behavior  
d) Change in usual pattern of drinking or drug use
14. Temporary physical and psychological symptoms that occur when use of an addictive substance is discontinued is  
a) Relapse    b) Compulsion    c) Withdrawal    d) Addiction
15. Chemicals that relay messages (impulses) from one nerve cell to another or to other cells are \_\_\_\_  
a) Enzymes    b) Neuro transmitters  
c) Synapses    d) Chemoreceptors
16. What is the definitions of overweight?  
a)  $BMI > 25 \text{ kg/m}^2$     b)  $BMI = 25 \text{ kg/m}^2$   
c)  $BMI 25 - 29.9 \text{ kg/m}^2$     d)  $BMI < 25 \text{ kg/m}^2$
17. Communication is a part of \_\_\_\_ skills.  
a) Soft    b) Hard    c) Short    d) Rough



30. Which of the following statement about amphetamines is true?  
a) They are not addictive  
b) They are stimulants  
c) They have no known side effects  
d) They cause drop in heart rate and respiration rate.
31. Alcohol based hand rub is the preferred method of hand hygiene.  
a) False                      b) True                      c) Not sure                      d) None of these
32. Is addiction a  
a) Mental disorder      b) Physical disorder      c) Both (a) & (b)                      d) None of these
33. Important roles of life are  
a) Feeling happy      b) Enjoy life                      c) Fighting diseases      d) All of these
34. Signs of mental illness are  
a) Abnormal thinking , perception and judgment  
b) Abnormal changes in feeling and memory  
c) Both (a) & (b)  
d) Abnormal changes in behavior towards others.
35. People suffering from mental diseases in India.  
a) 1 – 1.5 millions      b) 6 – 7 millions                      c) 10 – 15 millions                      d) 60 – 70 millions
36. Social health hazards includes.  
a) Stigma                      b) Anti – social behaviour  
c) Higher crime records                      d) All of these
37. Hyper obesity value in terms of BMI.  
a) > 25                      b) 40                      c) > 40                      d) < 40
38. Goals of communication.  
a) To inform , to persuade                      b) To inform, fear of offending  
c) To persuade, fear of offending                      d) None of these
39. In which year WHO articulated the right to health in its constitution?  
a) 2000                      b) 1946                      c) 1952                      d) 1987
40. Spiritual health refers  
a) Body composition      b) Mediation                      c) Exercising                      d) All of these
41. This causes maximum accumulation of fat in the liver.  
a) Meat & Egg      b) Alcohol                      c) Saturated fat                      d) Starch
42. Using abbreviations in communication leads to which type of communication barrier.  
a) Language / Linguistic                      b) Physical  
c) Cultural                      d) Organizational

43. Obesity occurs due to ; More consumption of  
a) Calorie                      b) Protein                      c) Vitamins                      d) None of these
44. What is intimately related?  
a) Body & health                      b) Body & mind  
c) Disease & Health                      d) Body & Spiritual values
45. What influences your health?  
a) Heredity                      b) Environment  
c) Physical environment                      d) All of these
46. A lack of judgment regarding whether a particular substance or behavior will be healthy of damaging to oneself is known as  
a) Denial                      b) Obsession                      c) Loss of control                      d) Compulsion
47. Which of the following diseases is transmitted by an insect vector?  
a) Malaria                      b) HIV/AIDS                      c) CTB                      d) None of these
48. Disease of the heart, joint and nervous system is called  
a) Degenerative diseases                      b) Mental diseases  
c) Communicable diseases                      d) Deficiency diseases
49. How often one must have a routine body checkup?  
a) Once in 10 years                      b) Twice a year                      c) Once a year                      d) Once in 5 year
50. How do puzzles help you?  
a) Physical exercise                      b) They are a time pass  
c) They help exercise the brain                      d) They help you learn numbers

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